

128Gb Based NAND Flash

128Gb, 256Gb, 512Gb, 1Tb and 2Tb NAND Flash Specification

Document Title

NAND Flash

Revision History

Revision No.	History	Date	Remark
0.1	- Draft Version	Aug. 2013	
0.2	- Corrected TSOP Pin assignent	Aug. 2013	
0.3	- Internal release	Dec. 2013	
0.4	- Corrected typo error On page 5 and 48 to correct the typo	Dec. 2013	
0.5	- Inserted a H27QDG8D2B8R product On page 6 and 10	Jan. 2014	
0.6	- Corrected ball assignments (316-Ball, site Y-6) On page 16	Jan. 2014	
0.7	- Corrected Read ID Operation - Corrected Some Typo	Feb. 2014	
0.8	- Added a eMLC lineup On page 5 and 6 - Determined Read ID for individual list of package type On page 45 and 69 - Corrected Some Typo for Table 5-16 On page 88	Feb. 2014	
0.9	- Corrected Some Typo On page 5	Feb.2014	
0.91	- Corrected Some Typo at Set/Get Parameter - Deleted unnecessary timing parameters	Mar.2014	
0.92	- Deleted Intelligent Copyback Operation	Mar.2014	
0.93	- Changed note of Figure 4.5 - Changed Figure 4.12 - Changed Figure 2 on page 103	Mar.2014	
0.94	- Removed cache program on status read	Mar. 2014	
0.95	- Changed Figure2,4 of Addendum - Fixed typo on Figure 20, 22,27 (tDBSY) - Defined tCDQSS note of Fingure 4-5 clearly - Added Read ID speed constraint on page 53	Apr. 2014	
0.96	- Modified Half page read on page 60	May. 2014	
0.97	- Corrected Some Typo for Figure 5-38	Jul. 2014	
0.98	- Corrected Some Typos on Timing Mode Transitions	Jul. 2014	
0.99	- Corrected Some Typos on Product list table	Jul. 2014	
1.0	- Corrected Some Typos on Power Down Sequence	Aug. 2014	

1.1	- Corrected Some Typos for Table 2-4	Aug. 2014	
1.2	- Corrected capacitances for Table 2-4	Aug. 2014	
1.3	- Fixed Typos - Modified Figure 2-6 (316 ball assignments) on page 16 - Added "ODT not supported" comments - Corrected SLC Address in Table 4-3 - Removed Cache Read contents	Aug. 2014	
1.4	- Added part number in the Table 1-1 - Added tWB comment on page 33 - Fixed Typos in the Table 5-15 - Using VREFQ instead VREF	Sep. 2014	
1.5	- Add valid block table - Modified Figure 5-28 and Figure 5-29 for Register Read Out mode - Fixed some typos - Remove tITC, it's the same as tFEAT. - Add Read ID information to the Table 5-7, 5-8	Oct. 2014	
1.6	- Change tBERS to 7ms(typ.) - Change busy time of 1st FFh to 5ms at Figure 4-2 - Add VREF pin description - Fixed typos	Nov. 2014	
1.7	- Remove unnecessary notes on page 5 - Fix the 316 ball map on page 17	Mar. 2015	
1.8	- Changed e-MLC tBERS to 11ms(typ.) on page5 - Inserted a AC parameter for tCDQSH value on page 32, 33 and 56.	Mar. 2015	
1.9	- Changed typo on page8	Jun. 2015	

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Key features

■ Multilevel Cell technology

■ NAND INTERFACE

- Toggle DDR Command Interface
- x8 bus width
- Multiplexed Command, Address and data signal port

■ Supply Voltage

Vcc : 2.7V ~ 3.6V, VccQ : 1.7V ~ 1.95V_All speed
 Vcc : 2.7V ~ 3.6V, VccQ : 2.7V ~ 3.6V_200Mbps under

■ Organization

- (16,384+1,664)bytes x 256pages x (2,048+44)blocks x 2plane

■ Page Read / Program Time

	e-MLC	c-MLC	channel-MLC
Page / Spare Size	16,384+1,664		
Block Size	4MB per Device		
P/E Cycles	10K	3K	1K
Data Retention	3-month	1-yr	1-yr
Random Read Time	75us (typ), 110us (max)		
Page Program Time	1.8ms (typ), 5ms (max)	1.5ms (typ), 5ms (max)	
Erase Time	11ms (typ), 15ms (max)	7ms (typ), 10ms (max)	

- DQ Burst Read : 80 mA max. (ICC4R)
- DQ Burst Program: 80 mA max. (IDD4W)
- BUS Idle : 5 mA max.
- Standby CMOS: 50 uA max.

■ Package

- 152Ball fBGA , Size : 14x18 mm²
- 316Ball fBGA, Size : 14x18 mm²

■ Product

- 16GByte: single die stack
- 32GByte: two-die stack
- 64GByte: four-die stack
- 128GByte: eight-die stack
- 256GByte: sixteen-die stack

PKG type				
	Channel	CE	RB#	Type
SDP	single	1	1	152Ball
DDP	Dual	2	2	152Ball
QDP	Quad	4	4	316Ball
QDP	Dual	4	4	152Ball
ODP	Quad	4	4	316Ball
ODP	Dual	4	4	152Ball
HDP	Quad	4	4	316Ball
HDP	Dual	4	4	152Ball

■ DQ performance

- Read cycle time :
 tRC = 5ns
 or tRC = 10ns
- Read/Write throughput per pin :400Mb/s or 200Mb/s

■ Single Die Operating Current

- Page Read : 50 mA max.
- Page Program : 50 mA max.

1. SUMMARY DESCRIPTION

The SKHynix NAND Flash is suited for high performance applications which use the batteries such as high-end mobile solution and PCs data storage solution. The 128Gb based NAND series has a toggle DDR interface.

Operates from a bidirectional DQS : SKHynix NAND Flash technology provides high-performance NAND Flash memory with an interface that supports up to 200Mbps or up to 400Mbps data read and write throughput.

H27Uxx8xxBxR-BCF NAND Flash has a fast interface, rated at 200Mbps, as compared to 40Mbps for legacy single data rate.
H27Qxx8xxBxR-BCC NAND Flash relatively already 40Mbps of speed of information processing of SDR NAND Flash are 10 times faster, amount to 400Mbps.

The high-speed bandwidth of SKHynix NAND Flash is supported to better support the ongoing shift toward advanced interfaces, as more mobile and PCs electronics devices requiring added performance and higher densities adopt interfaces such as embedded NAND solution series and Serial SSD solution series.

1.1. Product List

Table 1-1. Package product list

152-fBGA						
P/N	Density	Vcc/VccQ	I/O Speed	# of CE RB#	PACKAGE	Remark
H27QDG8T2B8R-BCF	16GB (128Gb)	3.3V / 1.8V	400Mb/s	1CE & 1R/B, single	152_fBGA	c-MLC
H27QEG8UDB8R-BCF	32GB (256Gb)	3.3V / 1.8V	400Mb/s	2CE & 2R/B, dual	152_fBGA	c-MLC
H27QFG8VEB8R-BCF	64GB (512Gb)	3.3V / 1.8V	400Mb/s	4CE & 4R/B, dual	152_fBGA	c-MLC
H27Q1T8YEB9R-BCF	128GB (1Tb)	3.3V / 1.8V	400Mb/s	4CE & 4R/B, dual	152_fBGA	c-MLC
H27Q2T8CEB9R-BCF ¹⁾	256GB (2Tb)	3.3V / 1.8V	400Mb/s	4CE & 4R/B, dual	152_fBGA	c-MLC
H27QDG822B8R-BCF ¹⁾	16GB (128Gb)	3.3V / 1.8V	400Mb/s	1CE & 1R/B, single	152_fBGA	e-MLC
H27QEG83DB8R-BCF ¹⁾	32GB (256Gb)	3.3V / 1.8V	400Mb/s	2CE & 2R/B, dual	152_fBGA	e-MLC
H27QFG84EB8R-BCF ¹⁾	64GB (512Gb)	3.3V / 1.8V	400Mb/s	4CE & 4R/B, dual	152_fBGA	e-MLC
H27Q1T85EB9R-BCF ¹⁾	128GB (1Tb)	3.3V / 1.8V	400Mb/s	4CE & 4R/B, dual	152_fBGA	e-MLC
H27Q2T87EB9R-BCF ¹⁾	256GB (2Tb)	3.3V / 1.8V	400Mb/s	4CE & 4R/B, dual	152_fBGA	e-MLC
H27QDG8D2B8R-BCF ²⁾	16GB (128Gb)	3.3V / 1.8V	400Mb/s	1CE & 1R/B, single	152_fBGA	Channal MLC
H27QEG8LUD8R-BCF	32GB (256Gb)	3.3V / 1.8V	400Mb/s	2CE & 2R/B, dual	152_fBGA	
H27QFGLVEB8R-BCF	64GB (512Gb)	3.3V / 1.8V	400Mb/s	4CE & 4R/B, dual	152_fBGA	
H27Q1TLYEB9R-BCF	128GB (1Tb)	3.3V / 1.8V	400Mb/s	4CE & 4R/B, dual	152_fBGA	

316-fBGA					
P/N	Density	Vcc/VccQ	I/O Speed	# of CE RB#	PACKAGE
H27QFG8VQB2R-BCF	64GB (512Gb)	3.3V / 1.8V	400Mb/s	4CE & 4R/B, quad	316Ball fBGA
H27Q1T8YQB3R-BCF	128GB (1Tb)	3.3V / 1.8V	400Mb/s	4CE & 4R/B, quad	316Ball fBGA
H27Q2T8CQB3R-BCF	256GB (2Tb)	3.3V / 1.8V	400Mb/s	4CE & 4R/B, quad	316Ball fBGA

Note 1) Please contact to SKhynix office for a product availability.

2) H27QDG8D2B8R-BCF product is available in the emerging-market such as Digital card and USB 3.0 assemblies for a high speed nand card.

1.2. Part number Information

H	2	7	x	x	x	8	x	x	B	x	R	-	B	C	F
1~3			4	5~6		7	8	9	10	11	12	13	14	15	16

#1 ~ #3: Memory

H27: SKHynix NAND Flash

#4: Power supply

U: 3.3V Vcc, 3.3V VccQ

Q: 3.3V Vcc, 1.8V VccQ

#5 ~ #6: Density

DG: 128Gb

EG: 256Gb

FG: 512Gb

1T: 1Tb

#7: Input/Output Bus width

8: x8 input/output

L : x8 and customized ECC

I : x8 and customized ECC

#8: Die stack

T: c-MLC SDP

U: c-MLC DDP

V: c-MLC QDP

Y: c-MLC ODP

2: e-MLC SDP

3: e-MLC DDP

4: e-MLC QDP

5: e-MLC ODP

D: SDP for NAND card

#9: Configuration

2: 1CE & 1R/B, single

D: 2CE & 2R/B, dual

E: 4CE & 4R/B, dual

Q: 4CE & 4R/B, quad

#10: Generation

M: 1st product

A: 2nd product

B: 3rd product

C: 4th product

D: 5th product

E: 6th product

#11: Package

T: T-SOP

8: 152-fBGA (12x18, t=1.0, D=550um)

9: 152-fBGA (12x18, t=1.35, D=550um)

2: 316-fBGA (14x18, t=1.0, D=550um)

3: 316-fBGA (14x18, t=1.35, D=550um)

D: Wafer (PGD-2)

#12: Package material

R: Green (Lead & Halogen free)

P: Lead free

A: Wafer

#14: Bad block

B: Included bad block

P: All good block

#15: Temperature

C: Commercial temp. product

I: Industrial temp. product

#16: Input/output speed

Blank: Legacy speed

C: 100MHz (200Mb/s)

F: 200MHz (400Mb/s)

G: 266MHz (533Mb/s)

1.3. Definitions and Abbreviations

DDR : Acronym for double data rate.

Address : The address is comprised of a column address with 2 cycles and a row address with 3 cycles. The row address identifies the page, block and LUN to be accessed. The column address identifies the byte within a page to access. The least significant bit of the column address shall always be zero.

Column : The byte location within the page register.

Row : Refer to the block and page to be accessed.

Page : The smallest addressable unit for the Read and the Program operations.

Block : Consists of multiple pages and is the smallest addressable unit for the Erase operation.

Page register : Register used to transfer data to and from the Flash Array.

Defect area : The defect area is where factory defects are marked by the manufacturer. Refer to the "*Factory Defect Mapping*".

Device : The packaged NAND unit. A device may contain more than a target.

LUN (Logical Unit Number) : The minimum unit that can independently execute commands and report status. There are one or more LUNs per CE.

Target : An independent NAND Flash component with its own CE signal.

SR[x] (Read Status) : SR refers to the status register contained within a particular LUN. SR[x] refers to bit x in the status register for the associated LUN. Refer to "*Read Status Operation*" for the definition of bit meanings within the status register.

1.4. Diagram Legend

Diagrams in the Toggle DDR datasheet use the following legend:

This legend shows the command data. Refer to the Table "*Basic Command Sets*" for more information about the command data.

Command

This legend shows the Address data. The address are comprised of 2 cycles column address and 3 cycles row address.

Address (C1 C2 R1 R2 R3)

C1 : Column address 1

C2 : Column address 2

R1 : Row address 1

R2 : Row address 2

R3 : Row address 3

This legend shows Host writing data(data input) to the device.

W-Data

This legend shows Host reading data(data output) from the device.

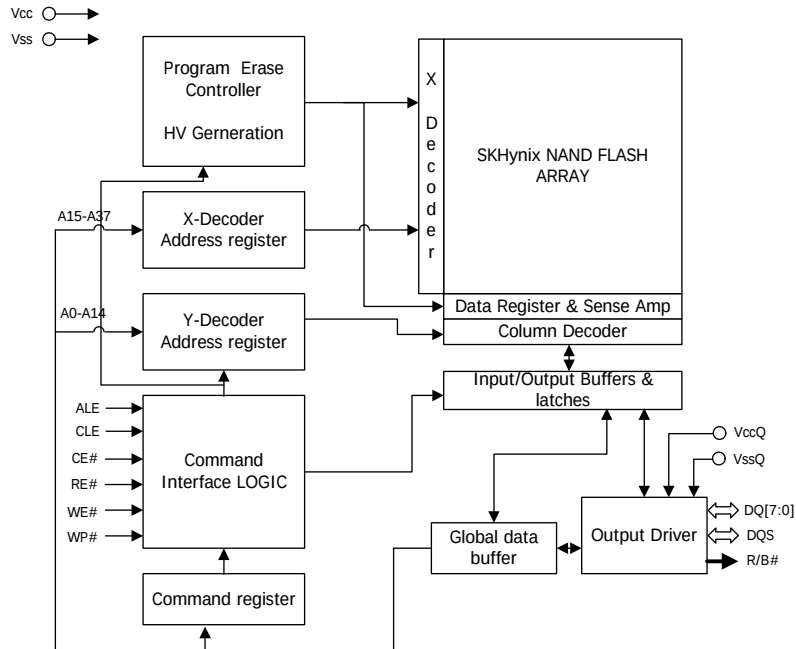
R-Data

This legend shows Host reading the status register within a particular LUN.

SR[x]

1.5. Block Diagram

Figure 1-1. NAND Flash Die Functional Block Diagram



2. Physical Interface and Measurement condition

2.1. Package Information

[152-Ball Pin Assignment(Top view)]

Figure 2-1. H27QDG8T2B8R, H27QDG8D2B8R: 1CE Single Channel ball assignments for 8-bit data access

	1	2	3	4	5	6	7	8	9	10	11	12	13
A	NC	NC	NC	NC						NC	NC	NC	NC
B	NC	NC	NC	NC						NC	NC	NC	NC
C	NC	NC	NC	NC						NC	NC	NC	NC
D	NC	NC	VccQ	VccQ	NC	Vss		Vcc	NC	VccQ	VccQ	NC	NC
E	NC	NC	VssQ	NC	VssQ	NC		NC	VssQ	NC	VssQ	NC	NC
F			NC	NC	NC	NC		NC	NC	NC	NC		
G			VssQ	VccQ	NC	NC		NC	NC	VccQ	VssQ		
H			NC	NC	NC	NC		NC	NC	NC	NC		
J			Vss	Vcc	R/B	NC		NC	NC	Vcc	Vss		
K			NC	NC	$\overline{CE}$	NC		VPP (or NU)	$\overline{WP}$	NC	NC		
L			VssQ	VccQ	NC	NC		CLE	ALE	VccQ	VssQ		
M			DQ7	DQ6	VREFQ	$\overline{WE}$		RE	$\overline{DQS}$	DQ1	DQ0		
N	NC	NC	VssQ	DQ5	VssQ	$\overline{RE}$		DQS	VssQ	DQ2	VssQ	NC	NC
P	NC	NC	VccQ	VccQ	DQ4	Vcc		Vss	DQ3	VccQ	VccQ	NC	NC
R	NC	NC	NC	NC						NC	NC	NC	NC
T	NC	NC	NC	NC						NC	NC	NC	NC
U	NC	NC	NC	NC						NC	NC	NC	NC

152 ball fBGA, (1-Ch, 1CE#)

Note 1) For H27QDG8D2B8R-BCF product, K-9 Pin site is a "Not Use".

Figure 2-2. H27QEG8UDB8R: 2CE Dual Channel ball assignments for 8-bit data access

	1	2	3	4	5	6	7	8	9	10	11	12	13
A	NC	NC	NC	NC						NC	NC	NC	NC
B	NC	NC	NC	NC						NC	NC	NC	NC
C	NC	NC	NC	NC						NC	NC	NC	NC
D	NC	NC	VccQ	VccQ	DQ3_1	Vss		Vcc	DQ4_1	VccQ	VccQ	NC	NC
E	NC	NC	VssQ	DQ2_1	VssQ	DQS_1		$\overline{RE}_1$	VssQ	DQ5_1	VssQ	NC	NC
F			DQ0_1	DQ1_1	$\overline{DQS}_1$	RE_1		$\overline{WE}_1$	VREFQ_1	DQ6_1	DQ7_1		
G			VssQ	VccQ	ALE_1	CLE_1		NC	NC	VccQ	VssQ		
H			NC	NC	$\overline{WP1}_1$	NC		NC	$\overline{CE0}_1$	NC	NC		
J			Vss	Vcc	R/ $\overline{B0}$	NC		NC	R/ $\overline{B1}$	Vcc	Vss		
K			NC	NC	$\overline{CE0}_0$	NC		VPP (or NU)	$\overline{WP}_0$	NC	NC		
L			VssQ	VccQ	NC	NC		CLE_0	ALE_0	VccQ	VssQ		
M			DQ7_0	DQ6_0	VREFQ_0	$\overline{WE}_0$		RE_0	$\overline{DQS}_0$	DQ1_0	DQ0_0		
N	NC	NC	VSSQ	DQ5_0	VssQ	$\overline{RE}_0$		DQS_0	VssQ	DQ2_0	VssQ	NC	NC
P	NC	NC	VccQ	VccQ	DQ4_0	Vcc		Vss	DQ3_0	VccQ	VccQ	NC	NC
R	NC	NC	NC	NC						NC	NC	NC	NC
T	NC	NC	NC	NC						NC	NC	NC	NC
U	NC	NC	NC	NC						NC	NC	NC	NC

152 ball fBGA, (2-Ch, 2CE#)

Figure 2-3. H27QFG8VEB8R, H27Q1T8YEB9R & H27Q2T8CEB9R: 4CE Dual Channel ball assignments for 8-bit data success

	1	2	3	4	5	6	7	8	9	10	11	12	13
A	NC	NC	NC	NC						NC	NC	NC	NC
B	NC	NC	NC	NC						NC	NC	NC	NC
C	NC	NC	NC	NC						NC	NC	NC	NC
D	NC	NC	VccQ	VccQ	DQ3_1	Vss		Vcc	DQ4_1	VccQ	VccQ	NC	NC
E	NC	NC	VssQ	DQ2_1	VssQ	DQS_1		$\overline{\text{RE}}_1$	VssQ	DQ5_1	VssQ	NC	NC
F			DQ0_1	DQ1_1	$\overline{\text{DQS}}_1$	RE_1		$\overline{\text{WE}}_1$	VREFQ_1	DQ6_1	DQ7_1		
G			VssQ	VccQ	ALE_1	CLE_1		NC	NC	VccQ	VssQ		
H			NC	NC	$\overline{\text{WP}}_1$	NC		$\overline{\text{CE}}_1_1$	$\overline{\text{CE}}_0_1$	NC	NC		
J			Vss	Vcc	R/ $\overline{\text{B}}_0$	R/ $\overline{\text{B}}_2$		R/ $\overline{\text{B}}_3$	R/ $\overline{\text{B}}_1$	Vcc	Vss		
K			NC	NC	$\overline{\text{CE}}_0_0$	$\overline{\text{CE}}_1_0$		VPP (or NU)	$\overline{\text{WP}}_0$	NC	NC		
L			VssQ	VccQ	NC	NC		CLE_0	ALE_0	VccQ	VssQ		
M			DQ7_0	DQ6_0	VREFQ_0	$\overline{\text{WE}}_0$		RE_0	$\overline{\text{DQS}}_0$	DQ1_0	DQ0_0		
N	NC	NC	VSSQ	DQ5_0	VssQ	$\overline{\text{RE}}_0$		DQS_0	VSSQ	DQ2_0	VssQ	NC	NC
P	NC	NC	VccQ	VccQ	DQ4_0	Vcc		Vss	DQ3_0	VccQ	VccQ	NC	NC
R	NC	NC	NC	NC						NC	NC	NC	NC
T	NC	NC	NC	NC						NC	NC	NC	NC
U	NC	NC	NC	NC						NC	NC	NC	NC

152 ball fBGA, (2-Ch, 4CE#)

Figure 2-4. TBD: 8CE Dual Channel ball assignments for 8-bit data access

	1	2	3	4	5	6	7	8	9	10	11	12	13
A	NC	NC	NC	NC						NC	NC	NC	NC
B	NC	NC	NC	NC						NC	NC	NC	NC
C	NC	NC	NC	NC						NC	NC	NC	NC
D	NC	NC	VccQ	VccQ	DQ3_1	Vss		Vcc	DQ4_1	VccQ	VccQ	NC	NC
E	NC	NC	VssQ	DQ2_1	VssQ	DQS_1		$\overline{RE}_1$	VssQ	DQ5_1	VssQ	NC	NC
F			DQ0_1	DQ1_1	$\overline{DQS}_1$	RE_1		$\overline{WE}_1$	VREFQ_1	DQ6_1	DQ7_1		
G			VssQ	VccQ	ALE_1	CLE_1		$\overline{CE3}_1$	$\overline{CE2}_1$	VccQ	VssQ		
H			NC	NC	$\overline{WP1}$	NC		$\overline{CE1}_1$	$\overline{CE0}_1$	NC	NC		
J			Vss	Vcc	R/ $\overline{B0}$	R/ $\overline{B2}$		R/ $\overline{B3}$	R/ $\overline{B1}$	Vcc	Vss		
K			NC	NC	$\overline{CE0}_0$	$\overline{CE1}_0$		VPP (or NU)	$\overline{WP}_0$	NC	NC		
L			VssQ	VccQ	$\overline{CE2}_0$	$\overline{CE3}_0$		CLE_0	ALE_0	VccQ	VssQ		
M			DQ7_0	DQ6_0	VREFQ_0	$\overline{WE}_0$		RE_0	$\overline{DQS}_0$	DQ1_0	DQ0_0		
N	NC	NC	VSSQ	DQ5_0	VssQ	$\overline{RE}_0$		DQS_0	VSSQ	DQ2_0	VssQ	NC	NC
P	NC	NC	VccQ	VccQ	DQ4_0	Vcc		Vss	DQ3_0	VccQ	VccQ	NC	NC
R	NC	NC	NC	NC						NC	NC	NC	NC
T	NC	NC	NC	NC						NC	NC	NC	NC
U	NC	NC	NC	NC						NC	NC	NC	NC

152 ball fBGA, (2-Ch, 8CE#)

Figure 2-5. BGA-152 ball spacing requirements (top & view, dimensions in millimeters)

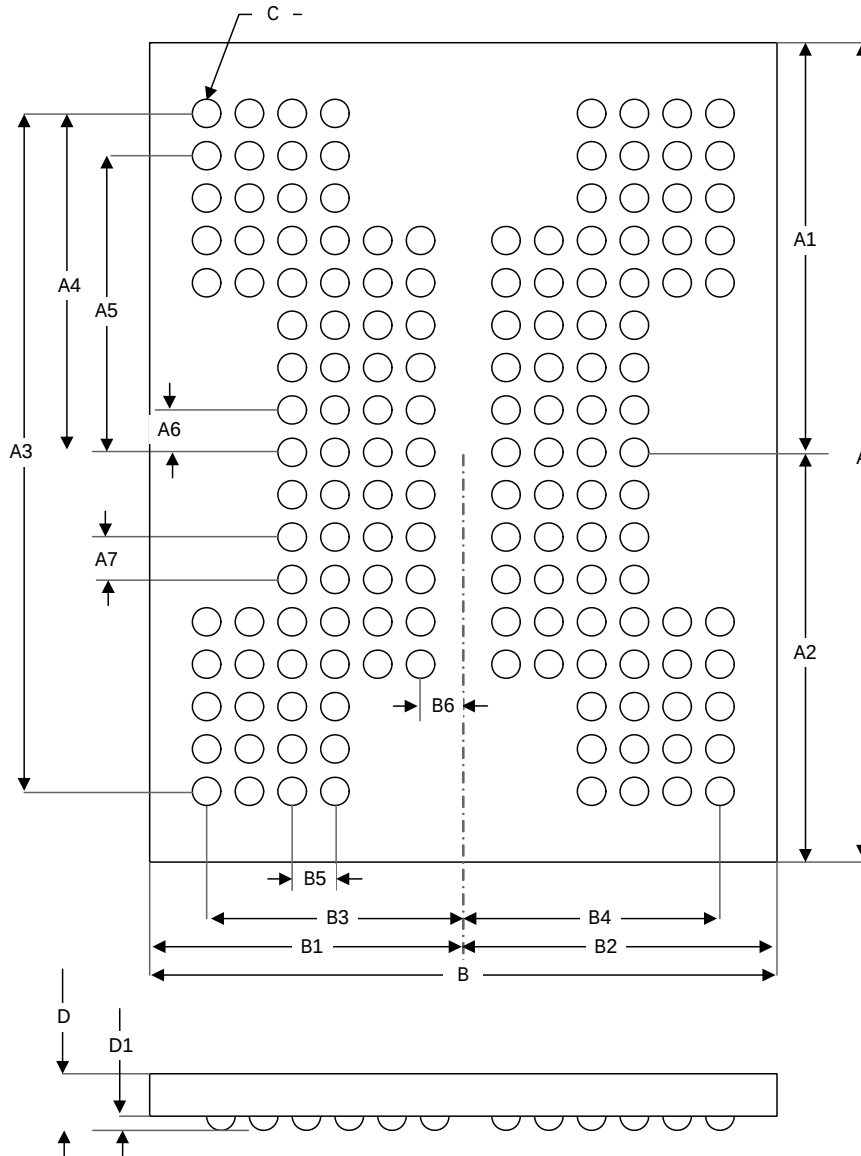


Figure 2-5. defines the ball spacing requirements for the 152-ball BGA package. The solder ball diameter is 0.55mm post reflow. The 152-ball BGA has one package size : 14mm x 18mm.

Table 2-1. 152-fBGA 14x18mm, Package Mechanical Data

Symbol	Milimeters			Symbol	Milimeters		
	Min	Typ	Max		Min	Typ	Max
A	17.90	18	18.10	B	13.90	14	14.10
A1	-	9	-	B1	-	7	-
A2	-	9	-	B2	-	7	-
A3	-	16	-	B3	-	6	-
A4	-	8	-	B4	-	6	-
A5	-	7	-	B5	-	1	-
A6	-	1	-	B6	-	1	-
A7	-	1	-				
Symbol	Milimeters			Symbol	Milimeters		
	Min	Typ	Max		Min	Typ	Max
C ^{Note1)}	0.50	0.55	0.60	D ^{Note1)}	0.8	0.9	1.0
				D ^{Note 2)}	1.15	1.25	1.35
				D1	0.27	0.32	0.37

Note 1) H27QDG8T2B8R, H27QEG8UDB8R and H27QFG8VEB8R

Note 2) H27Q1T8YEB6R and H27Q2T8CEB9R

Figure 2-6. H27QFG8VQB2R, H27Q1T8YQB3R & H27Q2T8CEB3R: $\overline{\text{4CE}}$ Quad-Channel, 316 ball assignments for 8-bit data access

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC
B	NC	NC	NC	VCCQ	VSSQ	VCCQ	VSSQ	VREF	VCC	VSS	VCC	VSS	VCC	NC	NC	NC
C	NC	NC	VSSQ	VCC	VSS	DQ7_2	DQ7_0			VCCQ	NC	NC	NC	VPP	NC	NC
D	NC	VCC	VSS	NC	NC	DQ6_2	DQ6_0			VSSQ	NC	NC	NC	VSS	VCC	NC
E	NC	VCCQ	VSSQ	NC	NC	DQ5_2	DQ5_0			NC	R/ $\overline{\text{B0}}_2$	NC	NC	NC	NC	NC
F	NC	NC	NC	VCCQ	VSSQ	DQ4_2	DQ4_0			NC	R/ $\overline{\text{B0}}_0$	NC	NC	NC	VSS	NC
G	NC	VCCQ	VSSQ	VCC	VSS	DQS_2	DQS_0			$\overline{\text{WP}}_0$	$\overline{\text{WP}}_2$	$\overline{\text{CE}}1_2$	NC	NC	VCC	NC
H	NC	VCC	VSS	VCCQ	VSSQ	$\overline{\text{DQS}}_2$	$\overline{\text{DQS}}_0$			CLE_0	CLE_2	$\overline{\text{CE}}1_0$	NC	NC	NC	NC
J	NC	NC	NC	NC	DQ3_2	DQ3_0	RE_2			RE_0	ALE_2	$\overline{\text{CE}}0_2$	NC	NC	VSS	NC
K	NC	VCCQ	VSSQ	NC	DQ2_2	DQ2_0	$\overline{\text{RE}}_2$			$\overline{\text{RE}}_0$	ALE_0	$\overline{\text{CE}}0_0$	NC	NC	VCC	NC
L	NC	NC	VREFQ	NC	DQ1_2	DQ1_0	$\overline{\text{WE}}_2$			$\overline{\text{WE}}_0$	DQ0_1	DQ0_3	NC	VSSQ	VCCQ	NC
M	NC	VCCQ	VSSQ	NC	DQ0_2	DQ0_0	$\overline{\text{WE}}_1$			$\overline{\text{WE}}_3$	DQ1_1	DQ1_3	NC	VREF	NC	NC
N	NC	VCC	NC	NC	$\overline{\text{CE}}0_1$	ALE_1	$\overline{\text{RE}}_1$			$\overline{\text{RE}}_3$	DQ2_1	DQ2_3	NC	VSSQ	VCCQ	NC
P	NC	VSS	NC	NC	$\overline{\text{CE}}0_3$	ALE_3	RE_1			RE_3	DQ3_1	DQ3_3	NC	NC	NC	NC
R	NC	NC	NC	NC	$\overline{\text{CE}}1_1$	CLE_3	CLE_1			$\overline{\text{DQS}}_1$	$\overline{\text{DQS}}_3$	VSSQ	VCCQ	VSS	VCC	NC
T	NC	VCC	NC	NC	$\overline{\text{CE}}1_3$	$\overline{\text{WP}}_3$	$\overline{\text{WP}}_1$			DQS_1	DQS_3	VSS	VCC	VSSQ	VCCQ	NC
U	NC	VSS	NC	NC	NC	R/ $\overline{\text{B0}}_1$	NC			DQ4_1	DQ4_3	VSSQ	VCCQ	NC	NC	NC
V	NC	NC	NC	NC	NC	R/ $\overline{\text{B0}}_3$	NC			DQ5_1	DQ5_3	NC	NC	VSSQ	VCCQ	NC
W	NC	VCC	VSS	NC	NC	NC	VSSQ			DQ6_1	DQ6_3	NC	NC	VSS	VCC	NC
Y	NC	NC	VPP	NC	NC	NC	VCCQ			DQ7_1	DQ7_3	VSS	VCC	VSSQ	NC	NC
AA	NC	NC	NC	VCC	VSS	VCC	VSS	VCC	VREF	VSSQ	VCCQ	VSSQ	VCCQ	NC	NC	NC
AB	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC

316 ball fBGA, ODP, 16DP (4-Ch, 8CE#)

Figure 2-7. BGA-316 ball spacing requirements (top & view, dimensions in millimeters)

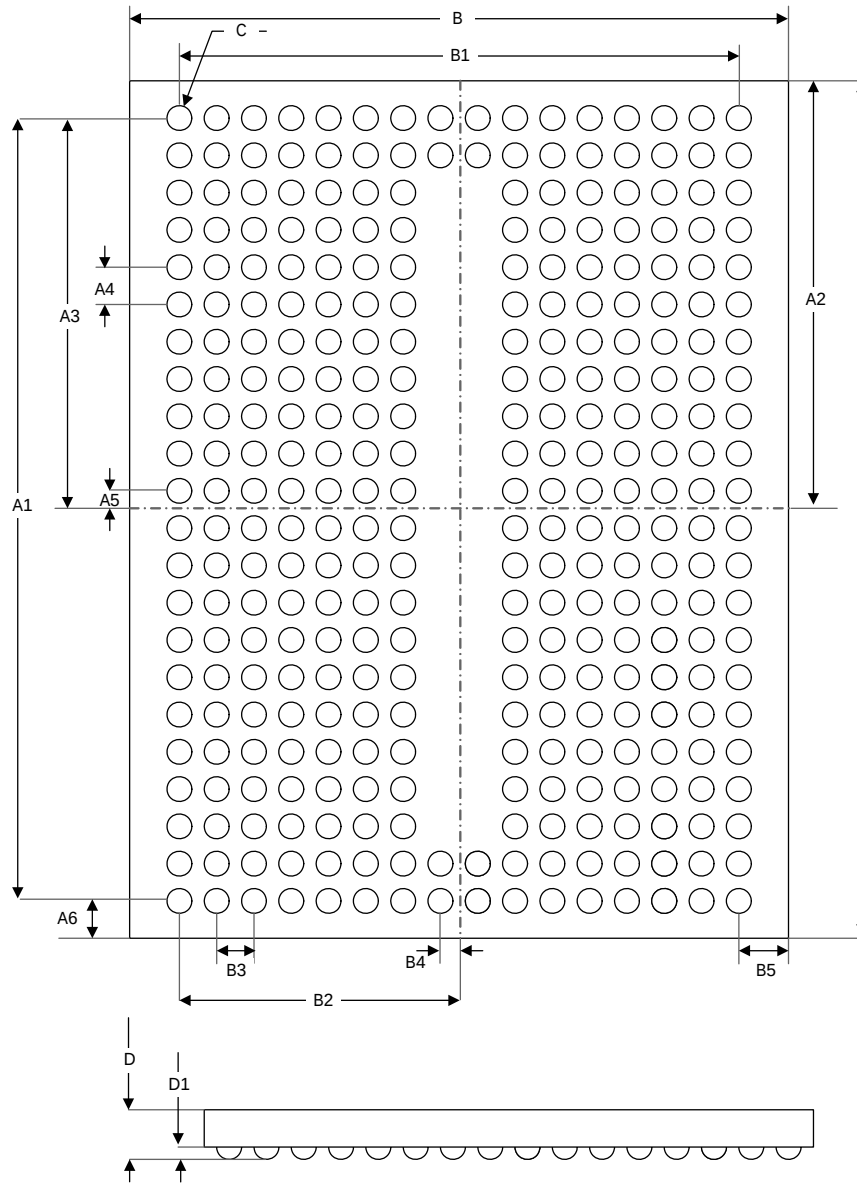


Figure 2-7 defines the ball spacing requirements for the 316-ball BGA package. The solder ball diameter is 0.50mm post reflow. The 316-ball BGA has one package size : 14mm x 18mm.

Table 2-2. BGA-316 ball 14mmx18mm, Package Mechanical Data

Symbol	Milimeters			Symbol	Milimeters		
	Min	Typ	Max		Min	Typ	Max
A	17.90	18.0	18.10	B	13.90	14.0	14.10
A1	-	16.8	-	B1	-	12.0	-
A2	-	9.0	-	B2	-	6.0	-
A3	-	8.4	-	B3	-	0.8	-
A4	-	0.8	-	B4	-	0.4	-
A5	-	0.4	-	B5	-	1.0	-
A6	-	0.6	-				
Symbol	Milimeters			Symbol	Milimeters		
	Min	Typ	Max		Min	Typ	Max
C	0.45	0.5	0.55	D	1.15	1.25	1.35
				D1	0.27	0.32	0.37

2.2. PIN DESCRIPTION

Pin Name	Description
DQ[7:0]	DATA INPUTS/OUTPUTS The DQ pins are used to input command, address and data, and to output data during read operations. The DQ pins float to high-z when the chip is deselected or when the outputs are disabled.
CLE	COMMAND LATCH ENABLE The CLE input controls the activating path for commands sent to the command register. When active high, commands are latched into the command register through the DQ ports on the rising edge of the $\overline{WE}$ signal.
ALE	ADDRESS LATCH ENABLE The ALE input controls the activating path for address to the internal address registers. Addresses are latched on the rising edge of $\overline{WE}$ with ALE high.
$\overline{CE}$	CHIP ENABLE The $\overline{CE}$ input is the device selection control. When the device is in the Busy state, $\overline{CE}$ high is ignored, and the device does not return to standby mode in program or erase operation.
$\overline{RE}$	Read Enable (True). RE_t The RE input is the serial data-out control, and when active, drives the data onto the DQ bus. Data is valid after t_{DQSRE} of rising edge & falling edge of RE, which also increments the internal column address counter by each one. RE is paired with differential signal RE(RE_c) to provide differential pair signaling to the system during reads.
RE	Read Enable Complement. RE_c The Read Enable Complement signal is the complementary signal to Read Enable.
$\overline{WE}$	WRITE ENABLE The $\overline{WE}$ input controls writes to the DQ port. Commands, addresses are latched on the rising edge of the $\overline{WE}$ pulse.
$\overline{WP}$	WRITE PROTECT The $\overline{WP}$ pin provides inadvertent program/erase protection during power transitions. The internal high voltage generator is reset when the $\overline{WP}$ pin is active low.
R/ $\overline{B}$	READY/BUSY OUTPUT The R/B output indicates the status of the device operation. When low, it indicates that a program, erase or random read operation is in process and returns to high state upon completion. It is an open drain output and does not float to high-z condition when the chip is deselected or when outputs are disabled.
DQS	DATA STROBE. DQS_t Data output is aligned with DQS falling/rising edge, and data inputs at DQS falling/rising.
$\overline{DQS}$	Data Strobe Complement. DQS_c The Data Strobe Complement signal is the complementary signal to Data Strobe.
Vcc	POWER VCC is the power supply for device.
VccQ	DQ POWER The VccQ is the power supply for input and/or output signals.
Vss	GROUND
VssQ	DQ GROUND The VssQ is the power supply ground
VREF	Voltage Reference, VREFQ This signal is used as an external voltage reference for input and DQ signals when VccQ 1.8V is selected. VREF must be used when data inputs/outputs is the speed of 400Mbps and over.
Vpp ³⁾	EXTERNAL HIGH VOLTAGE
NC	NO CONNECTION Lead is not internally connected.
NU	NOT USE Nothing should be connected with it.

NOTE :

- 1) Connect all Vcc and Vss pins of each device to common power supply outputs.
- 2) Do not leave all Vcc, VccQ, Vss and VssQ disconnected.
- 3) Vpp Pin: Depends on the use of customer. If you can't supplied External Vpp, this pin is NU.

Table 2-3. 152Ball fBGA Input/ Output capacitance (TA=25C, VCC=3.3V, f=100MHz)

Item	Symbol	Test Condition	Device		Min	Max	Unit
Input/Output Capacitance	CDQ	VIL=0V	H27QDG8T2B8R	128Gb (16GB)	-	6.7	pF
			H27QEG8UDB8R	256Gb (32GB)	-		
			H27QFG8VEB8R	512Gb (64GB)	-	10.0	
			H27Q1T8YEB9R	1Tb (128GB)	-	16.9	
			H27Q2T8CEB9R	2Tb (256GB)	-	32.7	
Input Capacitance	CIN	VIN=0V	H27QDG8T2B8R	128Gb (16GB)	-	6.4	
			H27QEG8UDB8R	256Gb (32GB)	-		
			H27QFG8VEB8R	512Gb (64GB)	-	9.2	
			H27Q1T8YEB9R	1Tb (128GB)	-	14.3	
			H27Q2T8CEB9R	2Tb (256GB)	-	27.3	

Note: Capacitance is periodically sampled and not 100% tested.

Table 2-4. 316Ball fBGA Input/ Output capacitance (TA=25C, VCC=3.3V, f=100MHz)

Item	Symbol	Test Condition	Device		Min	Max	Unit
Input/Output Capacitance	CDQ	VIL=0V	H27QFG8VQB2R	512Gb (64GB)	-	6.7	pF
			H27Q1T8YQB3R	1Tb (128GB)	-	10.0	
			H27Q2T8CQB3R	2Tb (256GB)	-	16.9	
Input Capacitance	CIN	VIN=0V	H27QFG8VQB2R	512Gb (64GB)	-	6.4	
			H27Q1T8YQB3R	1Tb (128GB)	-	9.2	
			H27Q2T8CQB3R	2Tb (256GB)	-	14.3	

Note: Capacitance is periodically sampled and not 100% tested.

2.3. Absolute Maximum DC Ratings

Stresses greater than those listing in Table of "*Absolute maximum rating*" may cause permanent damage to the device. This is a stress rating only. Operation beyond the operating conditions specified in Table of "*Absolute maximum rating*" is not recommended. Extended exposure beyond these conditions may affect device reliability.

Table 2-5. Absolute maximum DC ratings

Parameter	Symbol	Rating	Units
Voltage on any pin relative to VSS	VCC, VCCQ		-0.6 to +4.6
	VIN	VCCQ 3.3V ¹⁾	-0.6 to +4.6
		VCCQ 1.8V	-0.2 to +2.4
	VI/O	VCCQ 3.3V ¹⁾	-0.6 to +4.6
		VCCQ 1.8V	-0.2 to +2.4
Storage Temperature	TSTG	-65 to +100	degree
Soldering temperature	TSOLDER	260	degree

Note 1) 200Mb/s (max) product.

2.4. Operating temperature condition

Table 2-6. Operating Temperature Condition

Symbol	Parameter	Rating	Rproduct	Unit
TOPER	Operating Temperature Range for Commercial	0 to +70	H27xxx8xxBxR-BCx	degree
	Operating Temperature Range for Industrial ³⁾	-40 to 85 ³⁾	H27xxx8xxBxR-BIx ³⁾	

Note 1) Operating Temperature(TOPER) is the case surface temperature on the center/top side of the NAND.

2) The Normal Temperature Range specifies the temperatures where all NAND specifications will be supported. During operation, the NAND case temperature must be maintained between 0 to +70C for commercial under all operating conditions.

3) Please contact to SKHynix office for available product.

2.5. Recommended Operating Conditions

Table 2-7. Recommended Operating Condition

Symbol	Parameter		Normal Range			Unit
			Min	Typ.	Max	
VCC	Power Supply Voltage		2.7	3.3	3.6	V
VSS	Ground Voltage		0	0	0	
VCCQ	Supply Voltage for Input/Output signaling	1.8V I/O	1.7	1.8	1.95	
VCCQ	Supply Voltage for Input/Output signaling	3.3V I/O	2.7	3.3	3.6	
VSSQ	Ground Voltage for Input/Output signaling		0	0	0	
VPP	External high Voltage		11	12	13	

NOTE :

1) Vcc and VccQ be distinct and unique voltages. The device support one of the following Vcc/VccQ combinations,

Vcc = 3.3V, VccQ = 3.3V / Vcc = 3.3V, VccQ = 1.8V

All parameters, timing modes and other characteristics are related to the supported voltage combination.

2) The maximum external Vpp supply current per LUN is 5mA

2.6. AC Overshoot/Undershoot Requirements

The device may have AC overshoot or undershoot from VccQ and VssQ levels. Table for “AC Overshoot/Undershoot Specification” defines the maximum values that the AC overshoot or undershoot may attain. These values apply for 3.3V and 1.8V VccQ levels.

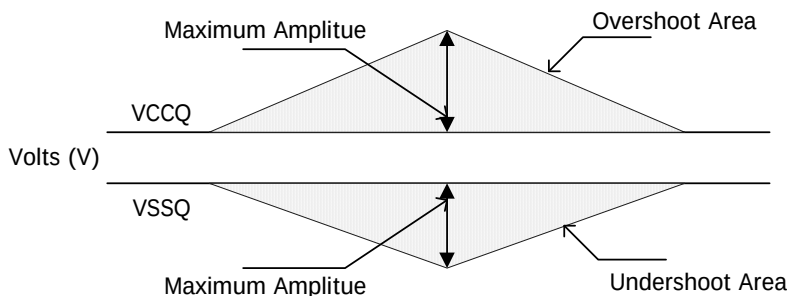
Table 2-8. AC Overshoot/Undershoot Specification

Parameter	128Gb MLC Toggle							Unit
	Maximum Value							
	~50	51~66	67~83	84~100	100~133	133~166	166~200	MHz
Max. peak amplitude allowed for overshoot area	1	1	1	1	1	1	1	V
Max. peak amplitude allowed for undershoot area	1	1	1	1	1	1	1	
Max. overshoot area above VccQ	3	2.25	1.8	1.5	1.13	0.9	0.75	V *ns
Max. undershoot area above VssQ	3	2.25	1.8	1.5	1.13	0.9	0.75	

NOTE :

1) This specification is intended for devices with no clamp protection and is guaranteed by design.

Figure 2-8. Overshoot/Undershoot Diagram



2.7. DC and Operating Characteristics

Table 2-9. DC & Operating Characteristics for VccQ=3.3V (Single Die)

Parameter	Symbol	Test Conditions	Min	Typ	Max	Units
Page Read Operation Current	ICC1	-	-	-	50	mA
Page Program Operation Current	ICC2	-	-	-	50	mA
Erase Operation Current	ICC3	-	-	-	50	mA
DQ Burst Read Current	ICC4R	$\overline{CE}=V_{IL}$, Half data switching	-	-	80	mA
	ICCQ4R	$\overline{CLOAD}=0pF$ (IOUT=0mA) $\overline{CE}=V_{IL}$, Half data switching tRC=10ns(100Mhz)	-	-	50	mA
I/O burst write current	ICC4W	$\overline{CE}=V_{IL}$, Half data switching	-	-	80	mA
	ICCQ4W	tDSC=10ns(100Mhz)	-	-	10	mA
Bus idle current	ICC5	-	-	-	10	mA

Parameter	Symbol	Test Conditions	Min	Typ	Max	Units
Standby current, CMOS	ISB (3)	$\overline{CE}=V_{CCQ}-0.2, \overline{WP}=0V/V_{CCQ}$	-	20	50	uA
Input Leakage Current	ILI	$V_{IN}=0$ to $V_{CCQ}(\max)$	-	-	+/-10	uA
Output Leakage Current	ILO	$V_{OUT}=0$ to $V_{CCQ}(\max)$	-	-	+/-10	uA
Output High Voltage Level	VOH	$I_{OH}=-400\mu A$	2.4	-	-	V
Output Low Voltage Level	VOL	$I_{OL}= 2.1mA$	-	-	0.4	V
Output Low Current(R/B)	$I_{OL}(R/\overline{B})$	$V_{OL}=0.4V$	8	10	-	mA

NOTE :

1) Typical value is measured at $V_{cc}=3.3V, T_A=25^{\circ}C$. Not 100% tested.

2) VOH and VOL should be available on these two conditions; Output Strength is nominal and $V_{ccQ}=3.3V$, R_{pd}/R_{pu} are all $V_{ccQ} \times 0.5$. If the driver strength settings are supported, Table of "DQ Driver Strength Settings" be used to derive the output driver impedance values.

3) For ICC4R the test conditions specify $I_{OUT} = 0$ mA.

Table 2-10. DC & Operating Characteristics for $V_{ccQ}=1.8V$ (Single Die)

Parameter	Symbol	Test Conditions	Min	Typ	Max	Units
Page Read Operation Current	ICC1	-	-	-	50	mA
Page Program Operation Current	ICC2	-	-	-	50	mA
Erase Operation Current	ICC3	-	-	-	50	mA
DQ Burst Read Current	ICC4R	$\overline{CE}=V_{IL}$, Half data switching	-	-	80	mA
	ICCQ4R	$\overline{C}_{LOAD}=0pF$ ($I_{OUT}=0mA$) $\overline{CE}=V_{IL}$, Half data switching $t_{RC}=5ns(200Mhz)$	-	-	50	mA
I/O burst write current	ICC4W	$\overline{CE}=V_{IL}$, Half data switching $t_{DSC}=5ns(200Mhz)$	-	-	80	mA
	ICCQ4W		-	-	10	mA
Bus idle current	ICC5	-	-	-	10	mA
Standby current, CMOS	ISB (3)	$\overline{CE}=V_{CCQ}-0.2, \overline{WP}=0V/V_{CCQ}$	-	20	50	uA
Input Leakage Current	ILI	$V_{IN}=0$ to $V_{CCQ}(\max)$	-	-	+/-10	uA
Output Leakage Current	ILO	$V_{OUT}=0$ to $V_{CCQ}(\max)$	-	-	+/-10	uA
Output High Voltage Level	VOH	$I_{OH}=-100\mu A$	$V_{CCQ}-0.1$	-	-	V
Output Low Voltage Level	VOL	$I_{OL}= 100\mu A$	-	-	0.1	V
Output Low Current(R/B)	$I_{OL}(R/\overline{B})$	$V_{OL}=0.2V$	3	4	-	mA
Vpp Idle current (Optional)	IPPI ⁴⁾	-	-	-	TBD	uA
Vpp Activecurrent (Optional)	IPPA ⁴⁾	-	-	-	TBD	mA

NOTE :

1) Typical value is measured at $V_{cc}=3.3V, T_A=25^{\circ}C$. Not 100% tested.

2) VOH and VOL should be available on these two conditions; Output Strength is nominal and $V_{ccQ}=1.8V$, R_{pd}/R_{pu} are all $V_{ccQ} \times 0.5$. If the driver strength settings are supported, Table of "DQ Driver Strength Settings" be used to derive the output driver impedance values.

3) For ICC4R the test conditions specify $I_{OUT} = 0$ mA.

4) IPP Idle current is IPP current measured when Vpp is supplied and Vpp is not enabled via Set Feature. IPP Active current is IPP current measured when Vpp is supplied and Vpp is enabled via Set Feature.

2.8. AC Test Condition

Table 2-11. Single Ended without VREFQ AC & DC input level

Parameter	Symbol	Min	Typ	Max	Units
DC Input high voltage	VIH (DC)	VCCQ * 0.7	-	VCCQ + 0.3	V
AC Input high voltage	VIH (AC)	VCCQ * 0.8	-	-	V
DC Input low voltage	VIL (DC)	-0.3	-	VCCQ * 0.3	V
AC Input low voltage	VIL (AC)	-	-	VCCQ * 0.2	V
NOTE: 1) VIL can undershoot to -0.3V and VIH can overshoot to VCCQ +0.3V for durations of 20 ns or less. 2) VREFQ shall be used at high interface speed above 200Mbps					

Table 2-12. Single Ended with VREFQ AC & DC input level

Parameter	Symbol	Min	Typ	Max	Units
DC Input high voltage	VIH (DC)	VREFQ + 0.15	-	VCCQ + 0.3	V
AC Input high voltage	VIH (AC)	VREFQ + 0.3	-	-	V
DC Input low voltage	VIL (DC)	-0.3	-	VCCQ * 0.3	V
AC Input low voltage	VIL (AC)	-	-	VREFQ - 0.3	V
Reference Voltage	VREFQ(DC)	VCCQ * 0.49		VCCQ * 0.51	V
NOTE: 1) VREFQ is used only for VCCQ 1.8V					

Table 2-13. DC Electrical Characteristics for ODT (not supported)

Parameter	Setting	Min	Typ	Max	Units	Note
RTT effective impedance value for:	50 ohm	32.5	50	67.5	ohms	1
	75 ohm	48.7	75	101.3		
	100 ohm	65	100	135		
	150 ohm	97.5	150	202.5		
VM deviation with respect to VCCQ/2	ΔVM	-7	-	7	%	

NOTE 1. Test conditions for RTT measurement.

2.8.1. VREFQ Tolerance

The DC-tolerance and AC-noise limits for the reference voltages. Figure for “VREFQ(DC) tolerance and VREFQ AC-noise limits” shows a valid reference voltage VREFQ(t) as a function of time.

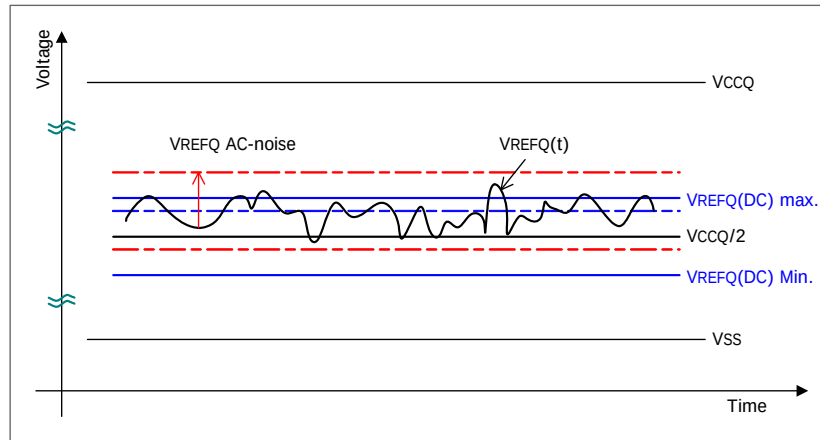
VREFQ(DC) is the linear average of VREFQ(t) over a very long period of time (e.g. 1sec). This average has to meet the min/max requirements in Table of “Single Ended with VREFQ AC & DC input level”.

VREFQ(t) may temporarily deviate from VREFQ(DC) by no more than $\pm 1\% V_{CCQ}$

The voltage levels for setup and hold time measurements $V_{IH}(AC)$, $V_{IH}(DC)$, $V_{IL}(AC)$ and $V_{IL}(DC)$ are dependent on VREFQ.

“VREFQ” shall be understood as VREFQ(DC), as defined in Figure for “VREFQ(DC) tolerance and VREFQ AC-noise limits”.

Figure 2-9. VREFQ(DC) tolerance and VREFQ AC-noise limits



2.8.2. Differential Input/Output AC Characteristics (VCCQ=1.8V only)

Table 2-14. Differential AC/DC Input Logic Level

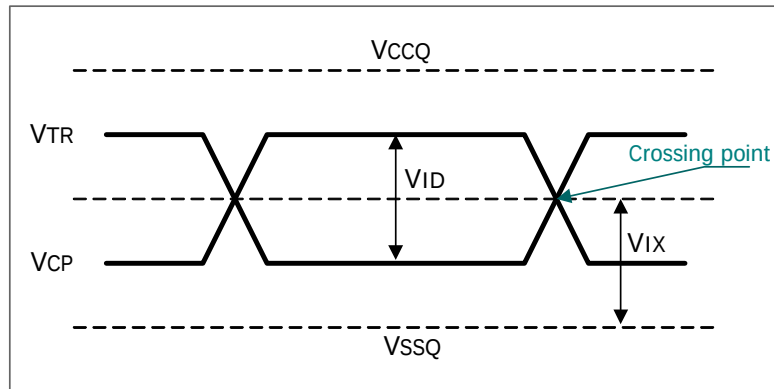
Parameter	Symbol	Min	Max	Units
AC differential input voltage 1)	$V_{ID}(DC)$	0.6	$V_{CCQ} + 0.6$	V
AC differential cross point voltage 2)	$V_{IX}(AC)$	$V_{CCQ} * 0.5 - 0.175$	$V_{CCQ} * 0.5 + 0.175$	V
DC differential input voltage	$V_{ID}(DC)$	0.3	$V_{CCQ} + 0.3$	V
AC differential output cross point voltage	$V_{OX}(AC)$	$V_{CCQ} * 0.5 - 0.20$	$V_{CCQ} * 0.5 + 0.20$	V

NOTE:

1) $V_{ID}(AC)$ specifies the input differential voltage $|V_{TR} - V_{CP}|$ required for switching, where V_{TR} is the "true" input signal and V_{CP} is the "complementary" input signal. The minimum value is equal to $V_{IH}(AC) - V_{IL}(AC)$.

2) The typical value of $V_{IX}(AC)$ is expected to be about $0.5 * V_{CCQ}$ of the transmitting device and $V_{IX}(AC)$ is expected to track variations in V_{CCQ} . $V_{IX}(AC)$ indicates the voltage at which differential input signals must cross.

Figure 2-10. Differential signal levels



2.8.3. DQ Driver Strength

The device may be configured with multiple driver strengths with 'SET FEATURE' command. There are Underdrive, Nominal options. The Toggle DDR supports all two driver strength settings. Devices that support driver strength settings shall comply with the output driver requirements in this section. A device is only required to meet driver strength values for either 3.3V VccQ or 1.8V VccQ, and is not required to meet driver strength values for both 3.3V VccQ and 1.8V VccQ.

Table 2-15. DQ Driver Strength Settings

Setting	Driver Strength	VccQ
Normal	1.0x = 35 Ohms	3.3V
Overdrive 1	1.4x = 25 Ohms	
Overdrive 2	2.0x = 18 Ohms	
Underdrive	0.7x = 50 Ohms	
Normal	1.0x = 35 Ohms	1.8V
Overdrive 1	1.4x = 25 Ohms	
Overdrive 2	2.0x = 18 Ohms	
Underdrive	0.7x = 50 Ohms	

The impedance values corresponding to several different VccQ values are defined in Table of "Output Driver Strength Impedance Values" for 3.3V and 1.8V VccQ. The test conditions that shall be used to verify the impedance values are specified in Table of "Testing Conditions for Impedance Values". The terms TOPER(Min) and TOPER(Max) are in reference to the minimum and maximum operating temperature defined for the device.

Table 2-16. Testing Conditions for Impedance Values

Condition	Temperature	VCCQ(3.3V)	VCCQ(1.8V)	Process
Minimum Impedance	TOPER (Min) degrees Celsius	3.6V	1.95V	Fast - fast
Nominal Impedance	25 degrees Celsius	3.3V	1.8V	Typical
Maximum Impedance	TOPER (Max) degrees Celsius	2.7V	1.7V	Slow-slow

Table 2-17. Output Driver Strength Impedance Values

Output Strength	Rpd/ Rpu	VOUT to VssQ	Minimum		Nominal		Maximum		Units
			VCCQ 3.3V	VCCQ 1.8V	VCCQ 3.3V	VCCQ 1.8V	VCCQ 3.3V	VCCQ 1.8V	
Nominal	Rpd	0.2 X VCCQ	12.8	16.0	32.0	26.7	58.0	47.5	ohm
		0.5 X VCCQ	18.0	21.0	35.0	35.0	70.0	61.0	
		0.8 X VCCQ	23.0	26.5	40.0	47.0	95.0	85.3	
	Rpu	0.2 X VCCQ	23.0	26.5	40.0	47.0	95.0	85.3	
		0.5 X VCCQ	18.0	21.0	35.0	35.0	70.0	61.0	
		0.8 X VCCQ	12.8	16.0	32.0	26.7	58.0	47.5	
Overdrive 1	Rpd	0.2 X VCCQ	9.3	11.4	22.3	19	40	35	
		0.5 X VCCQ	12.6	15	25	25	50	44	
		0.8 X VCCQ	16.3	18.2	29	34	68	61	
	Rpu	0.2 X VCCQ	16.3	18.2	29	34	68	61	
		0.5 X VCCQ	12.3	15	25	25	50	44	
		0.8 X VCCQ	9.3	11.4	19	19	40	35	
Overdrive 2	Rpd	0.2 X VCCQ	7	8.2	16.2	13.5	28.7	27	
		0.5 X VCCQ	9	10.8	18	18	36	31.5	
		0.8 X VCCQ	11.8	13.2	21	23.5	50	42.7	
	Rpu	0.2 X VCCQ	11.8	13.2	21	23.5	50	42.7	
		0.5 X VCCQ	9	10.8	18	18	36	31.5	
		0.8 X VCCQ	7	8.2	14	13.5	28.7	27	
Underdrive	Rpd	0.2 X VCCQ	18.4	24.0	45.0	40.0	80.0	72.0	
		0.5 X VCCQ	25.0	30.0	50.0	50.0	100.0	87.0	
		0.8 X VCCQ	32.0	38.0	57.0	67.0	136.0	122.0	
	Rpu	0.2 X VCCQ	32.0	38.0	57.0	67.0	136.0	122.0	
		0.5 X VCCQ	25.0	30.0	50.0	50.0	100.0	87.0	
		0.8 X VCCQ	18.4	24.0	45.0	40.0	80.0	72.0	

Table 2-18. Pull-up and Pull-down Output Impedance Mismatch

Output Strength	VCCQ 3.3V		VCCQ 1.8V		Units
	Minimum	Maximum	Minimum	Maximum	
Norminal	0	12.3	0	6.2	ohm
Overdrive 1	0	8.8	0	4.4	
Overdrive 2	0	6.3	0	3.2	
Underdrive	0	17.5	0	8.8	

NOTE :

- 1) Mismatch is the absolute value between pull-up and pull-down impedances. Both are measured at the same temperature and voltage.
- 2) Test conditions: $V_{CCQ} = V_{CCQ}(\min)$, $V_{out} = V_{CCQ} \times 0.5$, $T_A = T_{OPER}$

2.8.4. Input/Output Slew rate

The input slew rate requirements that the device shall comply with are defined in Table of “Derating factor” and Table of “Input Slew rate”. The output slew rate requirements that the device shall comply with are defined in Table of “Output Slew Rate Requirements”. The testing conditions that shall be used to verify the input slew rate are listed in Table for “Testing Conditions for Input Slew Rate” and “Testing Conditions for Output Slew Rate”.

Table 2-19.Derating factor

Input Slew Rate	Normal Vcc Range				Units
	200Mbps		400Mbps		Units
	VCCQ 3.3V	VCCQ 1.8V	VCCQ 1.8V	Small swing	
2.0V/ns	-	-	-	-	Ps
1.5V/ns	-	-	-	-	
1.0V/ns	0	0	0	0	
0.8V/ns	150	75	75	30	
0.6V/ns	350	150	150	75	

NOTE :

- 1) Derating factor is required when DQ slew rate is lower than minimum slew rate while the slew rate of DQS meets the minimum slew rate.

Table 2-20.Input Slew rate

VCCQ	VIL	VIH	Normal Vcc Range	
			Min	
			200Mbps	400Mbps
3.3V	$V_{CCQ} * 0.2$	$V_{CCQ} * 0.8$	1.0V/ns	N/A
1.8V	$V_{CCQ} * 0.2$	$V_{CCQ} * 0.8$	1.0V/ns	1.0V/ns
Small swing	$V_{REFQ} - 250mV$	$V_{REFQ} + 250mV$	-	1.0V/ns

Table 2-21. Testing Conditions for Input Slew Rate

Parameter	Value
Positive Input Transition	VIL (DC) to VIH (AC)
Negative Input Transition	VIH (DC) to VIL (AC)

Table 2-22. Output Slew Rate Requirements

Parameter	VCCQ 3.3V		VCCQ 1.8V		Units
	Minimum	Maximum	Minimum	Maximum	
Norminal	1.2	7.0	0.75	4.0	V/ns
Overdrive 1	1.5	9.0	0.85	5.0	
Overdrive 2	1.5	10.0	1.0	5.5	
Underdrive	1.0	5.5	0.60	4.0	

NOTE :

1) Measured with a test load of 5pF connected to VssQ.

2) The ratio of pull-up slew rate to pull-down slew rate is specified for the same temperature and voltage, over the entire temperature and voltage range.

For a given output, it represents the maximum difference between pull-up and pull-down drivers due to process variation.

Table 2-23. Testing Conditions for Output Slew Rate

Parameter	Value
VOL (DC)	VCCQ * 0.3
VOH (DC)	VCCQ * 0.7
VOL (AC)	VCCQ * 0.2
VOH (AC)	VCCQ * 0.8
Positive Output Transition	VOL (DC) to VOH (AC)
Negative Output Transition	VOH (DC) to VOL (AC)
tRISE ⁽¹⁾	Time during Rising Edge from VOL (DC) to VOH (AC)
tFALL ⁽¹⁾	Time during Falling Edge from VOH (DC) to VOL (AC)
Output Slew Rate Rising Edge	(VOH (AC) - VOL (DC)) / tRISE
Output Slew Rate Falling Edge	(VOH (DC) - VOL (AC)) / tFALL
Output Capacitive load	CL = 5pF

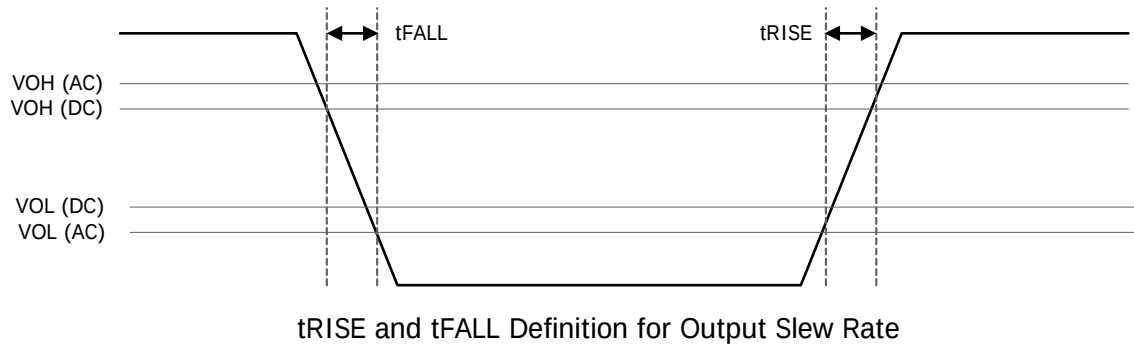
NOTE :

1) Refer to Figure of “tRISE and tFALL Definition for Output Slew Rate”.

2) Output slew rate is verified by design and characterization. It may not be subject to production test.

3) The minimum slew rate is the minimum of the rising edge and the falling edge slew rate. The maximum slew rate is the maximum of the rising edge and the falling edge slew rate.

Figure 2-11. tRISE and tFALL Definition for Output Slew Rate



2.9. On-Die Termination (Not Supported)

2.9.1. On-Die Termination

ODT is a feature that enables a NAND Flash device to turn termination resistance on or off for each Data Input/Output, DQS, and DQS pin. This feature is available for V_{cc} 3.3V & V_{ccQ} 1.8V operation only. The ODT feature is designed to improve the signal integrity of the memory channel by enabling the NAND Flash device controller to independently turn termination resistance on or off for a selected target. NAND Flash devices support only self-termination.

2.9.2. ODT Setting

The ODT setting is configured by the Set Features operation (see Table of "Toggle 2.0 Specific setting assignment" and "Definition of Toggle 2.0 specific setting" for additional information).

Figure 2-12. Set ODT Sequence

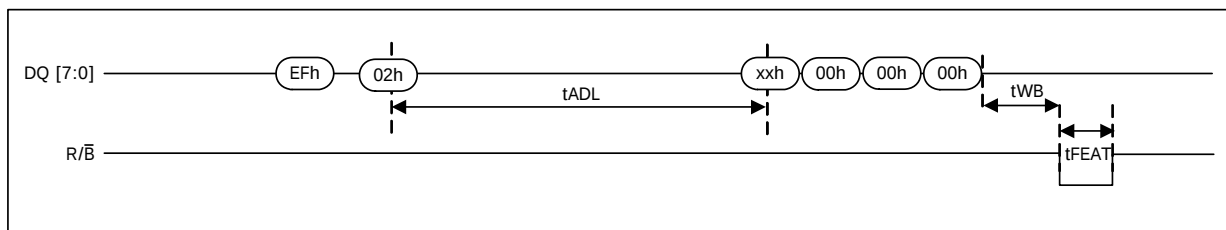
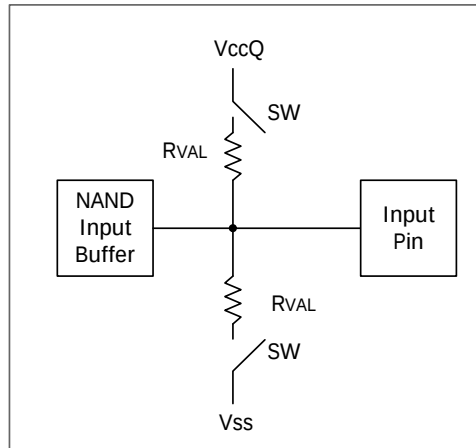


Figure 2-13. ODT Functional Representation



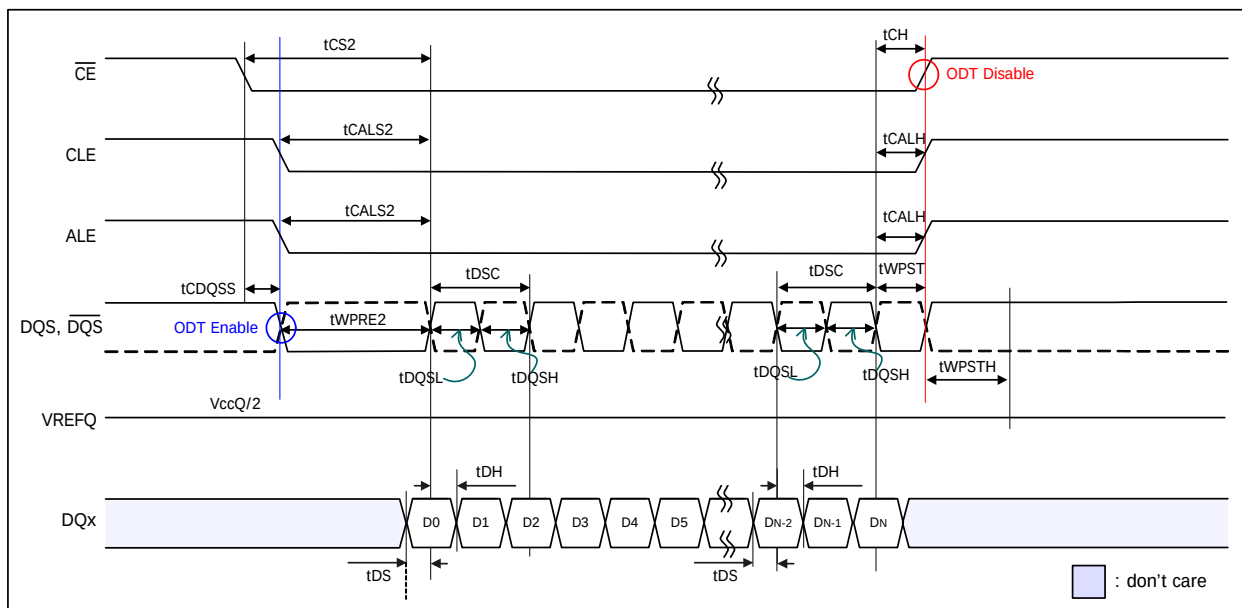
2.9.3. ODT On/Off During Read

ODT is not currently supported during Read operations.

2.9.4. ODT On/Off During Program

ODT is enabled within the Write preamble period as ALE, CLE, and DQS go low. ODT is disabled as $\overline{\text{CE}}$ goes high, or as either CLE or ALE go high while $\overline{\text{CE}}$ is low.

Figure 2-14. ODT On/Off During Program



2.10. AC and DC Input Measurement Levels

Table 2-24. AC Test condition

Parameter	H27xxG8xxBxR
Input Pulse Levels	VIL to VIH
Input Rise and Fall Times	1.0V/ns
Input and Output Timing Levels	VCCQ/2
Output Load	CL= 5pF

2.11. AC Timing Characteristics

2.11.1. Timing Parameters Description

Table 2-25. Toggle DDR Timing Parameters Description

Parameter	Description	Parameter	Description
tR	Data Transfer from Flash array to Register	tDS	Data Setup Time
tPROG	Program Time	tDVW	Output data valid window
tBERS	Erase Time	tFEAT	Busy time for Set Feature and Get Feature
tADL	Address to Data Loading Time	tQH	Output hold time from DQS
tAR	ALE Low to $\overline{\text{RE}}$ Low	tQHS	DQS hold skew factor
tCALH	CLE/ALE Hold Time	tRC	Read Cycle Time
tCALS	CLE/ALE Setup Time	tREH	$\overline{\text{RE}}$ High pulse width
tCALS2	CLE/ALE Setup Time (ODT Enable)	tRP	$\overline{\text{RE}}$ Low pulse width
tCAH	Command/Address Hold Time	tRPP	$\overline{\text{RE}}$ Low width for Read Status at power-up
tCAS	Command/Address Setup Time	tRPRE	Read Preamble
tCH	$\overline{\text{CE}}$ Hold Time	tRPRE2	Read Preamble (ODT Enable)
tCDQSH	DQS Setup Time for data input finish	tRPST	Read Postamble
tCDQSS	DQS Setup Time for data input start	tRPSTH	Read Postamble Hold Time
tCHZ	$\overline{\text{CE}}$ High to Output Hi-Z	tRR	Ready to $\overline{\text{RE}}$ High
tCLHZ	CLE High to Output Hi-Z	tRST	Device Resetting Time(Read/Program/Erase)
tCLR	CLE to $\overline{\text{RE}}$ Low	tWB	$\overline{\text{WE}}$ High to Busy
tCR	$\overline{\text{CE}}$ Low to $\overline{\text{RE}}$ Low	tWC	Write Cycle Time
tCRES	$\overline{\text{RE}}$ Set up Time	tWH	$\overline{\text{WE}}$ High pulse width
tCS	$\overline{\text{CE}}$ Setup Time	tWHR	$\overline{\text{WE}}$ High to $\overline{\text{RE}}$ Low
tCS2	$\overline{\text{CE}}$ Setup Time (ODT Enable)	tWHR2	$\overline{\text{WE}}$ High to $\overline{\text{RE}}$ Low for Random data out
tCSD	$\overline{\text{CE}}$ Disable to signal (CLE, ALE, $\overline{\text{WE}}$) don't care	tWPRE	Write Preamble
tDH	Data Hold Time	tWPRE2	Write Preamble (ODT Enable)

Parameter	Description	Parameter	Description
tcWAW	Command Write Cycle to Address Write Cycle Time for Random Data Input and Register Read Out mode	tWPST	Write Postamble
tdQSH	DQS Input High Pulse Width	tWPSTH	Write Postamble Hold Time
tdQSL	DQS Input Low Pulse Width	tWW	$\overline{WP}$ High/Low to $\overline{WE}$ low
tdQSQ	Output skew among data output and corresponding DQS	tDBSY	Dummy Busy Time for Multi-Plane Program
tdQSRE	$\overline{RE}$ to DQS and DQ delay	tWP	$\overline{WE}$ Low pulse Width
tdSC	Data Strobe Cycle Time		
tSPARAM	Set parameter time		
trPARAM	Get parameter time		

NOTE :

1) Commands (including read status/read status enhanced) shall not be issued until after tWB is complete.

2.11.2. Timing Parameters Table

Table 2-26. AC Timing Characteristics

Parameter	100MHz		133MHz		166MHz		200MHz		Unit
	Min	Max	Min	Max	Min	Max	Min	Max	
tADL	300	-	300	-	300	-	300	-	ns
tAR	10	-	10	-	10	-	10	-	ns
tCALH	5	-	5	-	5	-	5	-	ns
tCALS	15	-	15	-	15	-	15	-	ns
tCALS2	25	-	25	-	25	-	25	-	ns
tCAH	5	-	5	-	5	-	5	-	ns
tCAS	5	-	5	-	5	-	5	-	ns
tCDQSH	100	-	100	-	100	-	100	-	ns
tCDQSS	100	-	100	-	100	-	100	-	ns
tCH	5	-	5	-	5	-	5	-	ns
tCHZ	-	30	-	30	-	30	-	30	ns
tCLHZ	-	30	-	30	-	30	-	30	ns
tCLR	10	-	10	-	10	-	10	-	ns
tCR	10	-	10	-	10	-	10	-	ns
tCRES	10	-	10	-	10	-	10	-	ns
tCS	25	-	25	-	25	-	25	-	ns
tCS2	30	-	30	-	30	-	30	-	ns
tCSD	10	-	10	-	10	-	10	-	ns
tcWAW	300	-	300	-	300	-	300	-	ns
tdH	0.9	-	0.75	-	0.55	-	0.40	-	ns

Parameter	100MHz		133MHz		166MHz		200MHz		Unit
	Min	Max	Min	Max	Min	Max	Min	Max	
tDQSH	tDSC *0.45	-	tDSC *0.45	-	tDSC *0.45	-	tDSC *0.45	-	ns
tDQSL	tDSC *0.45	-	tDSC *0.45	-	tDSC *0.45	-	tDSC *0.45	-	ns
tDQSQ	-	0.8	-	0.6	-	0.5	-	0.4	ns
tDQSRE	5	25	5	25	5	25	5	25	ns
tDSC	10	-	7.5	-	6	-	5	-	ns
tDS	0.9	-	0.75	-	0.55	-	0.40	-	ns
tDVW	tDVW = tQH - tDQSQ								ns
tFEAT	-	1	-	1	-	1	-	1	us
tQH	tQH = min[tREH, tRP] - tQHS								ns
tQHS	-	0.8	-	0.6	-	0.5	-	0.4	ns
tRC	10	-	7.5	-	6	-	5	-	ns
tREH	tRC*0.45	-	tRC*0.45	-	tRC*0.45	-	tRC*0.45	-	ns
tRP	tRC*0.45	-	tRC*0.45	-	tRC*0.45	-	tRC*0.45	-	ns
tRPP	30	-	30	-	30	-	30	-	ns
tRPRE	15	-	15	-	15	-	15	-	ns
tRPRE2	25	-	25	-	25	-	25	-	ns
tRPST	tDQSRE + tRC*0.5	-	tDQSRE + tRC*0.5	-	tDQSRE + tRC*0.5	-	tDQSRE + tRC*0.5	-	ns
tRPSTH	25	-	25	-	25	-	25	-	ns
tRR	20	-	20	-	20	-	20	-	ns
tRST	10 / 30 / 200								us
tWB	-	100	-	100	-	100	-	100	ns
tWC	25	-	25	-	25	-	25	-	ns
tWH	11	-	11	-	11	-	11	-	ns
tWHR	120	-	120	-	120	-	120	-	ns
tWHR2	300		300		300		300		ns
tWP	11		11		11		11		ns
tWPRE	15		15		15		15		ns
tWPRE2	25		25		25		25		ns
tWPST	6.5		6.5		6.5		6.5		ns
tWPSTH	25	-	25	-	25	-	25	-	ns
tWW	100		100		100		100		ns
tSPARAM	100		100		100		100		ns
tRPARAM	100		100		100		100		ns

2.11.3. Read/Program / Erase Characteristics

Table 2-27. NAND Read/Program/Erase Characteristics

Parameter	Symbol	Min	Typ	Max	Units
Data Transfer from Flash array to Register	tR	-	75	110	us
Program Time	tPROG	-	1.5	5	ms
Dummy Busy Time for Multi-Plane setting	tDBSY	-	0.5	1	us
Number of Partial Program Cycles in the Same Page	Nop	-	-	1	cycle
Block Erase Time	tBERS	-	7	10	ms

NOTE :

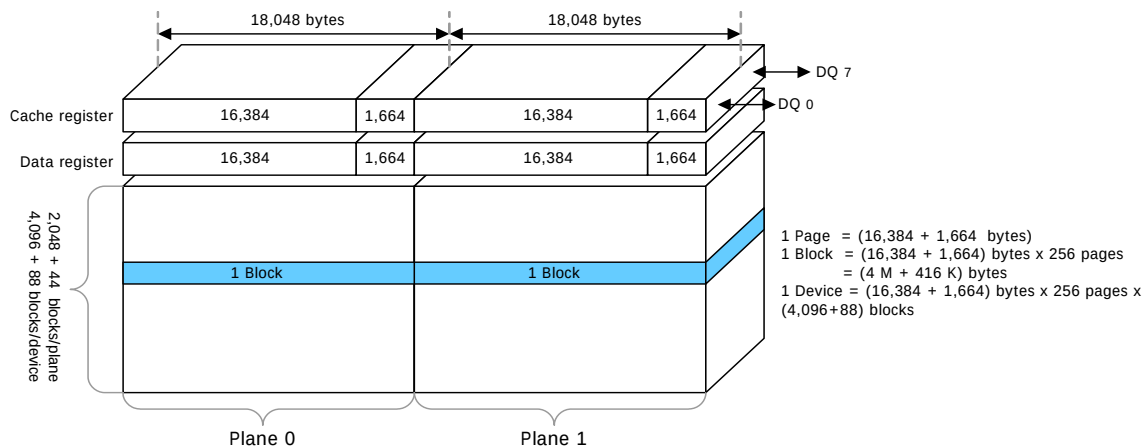
- 1) Typical program time is measured at Vcc=3.3V, TA=25C. Not 100% tested.
- 2) Typical Program time is defined as the time within which more than 50% of the whole pages are programed at 3.3V Vcc and 25C temperature.

3. Memory Organization

3.1. Memory Size

[Single chip based size]

Figure 3-1. Memory organization for 128Gb Logical Unit



NOTE :

- 1) 16GByte: single die stack
- 2) 32GByte: two-die stack that operates as two independant 128Gbit devices
- 3) 64GByte: four-die stack that operates as four independant 128Gbit devices
- 4) 128GByte: eight-die stack that operates as four independant 128Gbit devices

3.2. Addressing (I)

3.2.1. Single Die Addressing

Single die NAND of 128Gb read and program operations take 5 bytes of address.

Single die NAND Flash devices contain blocks made up of 256 pages, where each page contains 16,384 data bytes + 1,664 spare bytes. NAND Flash devices do not contain dedicated address pins. Addresses are loaded using a 5-cycle sequence as shown in Table for additional “Memory addressing” details.

Table 3-1. Memory addressing.

Bus Cycle	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7	Address information		
1 st Cycle	A0	A1	A2	A3	A4	A5	A6	A7	Column		A[14:0]
2 nd Cycle	A8	A9	A10	A11	A12	A13	A14	L ⁽¹⁾			
3 rd Cycle	A15	A16	A17	A18	A19	A20	A21	A22	Row	Page	A[22:15]
4 th Cycle	A23	A24	A25	A26	A27	A28	A29	A30		Plane	A[23]
5 th Cycle	A31	A32	A33	A34	A35	A36	A37	L ⁽¹⁾		Block	A[35:24]
										LUN	A[37:36]

Note

1. L must be set to Low.
2. The device ignores any additional address input cycle than required.

3.3. Target Memory Organization

A device contains one or more targets. A target is controlled by one $\overline{\text{CE}}$ signal. A target is organized into one or more logical units (LUNs).

A logical unit (LUN) is the minimum unit that can independently execute commands and report status. Specifically, separate LUNs may operate on arbitrary command sequences in parallel. For example, it is permissible to start a Page Program operation on LUN 0 and then prior to the operation’s completion to start a Read command on LUN 1. A LUN contains at least one page register and a Flash array. The number of page registers is dependent on the number of plane operations supported for the LUN. The Flash array contains a number of blocks.

A block is the smallest erasable unit of data within the Flash array of a LUN. There is no restriction on the number of blocks within the LUN. A block contains a number of pages.

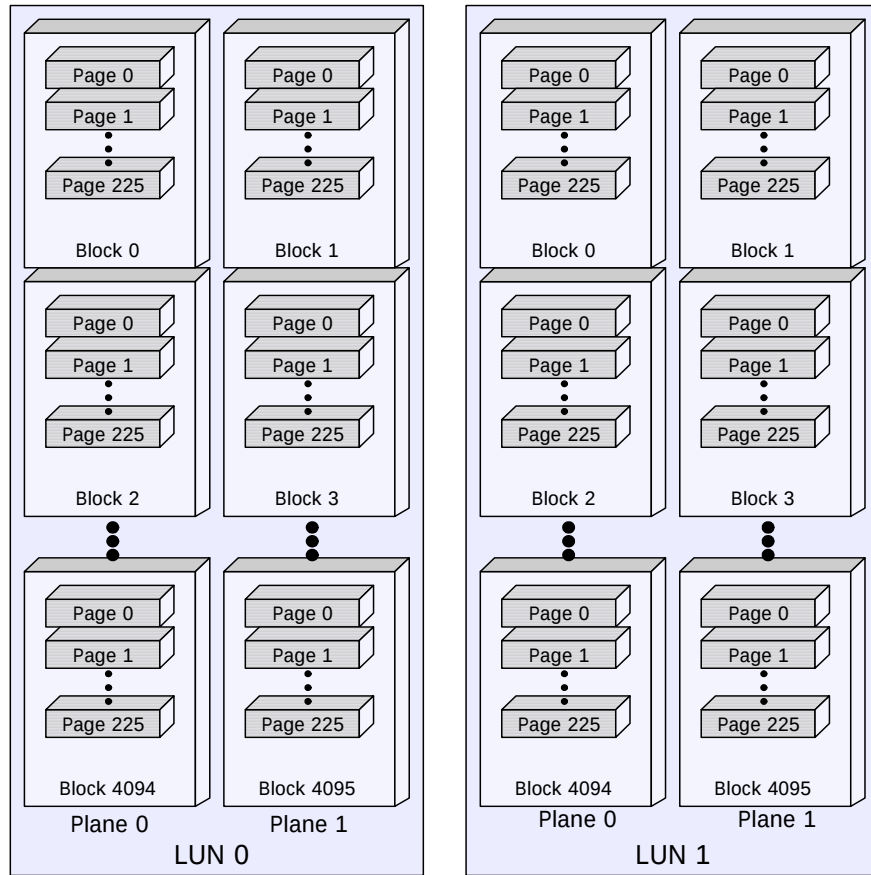
A page is the smallest addressable unit for read and program operations.

Each LUN shall have at least one page register. A page register is used for the temporary storage of data before it is moved to a page within the Flash array or after it is moved from a page within the Flash array.

The byte location within the page register is referred to as the column.

There are two mechanisms to achieve parallelism within this architecture. There may be multiple commands outstanding to different LUNs at the same time. To get further parallelism within a LUN, plane addressing may be used to execute additional dependent operations in parallel.

Figure 3-2. Target Memory Organization



3.4. Addressing (II)

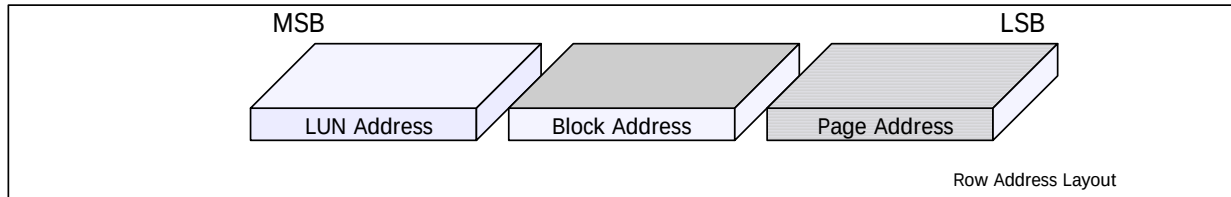
There are two address types used: the column address and the row address. The column address is used to access bytes within a page, i.e. the column address is the byte offset into the page. The least significant bit of the column address shall always be zero for a DDR interface, i.e. an even number of bytes is always transferred. The row address is used to address pages, blocks, and LUNs.

When both the column and row addresses are required to be issued, the column address is always issued first in one or more 8-bit address cycles. The row addresses follow in one or more 8-bit address cycles. There are some functions that may require only row addresses, such as Block Erase. In this case the column addresses shall not be issued.

For both column and row addresses the first address cycle always contains the least significant address bits and the last address cycle always contains the most significant address bits. If there are bits in the most significant cycles of the column and row addresses that are not used then they are required to be cleared to zero.

The row address structure is shown in Figure of "Row Address Layout" with the least significant row address bit to the right and the most significant row address bit to the left.

Figure 3-3. Row Address Layout



The page address is set by the least significant row address bits, and the LUN address is set by the most significant row address bit(s). The block address is between a page address and a LUN address.

A host shall not access an address of a page or block beyond maximum page address or block address.

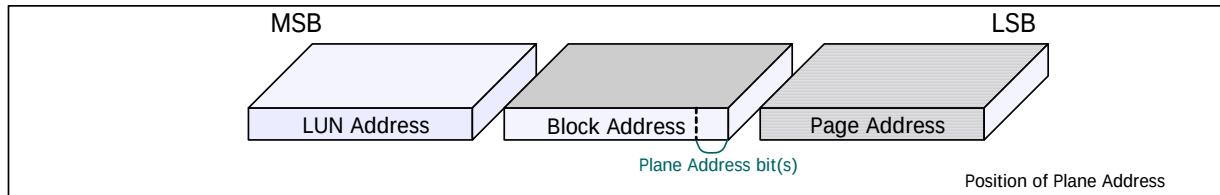
The Addressing of this device is shown in table “Memory addressing”.

3.4.1. Plane Addressing

The plane address comprises the lowest order bits of the block address as shown in Figure for “Position of Plane Address”. The plane address is used when performing a Two-plane command sequence on a particular LUN:

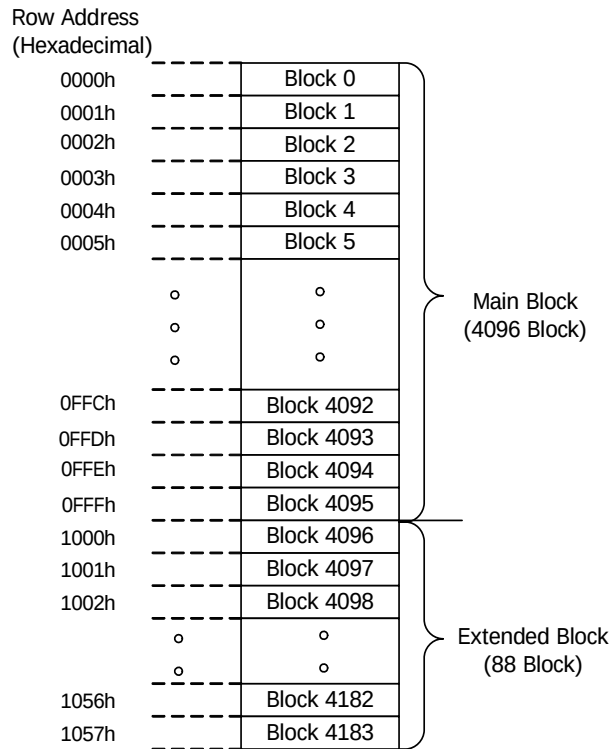
The plane address bit(s) shall be different within address setting sequences for the Two-plane-related operation, while the page address shall stay the same within address setting sequences for the multi-plane-related operation.

Figure 3-4. Position of Plane Address



3.4.2. Extended Block Arrangement

The device offer 88 extended blocks to increase valid blocks.



3.4.3. Valid Blocks

Table 3-2. The number of valid block

Prodduct	Density	Symbol	Min	Typ.	Max	Unit
128Gb (16GB) Single Die	128Gbit	NVB	4012	4096	4184	Blocks

Notes:

1. The 1st block is guaranteed to be a valid block at the time of shipment.
2. This single device has a maximum of 4184 valid blocks.
3. Invalid blocks are one that contains one or more bad bits. The device may contain bad blocks on shipment.

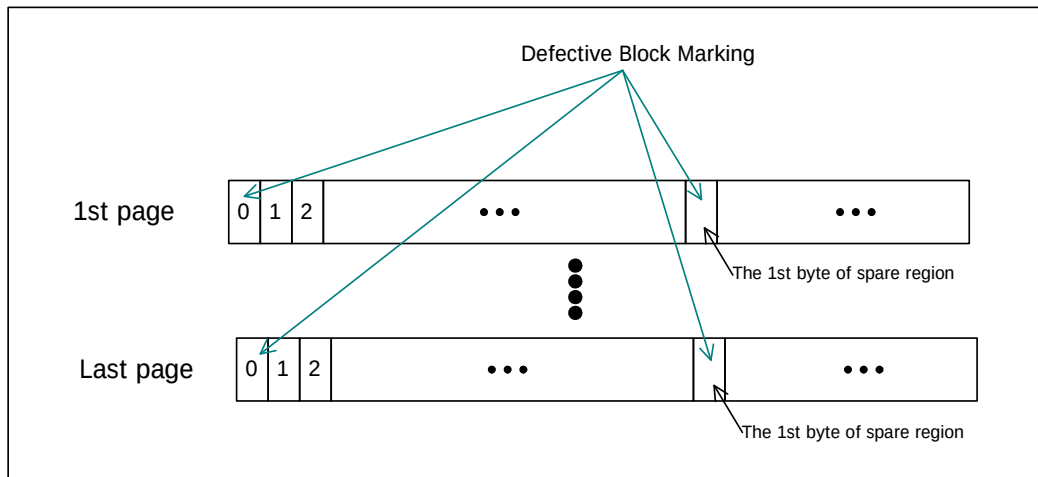
3.5. Factory Defect Mapping

The Flash array is not presumed to be pristine, and a number of defects that makes the blocks unusable may be present. Invalid blocks shall be sorted out from normal blocks by software.

3.5.1. Device Requirements

If a block is defective, the manufacturer shall mark the block as defective by setting the Defective Block Marking, as shown in Figure of “Area marked in first or last page of block indicating defect”, of the last page of the defective block to a value of non-FFh. The Defective Block Marking is located on the first byte of user data area or the first byte of spare data area in the pages within a block.

Figure 3-5. Area marked in first or last page of block indicating defect



3.5.2. Host Requirements

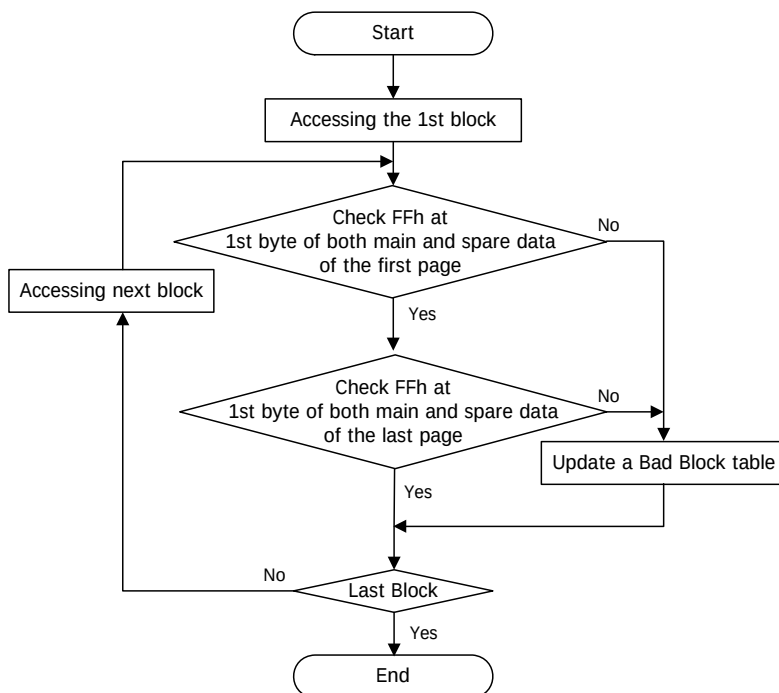
The host shall not erase or program blocks marked as defective by the manufacturer, and any attempt to do so yields indeterminate results.

Figure of “Flow chart to create initial invalid block table” outlines the flow chart how to create an initial invalid block table. It should be performed by the host to create the initial bad block table prior to performing any erase or programming operations on the target. The 1st byte of both main and spare region in non-defective blocks are read FFh with ECC enabled on the controller. A defective block is indicated by the majority of bits being read non-FFh in the Defective Block Marking location of either the first page or last page of the block. The host shall check the Defective Block Marking location of both the first and last past page of each block to verify the block is valid prior to any erase or program operations on that block.

NOTE :

Over the lifetime use of a NAND device, the Defective Block Marking of defective blocks may encounter read disturbs that cause bit changes. The initial defect marks by the manufacturer may change value over the lifetime of the device, and are expected to be read by the host and used to create a bad block table during initial use of the part.

Figure 3-6. Flow chart to create initial invalid block table



3.6. Addressing For Program Operation

Within a block, the pages must be programmed consecutively from the LSB (least significant bit) page of the block to MSB (most significant bit) pages of the block. Random page address programming is prohibited. In this case, the definition of LSB page is the LSB among the pages to be programmed. A starting page in an erased block for programming shall be page 0.

Table 3-2. Paired Page Address Information

Paired Page Address(1/4)		Paired Page Address(2/4)		Paired Page Address(3/4)		Paired Page Address(4/4)	
1st Group	2nd Group	1st Group	2nd Group	1st Group	2nd Group	1st Group	2nd Group
00h	02h	41h	44h	83h	86h	C5h	C8h
01h	04h	43h	46h	85h	88h	C7h	CAh
03h	06h	45h	48h	87h	8Ah	C9h	CCh
05h	08h	47h	4Ah	89h	8Ch	CBh	CEh
07h	0Ah	49h	4Ch	8Bh	8Eh	CDh	D0h
09h	0Ch	4Bh	4Eh	8Dh	90h	CFh	D2h
0Bh	0Eh	4Dh	50h	8Fh	92h	D1h	D4h
0Dh	10h	4Fh	52h	91h	94h	D3h	D6h
0Fh	12h	51h	54h	93h	96h	D5h	D8h
11h	14h	53h	56h	95h	98h	D7h	DAh

Paired Page Address(1/4)		Paired Page Address(2/4)		Paired Page Address(3/4)		Paired Page Address(4/4)	
1st Group	2nd Group	1st Group	2nd Group	1st Group	2nd Group	1st Group	2nd Group
13h	16h	55h	58h	97h	9Ah	D9h	DCh
15h	18h	57h	5Ah	99h	9Ch	DBh	DEh
17h	1Ah	59h	5Ch	9Bh	9Eh	DDh	E0h
19h	1Ch	5Bh	5Eh	9Dh	A0h	DFh	E2h
1Bh	1Eh	5Dh	60h	9Fh	A2h	E1h	E4h
1Dh	20h	5Fh	62h	A1h	A4h	E3h	E6h
1Fh	22h	61h	64h	A3h	A6h	E5h	E8h
21h	24h	63h	66h	A5h	A8h	E7h	EAh
23h	26h	65h	68h	A7h	AAh	E9h	ECh
25h	28h	67h	6Ah	A9h	ACh	EBh	EEh
27h	2Ah	69h	6Ch	ABh	A Eh	EDh	F0h
29h	2Ch	6Bh	6Eh	ADh	B0h	EFh	F2h
2Bh	2Eh	6Dh	70h	AFh	B2h	F1h	F4h
2Dh	30h	6Fh	72h	B1h	B4h	F3h	F6h
2Fh	32h	71h	74h	B3h	B6h	F5h	F8h
31h	34h	73h	76h	B5h	B8h	F7h	FAh
33h	36h	75h	78h	B7h	BAh	F9h	FCh
35h	38h	77h	7Ah	B9h	BCh	FBh	FEh
37h	3Ah	79h	7Ch	BBh	BEh	FDh	FFh
39h	3Ch	7Bh	7Eh	BDh	C0h		
3Bh	3Eh	7Dh	80h	BFh	C2h		
3Dh	40h	7Fh	82h	C1h	C4h		
3Fh	42h	81h	84h	C3h	C6h		

NOTE :

When program operation is abnormally aborted (ex. power-down, reset), not only page data under program but also paired page data may be damaged.

4. FUNCTION DESCRIPTION

4.1. DATA PROTECTION AND POWER TRANSITION SEQUENCE

4.1.1. Data Protection

The device is designed to offer protection from any involuntary program/erase during power transitions. An internal voltage detector disables all internal Program/Erase circuits when VCC is below about 2V. WP pin provides hardware protection and is recommended to be kept at VIL during power transitions.

Although two step command sequence for program/erase provides protection from any operations by unintentional command input, keeping CLE and ALE at VIL prevents any spurious commands asserted during power transitions.

It is highly recommended to keep $\overline{\text{CE}}$ at V_{IH} to prevent unnecessary current consumption during power transitions.

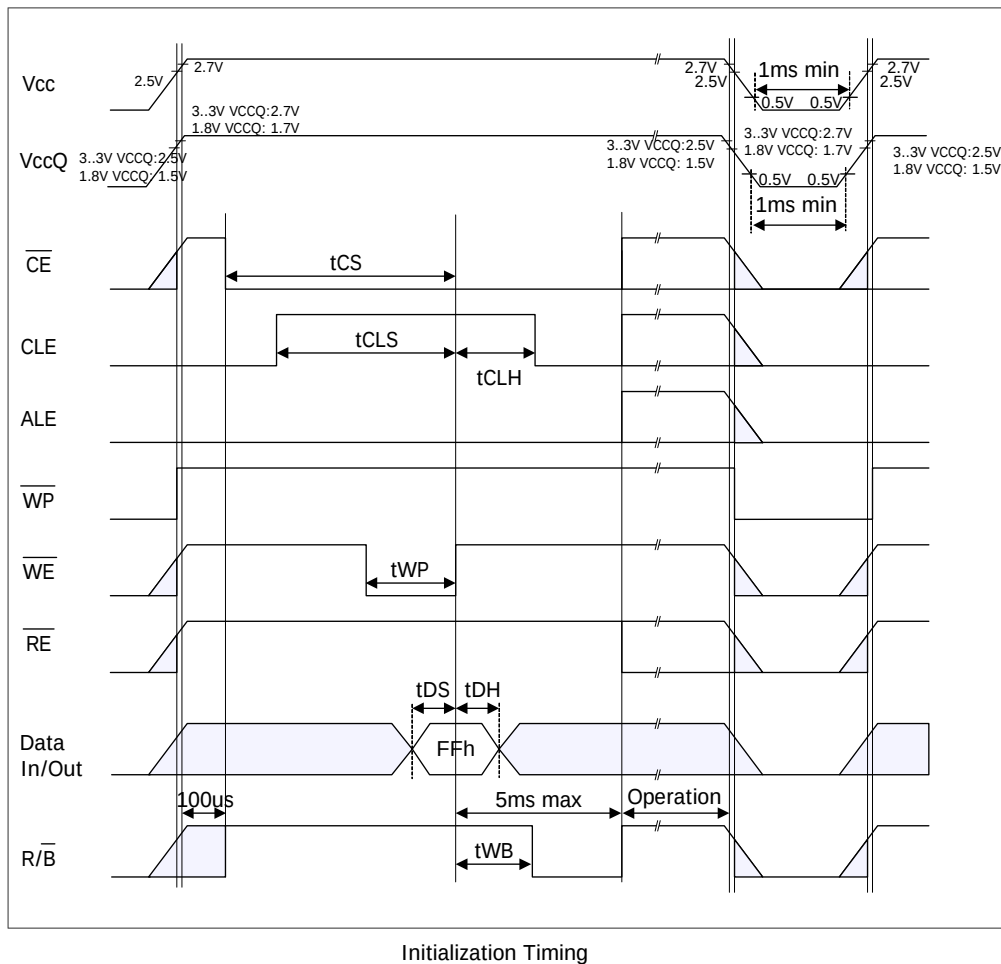
4.1.2. Power Up Sequence

For NAND devices that support VCCQ for Data Input/Output power supply, VCCQ must not exceed VCC during power up. The host must wait for R/B to be valid High before issuing Reset command (FFh) to initialize any targets that share same $\overline{\text{CE}}$. The R/B signal becomes valid after 100 μ s since both VCC and VCCQ reach 2.7V (1.7V for 1.8V VCCQ). The RESET command (FFh) must be issued to all targets as the first command after the NAND device is powered up and R/B becomes valid. Each target ($\overline{\text{CE}}$) will be busy for a maximum of 5ms after the RESET command (FFh) is issued. The RESET busy time can be monitored by polling R/B or issuing the READ STATUS(70h) command. Each NAND LUN(i.e. die) may draw less than 10mA over 1ms prior to the execution of the first RESET command (FFh) after the device is powered on. During the power up sequence including the RESET busy time, each LUN consumes a maximum current of 50mA.

4.1.3. Power Down Sequence

During power-down, VCCQ shall not exceed VCC.

Figure 4-1. Initialization Timing



This device defaults to legacy mode based on configuration. For details see "Data Interface / Timing Mode Transitions".

VCCQ setting method

The NAND default interface mode is set to 3.3V, SDR mode. Refer to the next table below when the Set parameter setting is available in VccQ. As for “set parameter” is given on Addendum 1.1.

Command Cycle	Address	Data	VccQ	Read ID
36h	61h	00h	3.3V	ADh, 3Ah, 14h, ABh, 42h, 4Ah
		40h	1.8V	ADh, 5Ah, 14h, ABh, 42h, 49h

4.2. Data Interface / Timing Mode Transitions

4.2.1. The following transitions between data interface are supported:

After the power supply is stable and VCC and VCCQ are within their normal operating range (2.7~3.3V), a Reset command (FFh) must be issued to initialize the device.

- SDR to DDR
- DDR to SDR

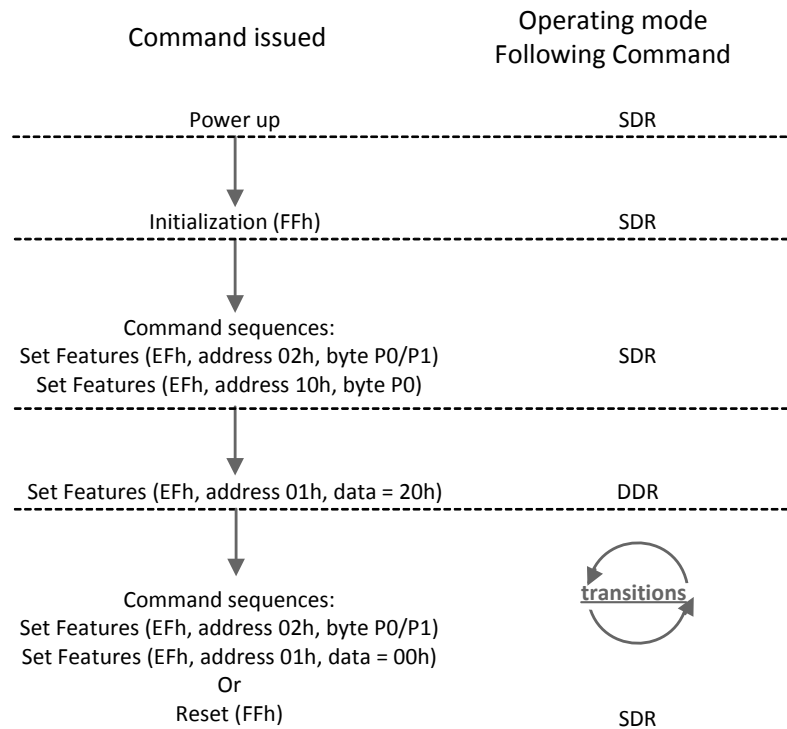
To change the data interface to DDR, the Set Features command is used. The Set Features command (EFh), Feature Address, and the four parameters are entered using the previously selected data interface. When issuing the Set Features command, the host shall drive the DQS signal high (if supported) during the entirety of the command (including parameter entry). After the fourth parameter is entered until the tFEAT time has passed the host shall not issue any commands to the device.

Prior to issuing any new commands to the device, the host shall transition $\overline{CE}$ high. The new data interface is active when the host pulls $\overline{CE}$ low.

For Toggle Mode (with differential clock signals DQS and/or $\overline{RE}$, ODT, and/or VREFQ), bytes P0 and P1 must first be loaded at Set Features register address 02h using the Set Features command and the desired values (see “Table 5-3. Toggle 2.0 specific setting assignment,” and “Table 5-4. Definition of Toggle 2.0 specific setting”). If the Set Features register address 02h is not loaded, it defaults to 00h, and the device operates as SDR mode.

4.2.2. SDR Transition from DDR

To transition from DDR to the SDR data interface, the host shall use the Reset (FFh) command. After the Reset is issued, the host shall not issue any commands to the device until after the tFEAT time has passed. Note that after the tFEAT time has passed, only status commands may be issued by the host until the Reset completes. After $\overline{CE}$ has been pulled high and then transitioned low again, the host should issue a Set Features to select the appropriate SDR timing mode.



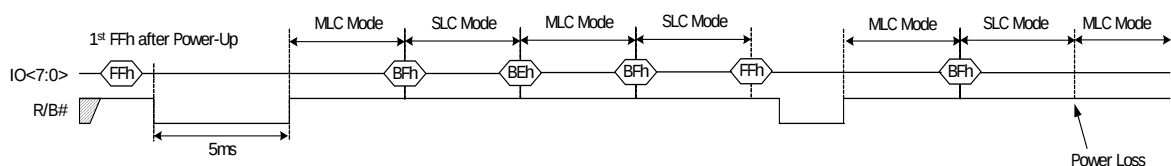
4.3. MLC to SLC mode transition (MLC2SLC)

MLC NAND flash supports two different cell types: MLC and SLC. It is provided with an additional feature which allows for a host to access one or more blocks in SLC mode. Program, read, and erase operation in SLC mode are much faster than standard MLC operations, and also the reliability is higher than normal MLC blocks.

The NAND supports two different cell types, MLC and SLC. The overhead to execute a SLC operation is small, just one command cycle to activate/deactivate SLC session. At power on, the device starts in MLC mode, and could be switched in SLC mode to providing MLC2SLC command (BFh). If a power loss occurs during SLC operation mode, the device will switch in MLC operation mode automatically. Either SLC2MLC command (BEh) or RESET (FFh) command could be used for to switch device in MLC operation mode. E/P/R within block boundary should be done in same mode. For example, read operation in SLC mode isn't allowed in the block programmed in MLC mode.

A block assigned to operate in on mode (MLC or SLC mode) should always stay to operate in the same mode during the life of the device. One command cycle is required to enter SLC session without busy state. After power-up, the device begins in MLC mode, and then MLC mode is switched to SLC mode with BFh command. Either BEh command or FFh command (reset) can be used to exit from MLC mode.

Figure 4-2. SLC Mode Transition Diagram



4.3.1. Difference between MLC and SLC Addressing

The difference between MLC addressing and SLC addressing exists because of the smaller number of pages in SLC mode. Please refer to tables as below.

Table 4-1. Memory SLC addressing.

Bus Cycle	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7	Address information		
1 st Cycle	A0	A1	A2	A3	A4	A5	A6	A7	Column		A[14:0]
2 nd Cycle	A8	A9	A10	A11	A12	A13	A14	L ⁽¹⁾			
3 rd Cycle	A15	A16	A17	A18	A19	A20	A21	L ⁽¹⁾	Row	Page	A[21:15]
4 th Cycle	A22	A23	A24	A25	A26	A27	A28	A29		Plane	A[22]
5 th Cycle	A30	A31	A32	A33	A34	A35	A36			Block	A[34:23]
										LUN	A[36:A35]

Note

1. L must be set to Low.
2. The device ignores any additional address input cycle than required.

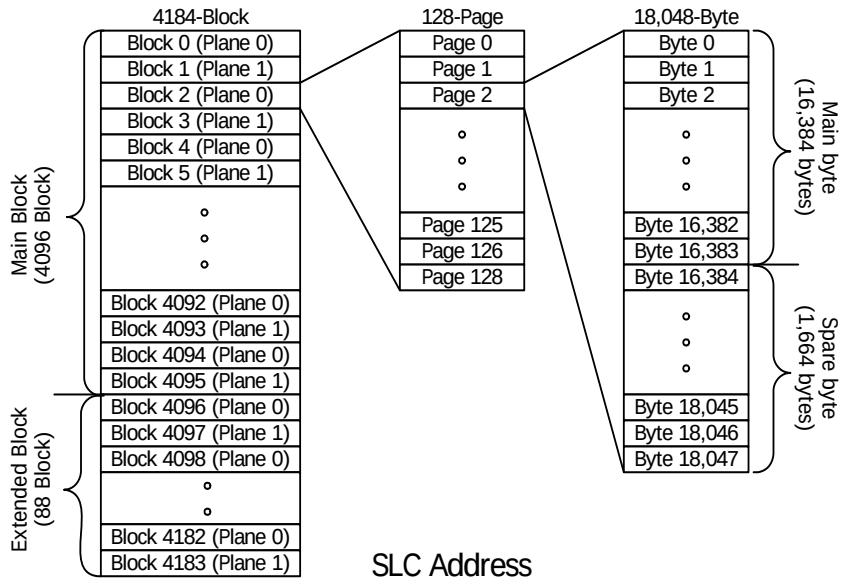
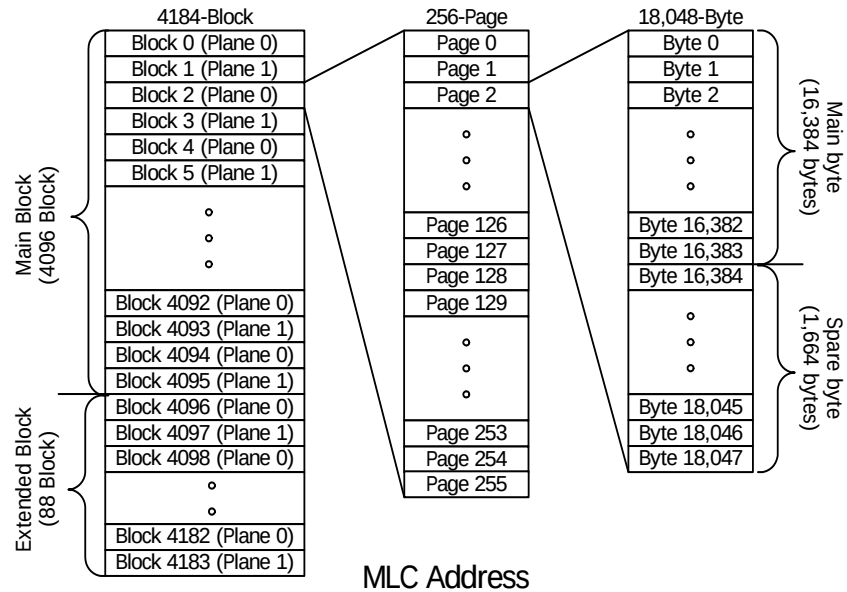


Table 4-2. SLC Performance

Parameter	Typical Value
tR	40us
tPROG	400us
tBERASE	5ms

4.4. Mode Selection

Table of “*Mode Selection*” describes the bus state for the Toggle DDR. Commands, addresses and data is all written through DQ's by bringing WE to low while CE is low.

Those are latched on the rising edge of WE. Command Latch Enable(CLE) and Address Latch Enable(ALE) are used to multiplex command and address respectively, via the DQ pins.

Host reads or writes data to the device using DQS signal. And data is latched on the falling and rising both edge of DQS on data input.

Table 4-3. Mode Selection

CLE	ALE	CE	WE	RE	DQS	WP	Mode	
H	L	L	Rising	H	X	X ¹⁾	Read Mode	Command Input
L	H	L	Rising	H	X	X		Address Input (5 cycles)
H	L	L	Rising	H	X	H	Write Mode	Command Input
L	H	L	Rising	H	X	H		Address Input (5 cycles)
L	L	L	H	H	falling / Rising	H	Data Input	
L	L	L	H	falling / Rising	falling / Rising	X	Data Output	
X	X	X	X	H	X	X	During Read(Busy)	
X	X	X	X	X	X	H	During Program(Busy)	
X	X	X	X	X	X	H	During Erase(Busy)	
X	X	X	X	X	X	L	Write Protect	
X	X	H	X	X	X	0V/VCC ²⁾	Stand-by	

NOTE :

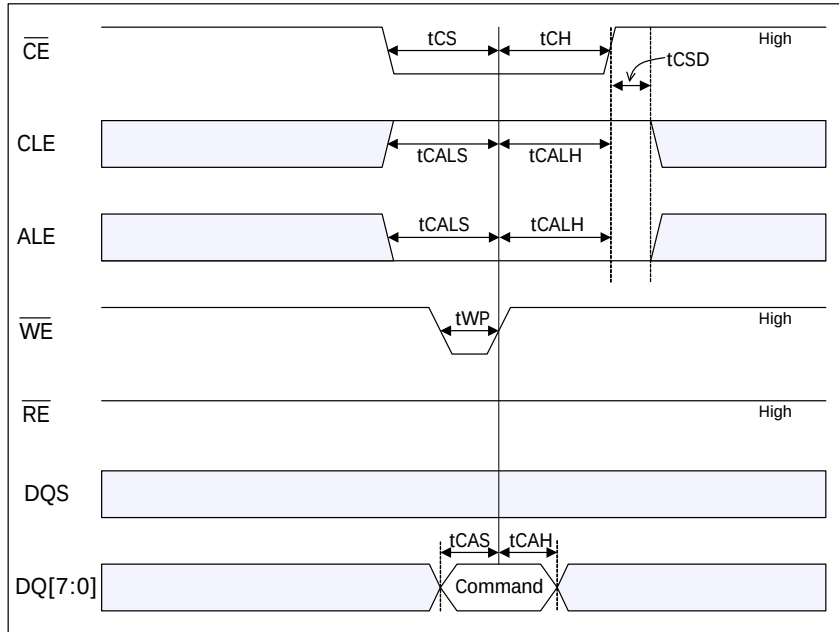
1) X can be VIL or VIH.

2) WP should be biased to CMOS high or CMOS low for standby.

4.5. General Timing

4.5.1. Command Latch Cycle

Figure 4-3. Command Latch Cycle

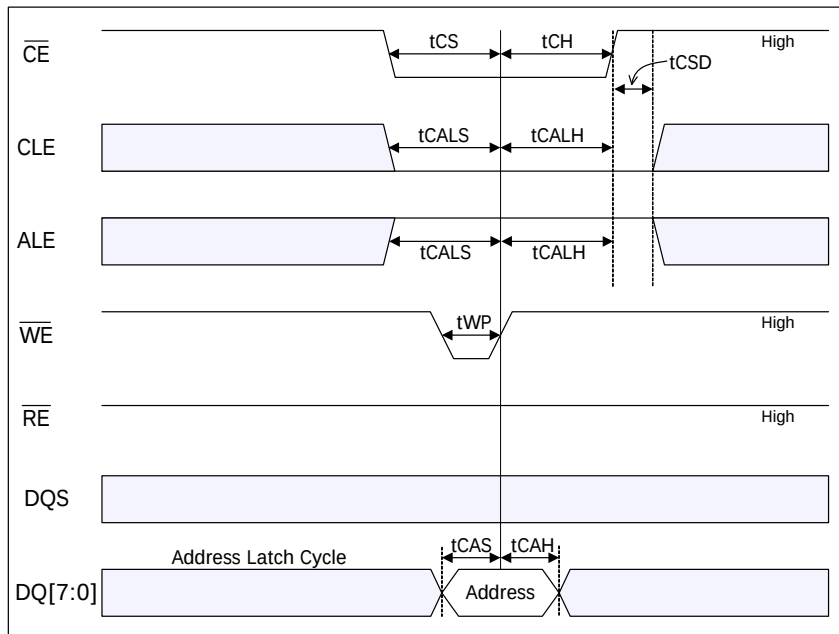


NOTE :

- 1) Command information is latched by $\overline{WE}$ going 'High' when $\overline{CE}$ is 'Low', CLE is 'High', and ALE is 'Low'.

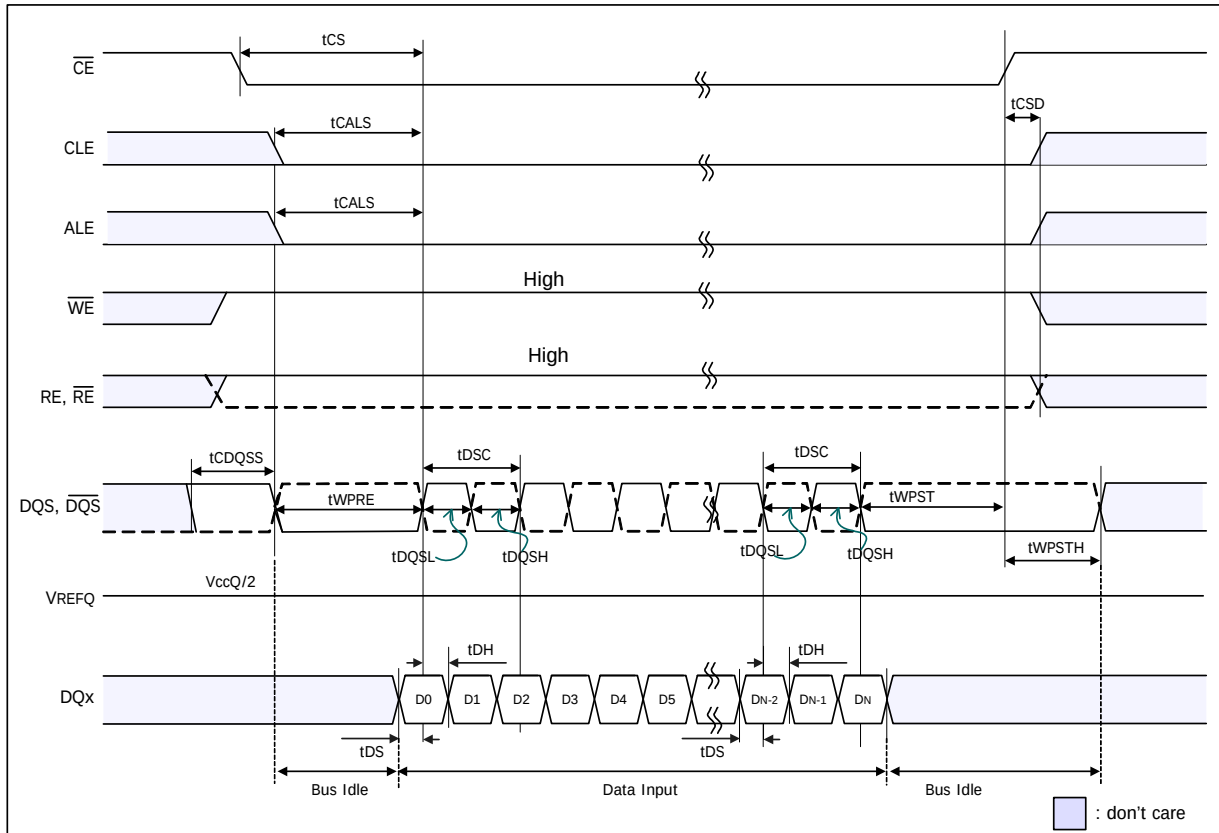
4.5.2. Address Latch Cycle

Figure 4-4. Address Latch Cycle



4.5.3. Basic Data Input Timing

Figure 4-5. Basic Data Input Timing

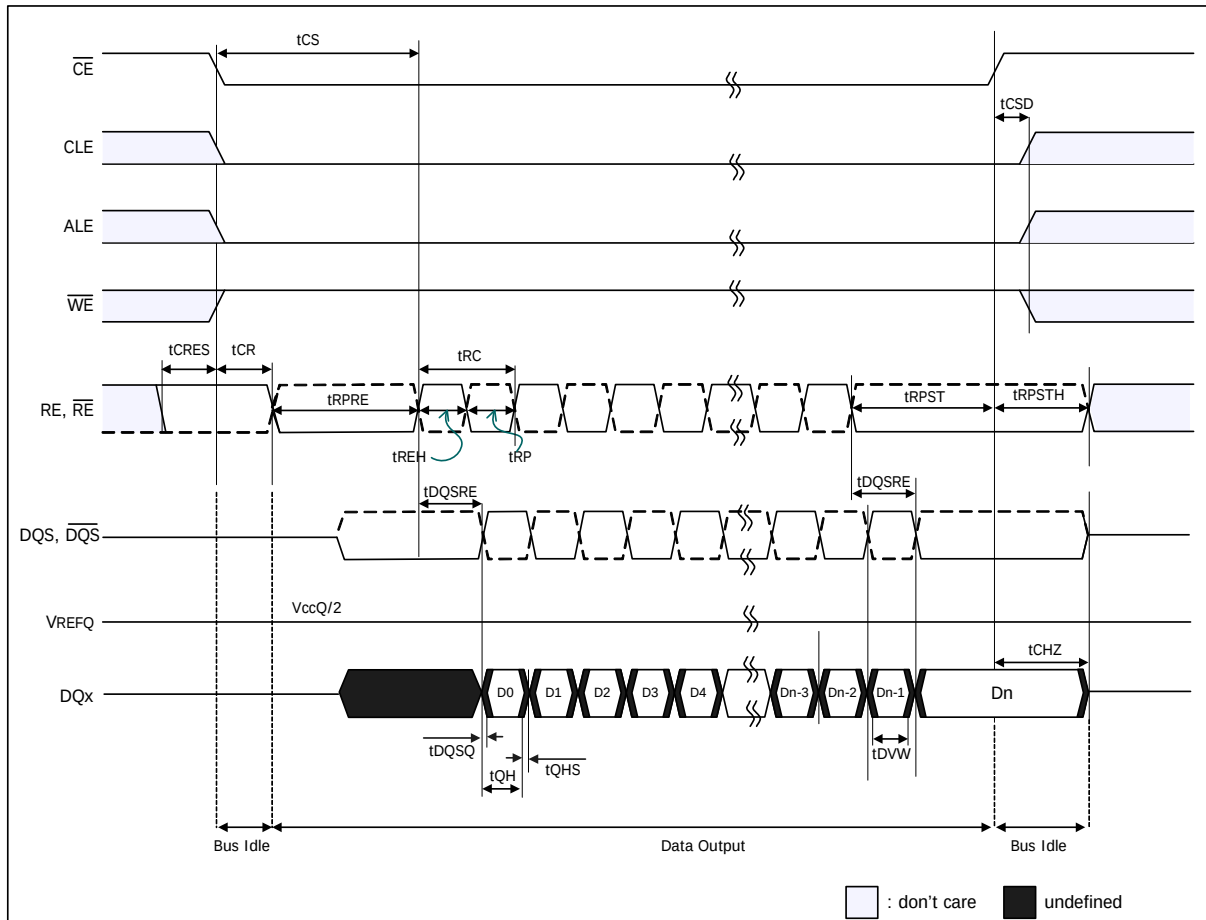


NOTE :

- 1) DQS, $\overline{DQS}$ and Data input buffers are turned-on when $\overline{CE}$ and DQS goes 'Low' and Data inputs begin with DQS, $\overline{DQS}$ toggling simultaneously.
- 2) ALE and CLE should not toggle during t_{WPST} period during t_{CALS} .
- 3) DQS and Data input buffers are turned-off if either CLE or $\overline{CE}$ goes 'High'.
- 4) t_{CDQSS} is defined from the last data input condition of the control signals such as $\overline{CE}$ falling, CLE and ALE.
It is minimum 100ns before the latest falling edge among $\overline{CE}$ falling, CLE and ALE.
- 5) DQS should be fixed a high or low during t_{CDQSS} and remained the status(high or low) before t_{WPST} .

4.5.4. Basic Data Output Timing

Figure 4-6. Basic Data Output Timing

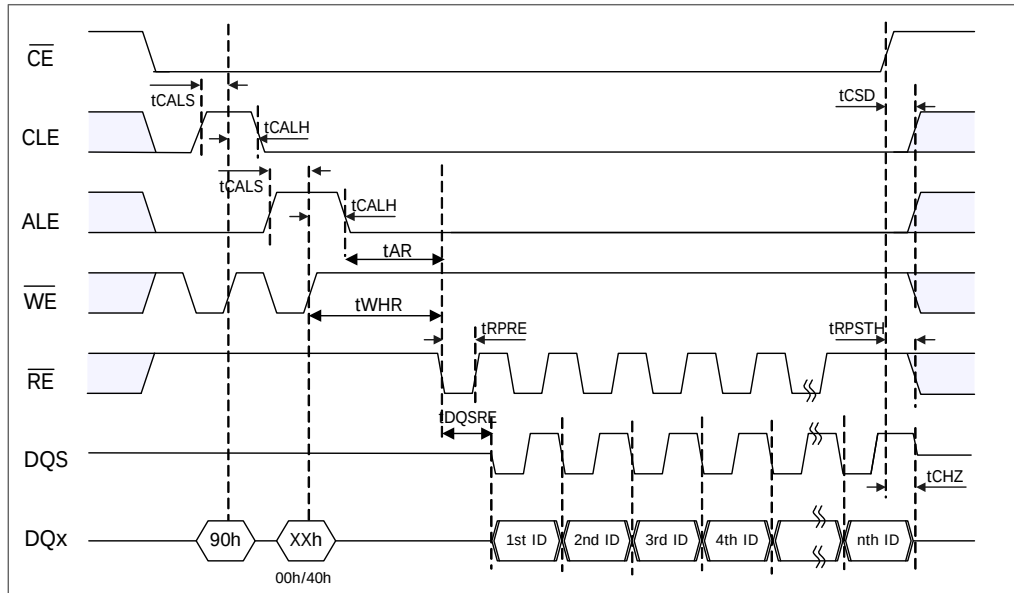


NOTE :

- 1) DQS, $\overline{DQS}$ and DQ drivers are turned-on when $\overline{CE}$ and $\overline{RE}$ goes Low for data out operation.
- 2) ALE and CLE should not toggle during t_{RPRE} period regardless of t_{CALS} .
- 3) DQS and DQ drivers turn from valid value to high-z if either CLE or $\overline{CE}$ goes high.
- 4) The least significant bit of the column address shall always be zero.
- 5) RE shall be high during t_{CRES}

4.5.5. Read ID Operation

Figure 4-7. Read ID Operation

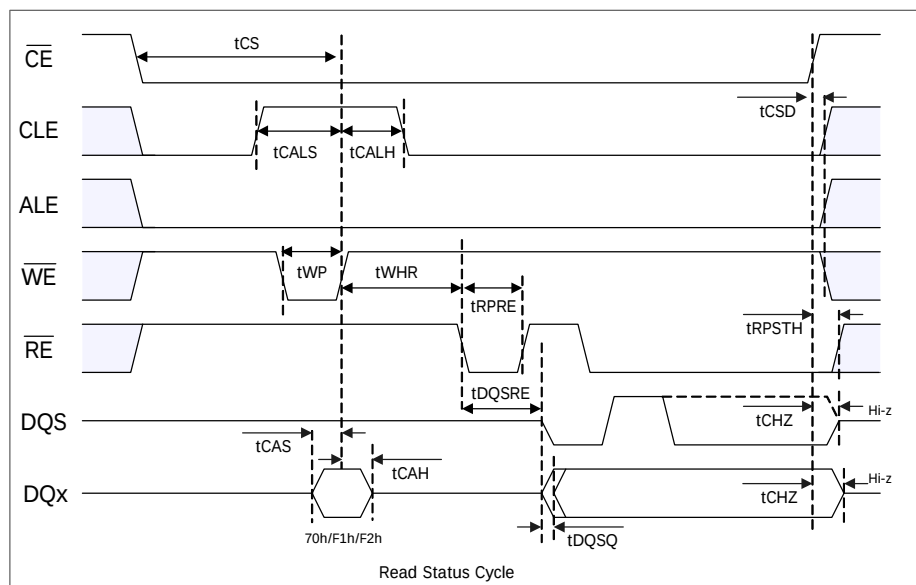


NOTE :

- 1) The Read ID is set by default value.
- 2) The max. speed of Read ID is 200Mbps.

4.5.6. Read Status Cycle

Figure 4-8. Read Status Cycle

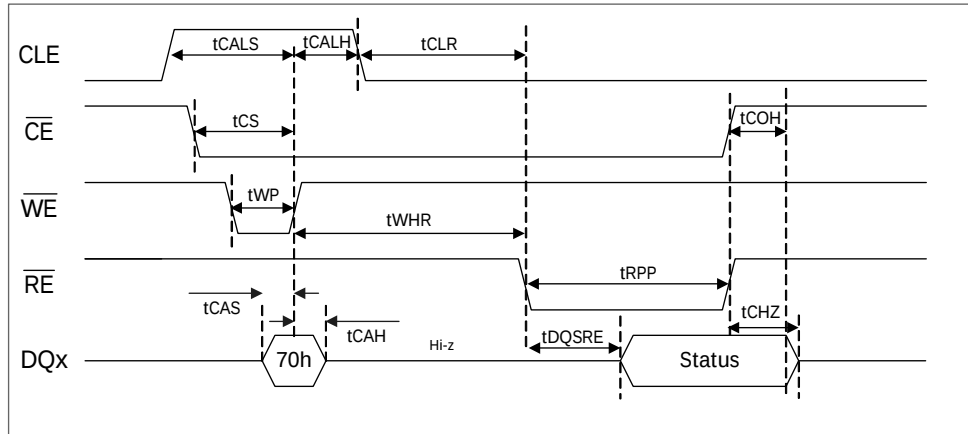


NOTE :

- 1) Even though toggle-mode NAND uses both low and high-going edges of DQS for reads, Read Status operation repeats same output until device status changes
- 2) $\overline{\text{DQS}}$ and Data out buffers turn from valid value to high-z when $\overline{\text{CE}}$ or CLE goes High
- 3) RE can toggle more than once.

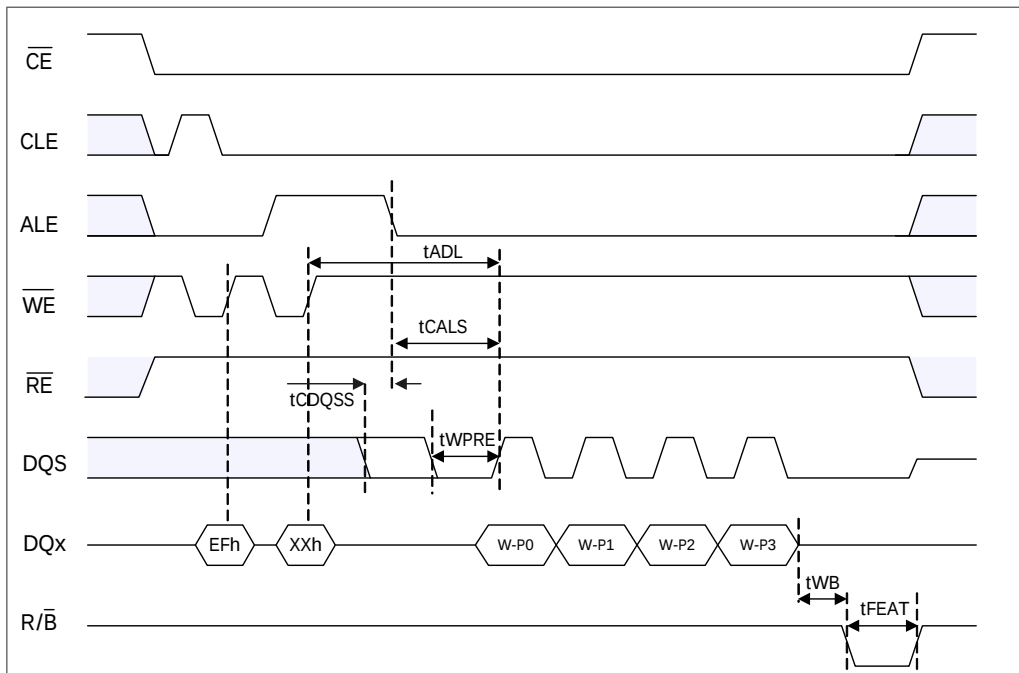
4.5.6.1. Read Status cycle before Toggle DDR setting at Initialization sequence by FFh command

Figure 4-9. Read Status cycle before Toggle DDR setting at Initialization sequence by FFh command



4.5.7. Set Feature

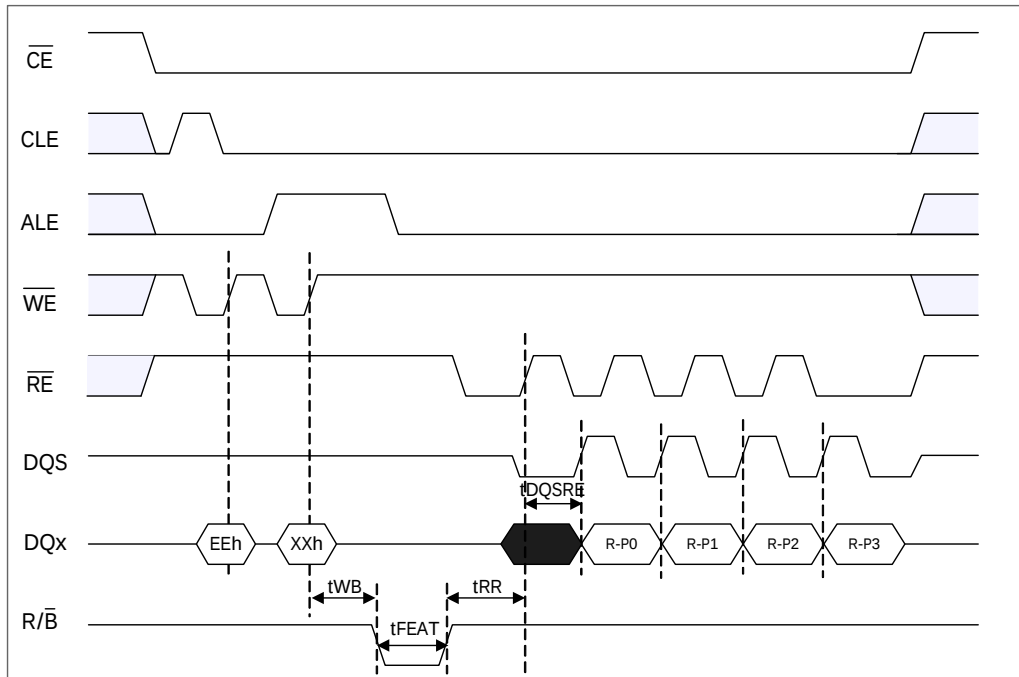
Figure 4-10. Set Feature



Note: Busy time is tFEAT when setting the timing mode.

4.5.8. Get Feature

Figure 4-11. Get Feature

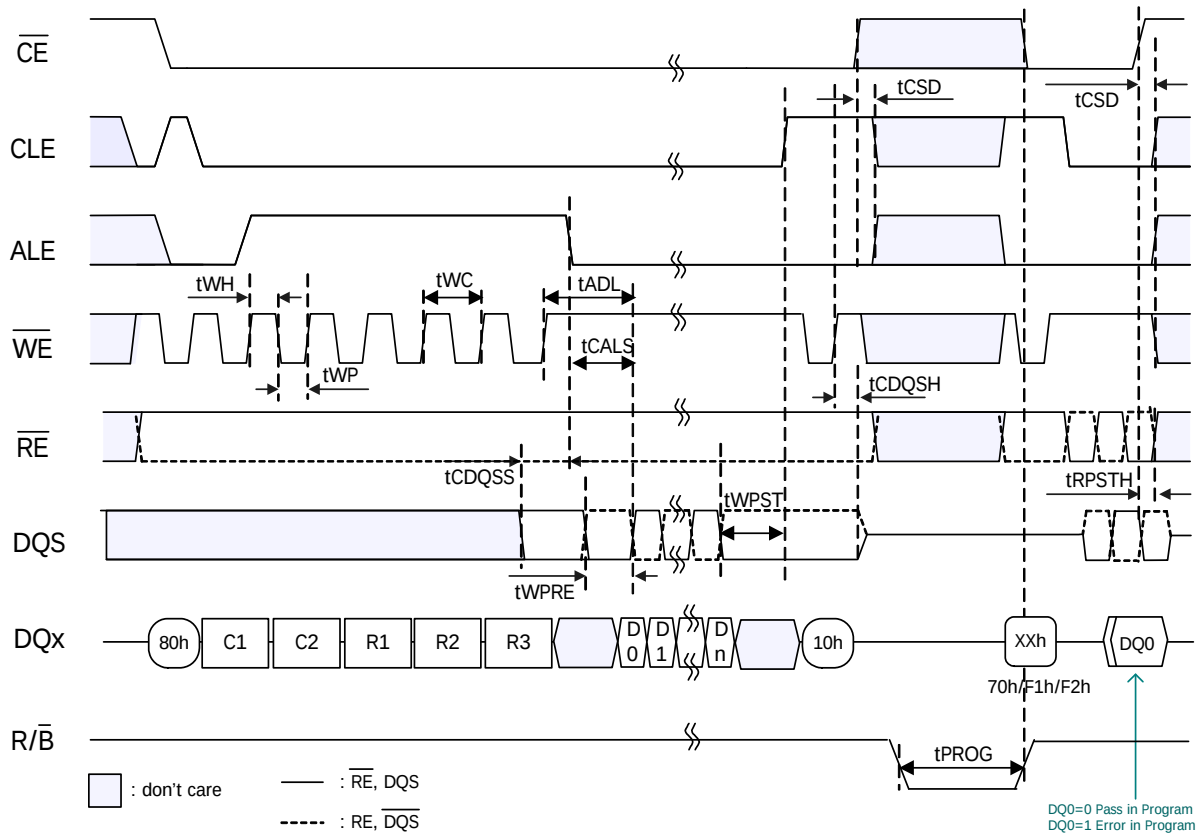


NOTE :

CE is once deasserted, it shall stay high over at least 125ns before it is asserted again.

4.5.9. Page Program Operation

Figure 4-12. Page Program Operation



NOTE :

$\overline{CE}$ is once deasserted, it shall stay high over at least 125ns before it is asserted again.

5. Memory Operation

5.1. BASIC OPERATION

[Basic Command Sets]

Toggle DDR NAND Flash Memory has addresses multiplexed into 8 I/Os. Command, address and data is all written through DQ[7:0] by bringing WE to low while CE is low. Those are latched on the rising edge of WE. Command Latch Enable(CLE) and Address Latch Enable(ALE) are used to multiplex command and address respectively, via the DQ[7:0] pins. Commands which apply to a specific page or block typically have a second command and ones that apply to a target or a LUN have a first command only.

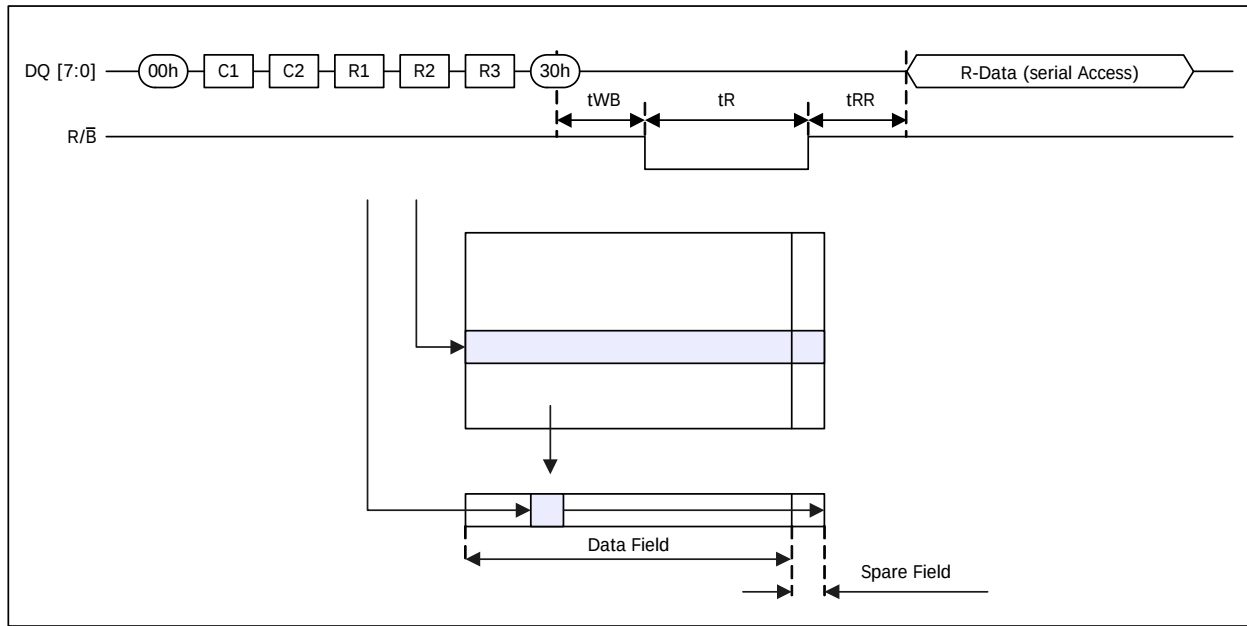
Basic Command sets

Command	Primary or Secondary	1st Set	Address Cycle	2nd Set	Acceptable while Accessed LUN is Busy	Acceptable while Other LUNs are Busy
Page Read	Primary	00h	5	30h	-	Y
Page Program	Primary	80h	5	10h	-	Y
Block Erase	Primary	60h	3	D0h	-	Y
Read for Copy-Back	Primary	00h	5	35h	-	Y
Copy-Back Program	Primary	85h	5	10h	-	Y
Random Data Input	Primary	85h	2	-	-	Y
Random Data Output	Primary	05h	2	E0h	-	Y
Set Features	Primary	EFh	1	-	-	-
Get Features	Primary	EEh	1	-	-	-
Read ID	Primary	90h	-	-	-	-
Read Status	Primary	70h	-	-	Y	Y
Reset	Primary	FFh	-	-	Y	Y
Reset LUN	Primary	FAh	3	-	Y	Y
Half Page Read	Primary	00h	5	2Dh/2Eh	-	Y

5.1.1. Page Read Operation

The Page Read function reads a page of data identified by row address for the selected LUN. The page of data is made available to be read from the page register starting at the specified column address. Figure of “*Page Read Sequence*” defines the Page Read behavior and timings. Reading beyond the end of a page results in indeterminate values being returned to the host.

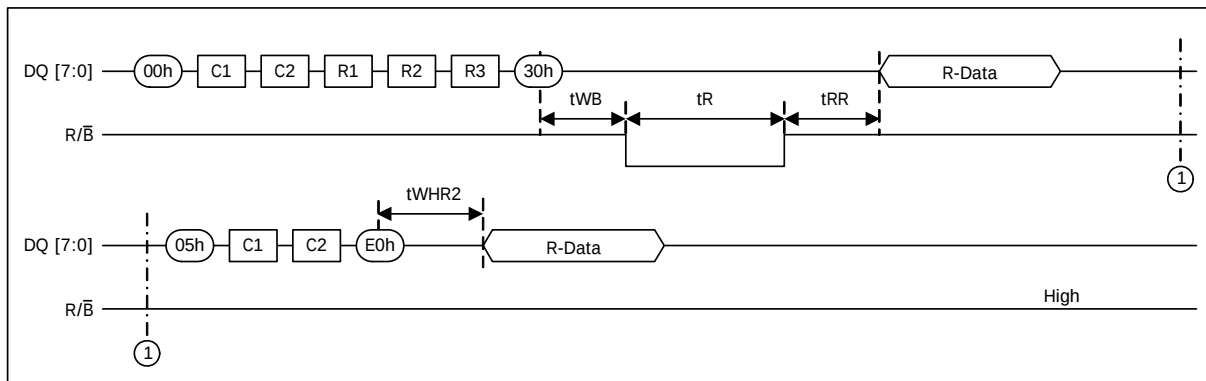
Figure 5-1. Page Read Sequence



5.1.1.1. Page Read Operation with Random Data Output

The Random Data Output function changes the column address from which data is being read in the page register for the selected LUN. The Random Data Output command shall only be issued when the LUN is in a read idle condition. Figure of “*Page Read with Random Data Output Sequence*” defines the Random Data Output behavior and timings. The host shall not read data from the LUN until t_{WHR2} (ns) after the second command (i.e. E0h) is written to the LUN.

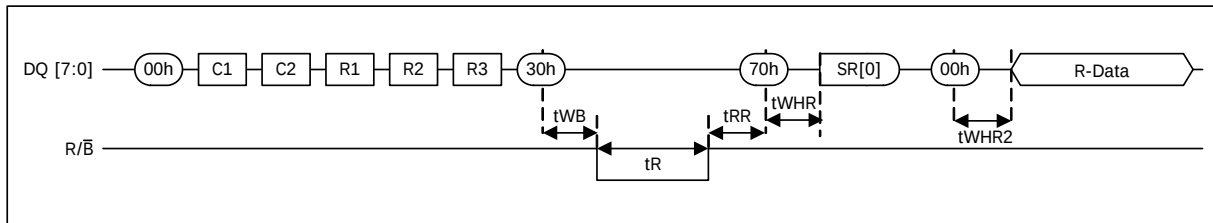
Figure 5-2. Page Read with Random Data Output Sequence



5.1.1.2. Data Out After Read Status

While monitoring the read status to determine when the t_R (transfer from Flash array to a page register) is complete, the host shall re-issue the 00h command to start reading data. Issuing the 00h command will cause data to be returned starting at the selected column address.

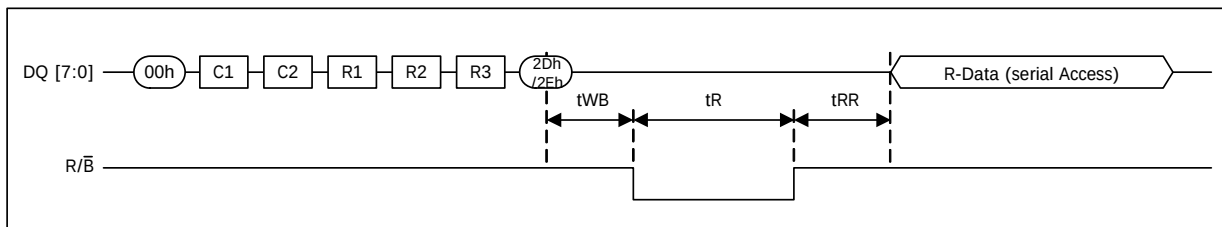
Figure 5-3. Data Out After Read Status Sequence



5.1.2. Fast 8KB Read (Fast Half page Read)

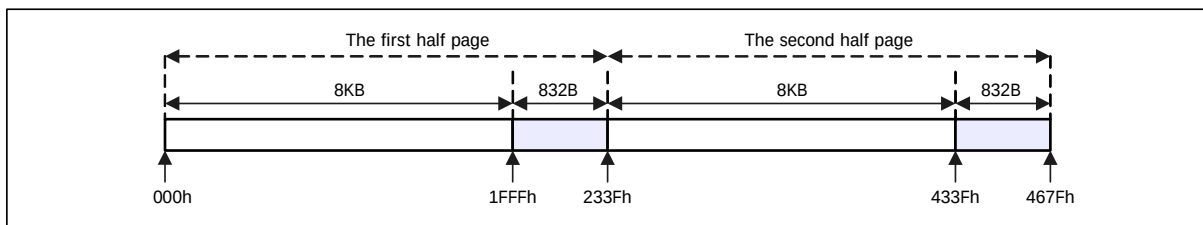
The Fast Half Read (8KB read) function transfers only half page data in NAND array to page register. This function helps applications maximize random read or write throughput. Fast Half Read operation supports only a single plane based Read operation. Since Fast Half Read function transfers only half page, to transfer other side half page data, another 8KB Read operation is required.

Figure 5-4. Fast Half READ sequence



There are the first half page or the second half page which is determined by confirm command and column address. The first half page data is transferred when the confirm command 2Dh is issued with column address between 0000h and 233Fh. The second half page data is transferred when the confirm command 2Eh is issued with column address between 2340h to 467Fh. Figure of "Page layout for Fast Half READ" represents the page layout when Fast 8KB Read is used.

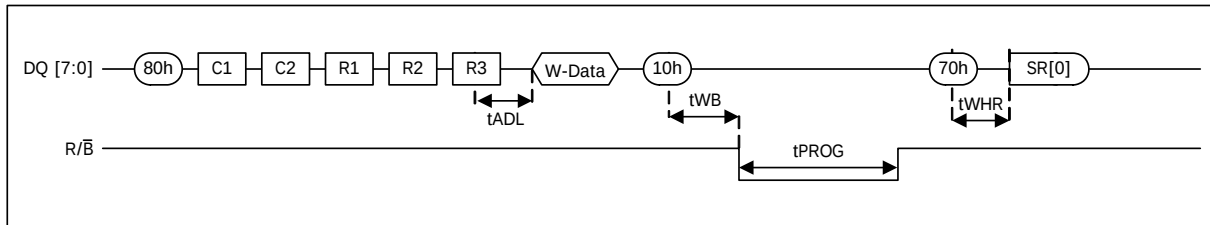
Figure 5-5. Page layout for Fast Half READ



5.1.3. Page Program Operation

The device is programmed basically on a page basis, and each page shall be programmed only once before being erased. The addressing order shall be sequential within a block. The contents of the page register are programmed into the Flash array specified by row address. SR[0] is valid for this command after SR[6] transitions from zero to one until the next transition of SR[6] to zero. Figure of “Page Program Sequence” defines the Page Program behavior and timings. Writing beyond the end of the page register is undefined.

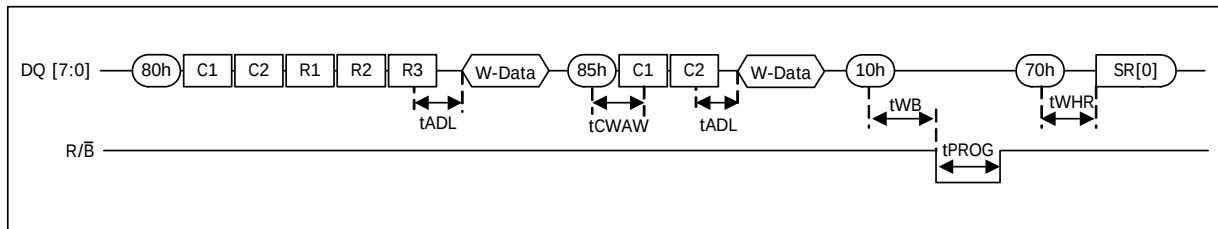
Figure 5-6. Page Program Sequence



5.1.3.1. Program Operation with Random Data Input

The device supports random data input in a page. The column address for the next data, which will be written, may be changed to the address using Random data input command(i.e. 85h). Random data input may be operated multiple times without limitation.

Figure 5-7. Program operation with Random Data Input Sequence

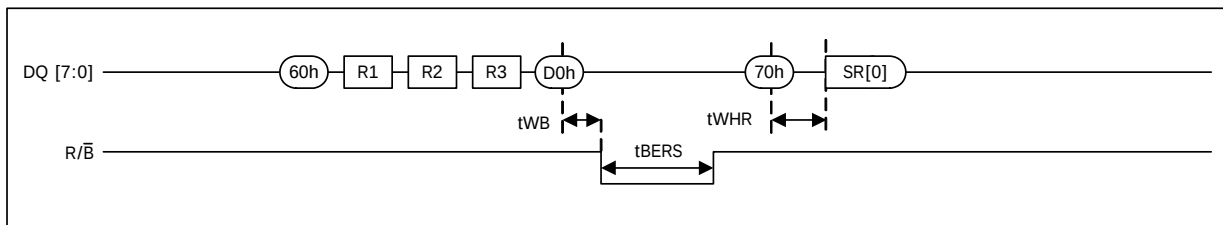


5.1.4. Block Erase Operation

The Block Erase operation is done on a block basis. Only three cycles of row addresses are required for Block Erase operation and a page address within the cycles is ignored while plane and block address are valid.

After Block Erase operation passes, all bits in the block shall be set to one. SR[0] is valid for this command after SR[6] transitions from zero to one(i.e. the selected LUN is ready) until the LUN goes in busy state by a next command. Figure of “Block Erase Sequence” defines the Block Erase behavior and timings.

Figure 5-8. Block Erase Sequence

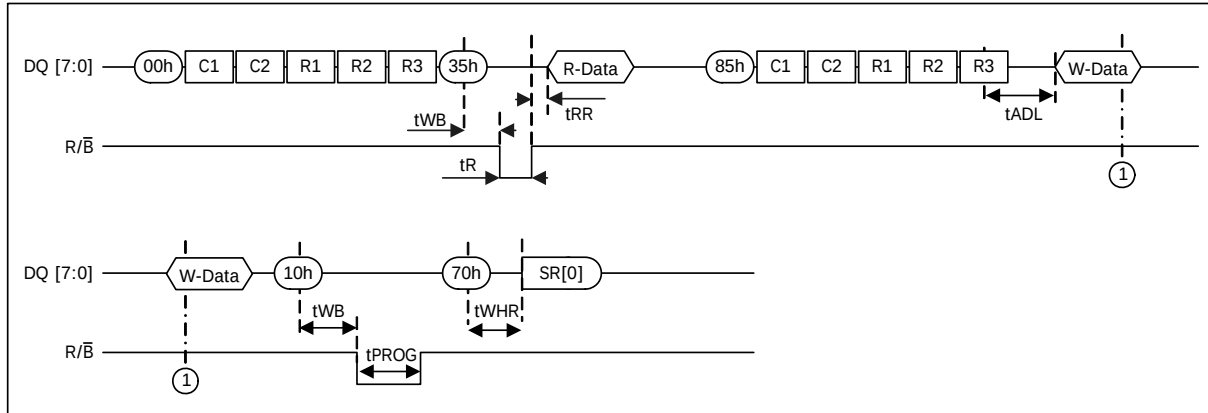


5.1.5. Copy-Back Program Operation

The Copy-Back Program with Read for Copy-Back is configured to efficiently rewrite data stored in a page without data re-loading when no error within the page is found. Since the time-consuming re-loading cycles are removed, copy-back operation helps the system performance improve. The benefit is especially obvious when a part of a block is updated and the rest of the block also needs to be copied to the newly assigned free block.

The Copy-Back operation consists of 'Read for Copy-Back' and 'Copy-Back Program'. A host reads a page of data from a source page using 'Read for Copy-Back' and copies read data back to a destination page on the same LUN by 'Copy-Back Program' command. Copy-Back Program Operation shall work only within the same plane. Figure of "Copy-Back Program Sequence" defines the Copy-Back Program behavior and timings.

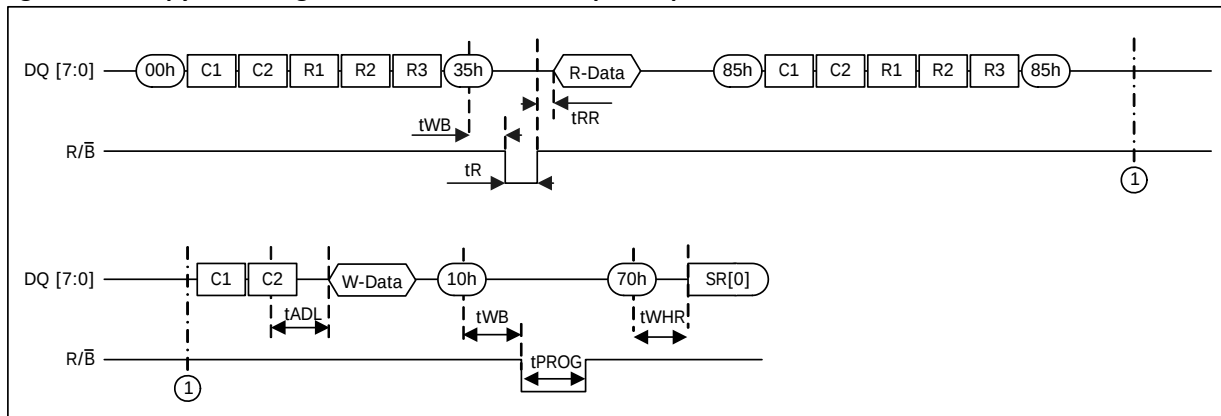
Figure 5-9. Copy-Back Program Sequence



5.1.5.1. Copy-Back Program Operation with Random Data Input

After a host completes to read data from a page register, the host may modify data using Random Data Input command if required. Figure of "Copy-Back Program with Random Data Input Sequence" defines the Copy-Back Program with Random Data Input behavior and timings.

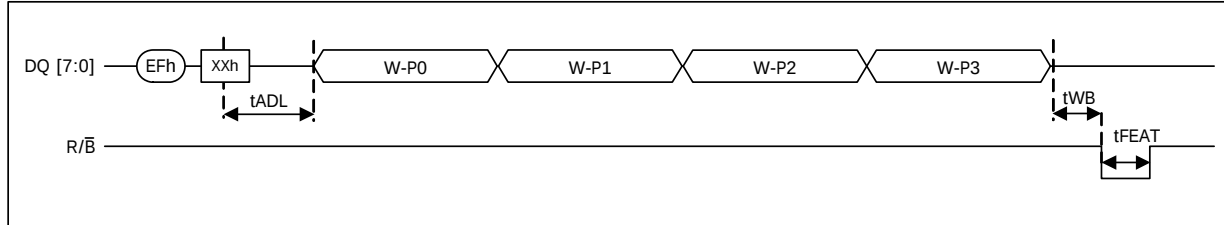
Figure 5-10. Copy-Back Program with Random Data Input Sequence



5.1.6. Set Feature Operation

Users may set particular features using 'Set Feature' operation. Once a feature is set by users, it shall not be changed until the device is powered off or setting is changed by users. Figure of "Set Feature Sequence" defines the Set Features behavior and timings and Table 5-1 defines features that users can change. Note that FFh command is not allowed during SET FEATURE sequence. These settings are not retained across Reset (FFh) and the power-on values are reverted to.

Figure 5-11. Set Feature Sequence



NOTE :

The feature-setting shall work on lower than 100MHz.

Table 5-1. Set feature addresses

1st Cycle	2nd Cycle	Description
EFh	01h	Timing mode setting
	02h	Toggle 2.0 specific setting
	10h	Driver strength setting
	30h	External VPP setting

5.1.6.1. Timing Mode setting (01h)

The Data Interface setting among the Timing Mode setting is not retained across Reset (FFh); after a Reset (FFh) the Data Interface shall be SDR.

The PC setting is retained across Synchronous Reset (FCh), and Reset LUN (FAh) commands. If the Reset (FFh) command is issued when the Data Interface is configured as Toggle mode, then the host shall use the SDR data interface until a new data interface is selected with Set Features. Hosts shall only set a timing mode that is explicitly shown as supported in the Read Parameter Page.

The results of the host using Set Features to transition from the Toggle mode to the SDR is indeterminate. To transition to the SDR data interface, the host should use the Reset (FFh) command.

Table 5-2. Timing mode setting assignment

	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0
P0	Reserved	PC	Data interface		Reserved			
P1	Reserved							
P2	Reserved							
P3	Reserved							

NOTE : P1, P2 and P3 are reserved and shall be written with 00h.

	Description
--	-------------

PC(Program Clear)	when the Program (80h) command is received, 0 : the page register(s) for each LUN that is part of the target is cleared. 1 : only the page register for the LUN that is selected by command is cleared.
Data Interface	00 : SDR (default power-on value) 10 : Toggle mode (NV_DDR2 mode) others : Reserved

5.1.6.2. Toggle 2.0 specific setting (02h)

This setting is required in order to use reference voltage and complementary signal and ODT. DQS latency cycle can also be configured by this SET FEATURE operation to read the first valid data correctly.

When VREFQ signal or complementary signals are set, those signals shall be applied before SET FEATURE sequence. The setting is done at the rising edge of R/B, thus the signals are used to check ready by READ STATUS operation.

Table 5-3. Toggle 2.0 specific setting assignment

	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0
P0	ODT (with RTT value) for Program				Reserved	$\overline{RE}$	$\overline{DQS}$	VREF
P1	# of Latency DQS cycle for Program				# of Latency DQS cycle for Read			
P2	ODT(with RTT value) for Read				Reserved	Reserved	ODT_RD	ODT_PR
P3	Reserved							

NOTE :

1) N/A, and reserved shall be written with 00h.

2) When differential signaling (i.e. RE and DQS) is used, VREFQ shall be set to be enabled for DQ.

Table 5-4. Definition of Toggle 2.0 specific setting

	Description
VREFQ	0 : Disabled (dafault) 1 : Enabled
$\overline{DQS}$	0 : Disabled (default) 1 : Enabled
RE_c	0 : Disabled (default) 1 : Enabled
ODT (DQ, DQS, and $\overline{DQS}$) for Program (Not Supported)	0000: Disabled (default) 0001: Enabled with 150 ohm 0010: Enabled with 100 ohm 0011: Enabled with 75 ohm 0100: Enabled with 50 ohm
ODT (DQ, DQS, and $\overline{DQS}$) for Read (Not Supported)	0000: Disabled (default) 0001: Enabled with 150 ohm 0010: Enabled with 100 ohm 0011: Enabled with 75 ohm 0100: Enabled with 50 ohm
# of Latency DQS cycle (READ)	0000 : No latency DQS cycle (default) 0001 : One latency DQS cycle 0010 : Two latency DQS cycle

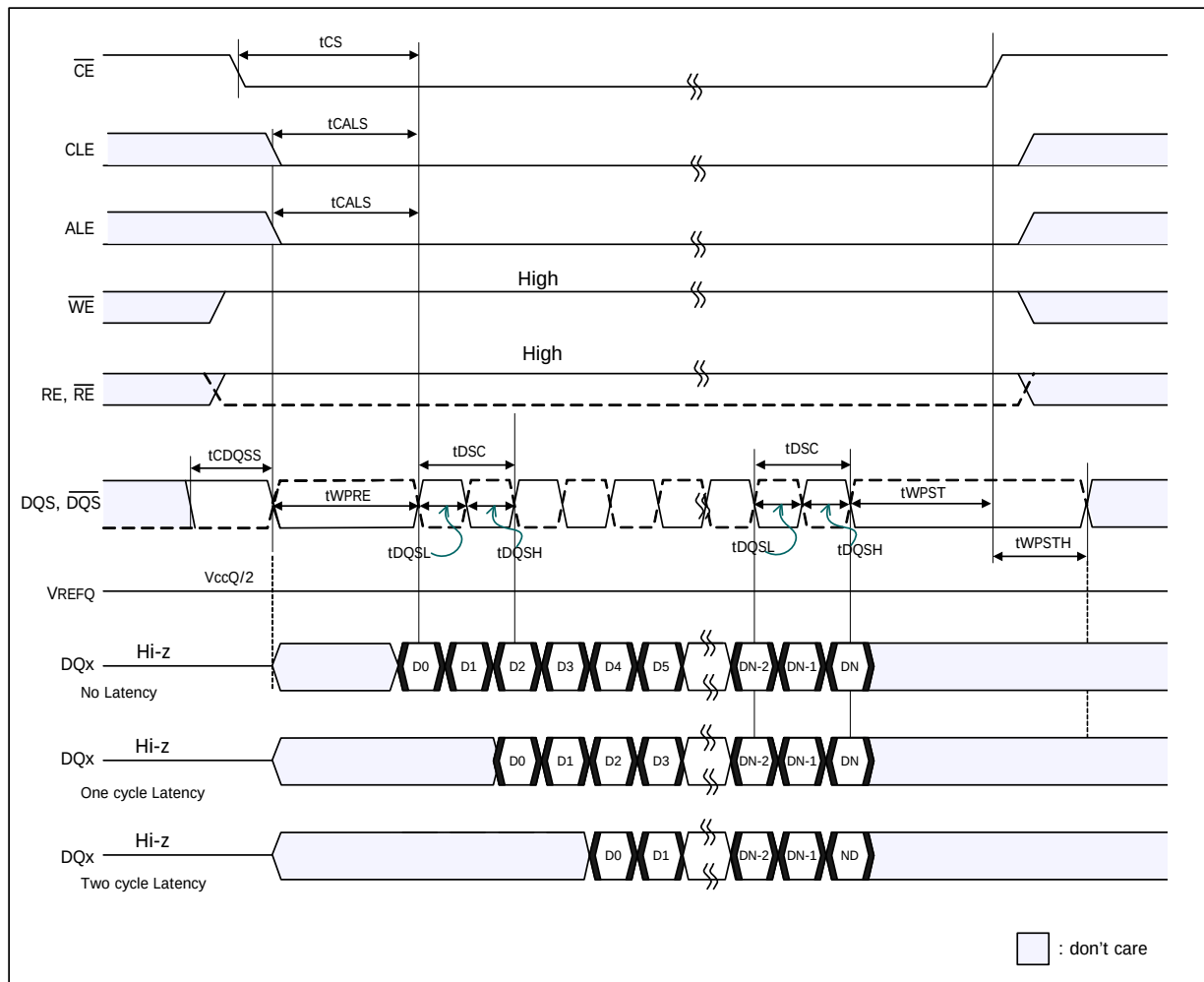
	Description
# of Latency DQS cycle (WRITE)	0000 : No latency DQS cycle (default) 0001 : One latency DQS cycle 0010 : Two latency DQS cycle

NOTE :

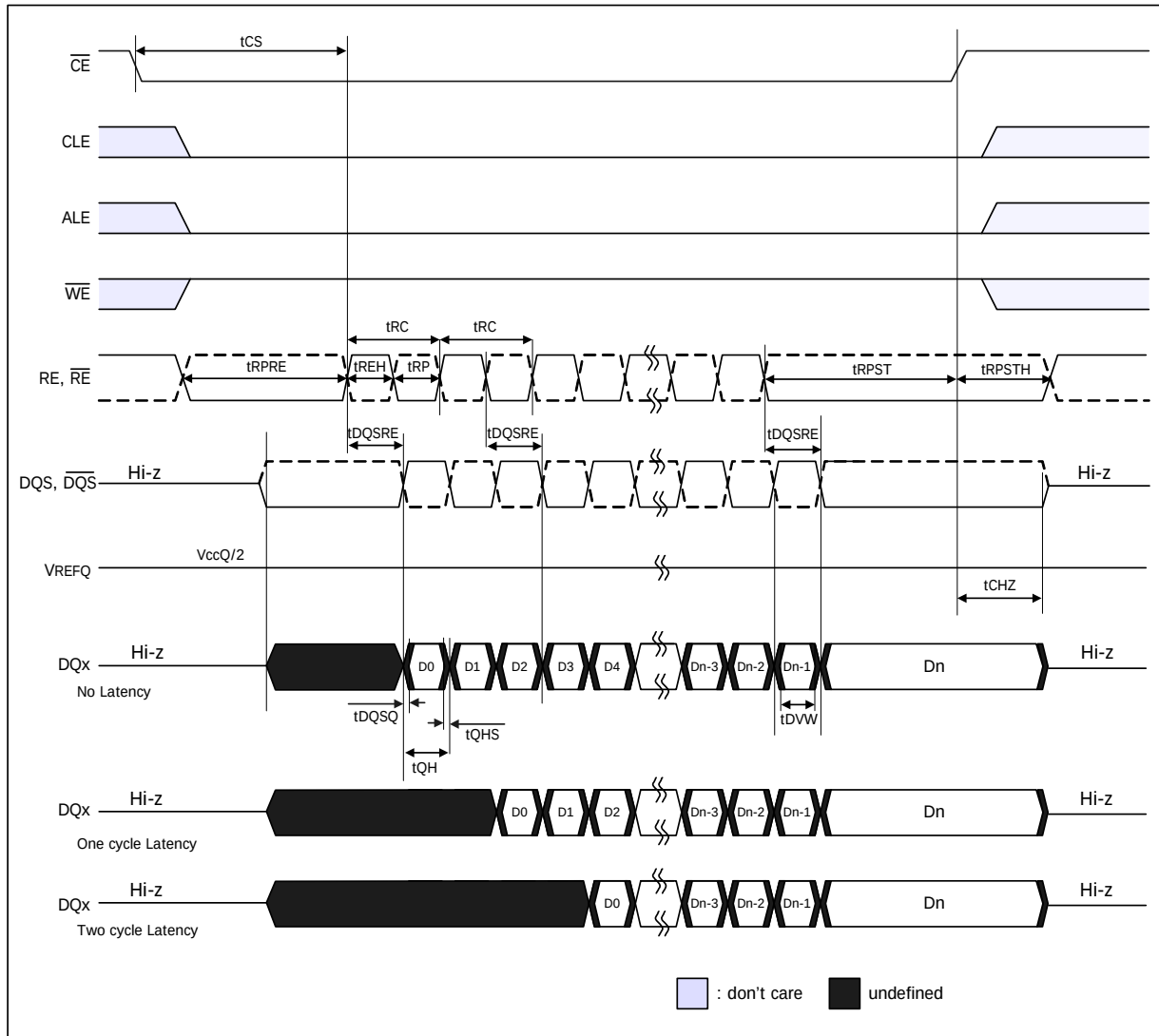
If DQS latency be enabled while the Toggle interface is active, then DQS latency shall be used for all subsequent commands after the Set Features is complete.

Figure 5-12. Example of DQS latency (Write / Read)

Write



Read



5.1.6.3. Driver strength setting (10h)

Driver strength is configured according to the P0 value.

Table 5-5. Definition of Driver strength setting

P0 Value	Description
00h ~ 01h	Reserved
02h	Driver Multiplier : Underdriver
03h	Reserved
04h	Driver Multiplier : 1 (default)
05h	Reserved

P0 Value	Description
06h	Driver Multiplier : Overdriver 1
07h	Reserved
08h	Driver Multiplier : Overdriver 2
09h ~ FFh	Reserved

NOTE:

P1, P2 and P3 are reserved and shall be written with 00h.

5.1.6.4. External VPP (30h)

External high voltage(i.e. typical 12V) feature offers power saving on program and read operations. The external high voltage shall be supplied prior to the feature setting and it shall persist within 11V to 12V until it is set to default(i.e. off). The maximum external Vpp supply current per LUN is 5mA.

Table 5-6. Definition of External Vpp

P0 Value	Description
00h	OFF(default)
01h	ON
02h ~ FFh	Reserved

NOTE :

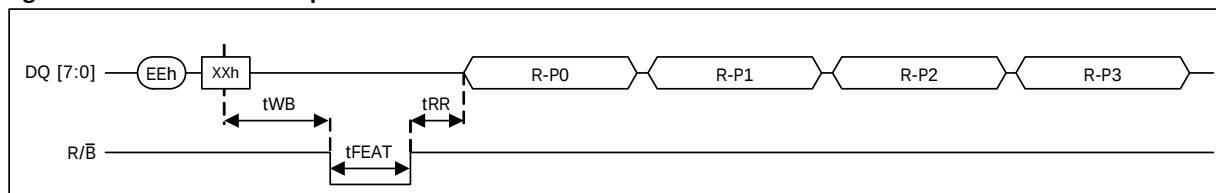
P1, P2 and P3 are reserved and shall be written with 00h.

5.1.7. Get Feature Operation

Users find how the target is set through 'Get Feature' command. The function shall return the current setting information. If a host starts to read the first byte of data (i.e. P0 value), the host shall complete reading all four bytes of data before issuing another command (including Read Status or Read Status Enhanced). Get Feature operation shall work on lower than 133Mbps. Figure of "Get Feature Sequence" defines the Get Features behavior and timings.

If Read Status (or Read Satus Enhanced) is used to monitor whether the tFEAT time is complete, the host shall issue Read command (i.e. 00h) to read P0-P1-P2-P3.

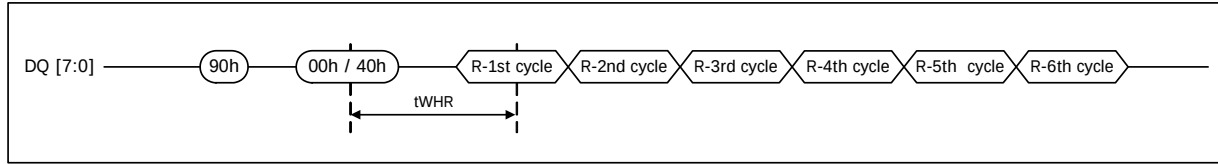
Figure 5-13. Get Feature Sequence



5.1.8. Read ID Operation

The ID of a target is read by command 90h followed by 00h or 40h address. Read ID operation shall work on lower than 133Mbps. Figure of “Read ID Sequence” defines Read ID operation behavior and timings.

Figure 5-14. Read ID Sequence



5.1.8.1. 00h Address ID Definition

Users can read six bytes of ID containing manufacturer code, device code and architecture information of the target by command 90h followed by 00h address. The command register remains in Read ID mode until another command is issued.

Table 5-7. 00h address ID cycle

Device	Density	VccQ	1st Cycle	2nd Cycle	3rd Cycle	4th Cycle	5th Cycle	6th Cycle
H27QDG8x2B8R-BCF	16GB	1.8V	ADh	5Ah	14h	ABh	42h	49h
H27QEGxxDB8R-BCF	32GB							
H27QFGxxxBxR-BCF	64GB							
H27Q1TxxxBxR-BCF	128GB							
H27Q2T8xxBxR-BCF	256GB							

Table 5-8. 00h address ID cycle before setting VccQ 1.8V

Device	Density	VccQ	1st Cycle	2nd Cycle	3rd Cycle	4th Cycle	5th Cycle	6th Cycle
H27QDG8x2B8R-BCF	16GB	3.3V	ADh	3Ah	14h	ABh	42h	4Ah
H27QEGxxDB8R-BCF	32GB							
H27QFGxxxBxR-BCF	64GB							
H27Q1TxxxBxR-BCF	128GB							
H27Q2T8xxBxR-BCF	256GB							

Table 5-9. Read ID Definition - Address ID cycle

	Description
1st Byte	Maker Code
2nd Byte	Device Code
3rd Byte	Internal Number, Cell Type, Number of Program pages, Cache program
4th Byte	Page size (without Spare area), Block size (without Additional Block), Redundant Area size
5th Byte	Plane Number, ECC Level
6th Byte	NAND Technology, Interface

5.1.8.2. 40h Address ID Definition

Toggle DDR NAND also provide a six bytes of JEDEC standard signature ID. Users can read the ID by command 90h followed by 40h address. Any data returned after the six bytes of JEDEC standard signature is considered reserved for future use.

Table 5-10. 40h Address ID Cycle

1st Cycle	2nd	3rd Cycle	4th Cycle	5th Cycle
4Ah	45h	44h	45h	43h

Table 5-11. 40h Address ID Cycle

Cycle	Description	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0
1st	J	0	1	0	0	1	0	1	0
2nd	E	0	1	0	0	0	1	0	1
3rd	D	0	1	0	0	0	1	0	0
4th	E	0	1	0	0	0	1	0	1
5th	C	0	1	0	0	0	0	1	1

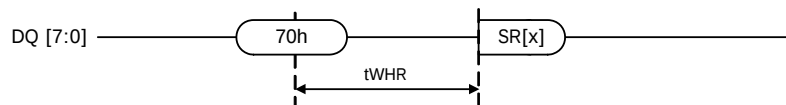
5.1.9. Read Status Operation

In the case of non-multi-plane operations, the Read Status function retrieves a status value for the last operation issued. If multiple multi-plane operations are in progress on a single LUN, then Read Status returns the composite status value. Specifically, Read Status shall return the combined status value of the independent status register bits according to Table for “Read Status Definition” Figure of “Read Status Sequence” defines the Read Status behavior and timings.

Table 5-12. Read Status Definition

	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7
Definition of value	Pass : “0” Fail : “1”	Pass : “0” Fail : “1”	Reserved	Reserved	Reserved	Busy : “0” Ready : “1”	Busy : “0” Ready : “1”	Protected : “0” Not Protected : “1”
Block Erase	Pass/Fail	Not Use	Not Use	Not Use	Not Use	Not Use	Busy/Ready	Write Protect
Page Program	Pass/Fail	Not Use	Not Use	Not Use	Not Use	Not Use	Busy/Ready	Write Protect
Read	Not Use	Not Use	Not Use	Not Use	Not Use	Not Use	Busy/Ready	Write Protect

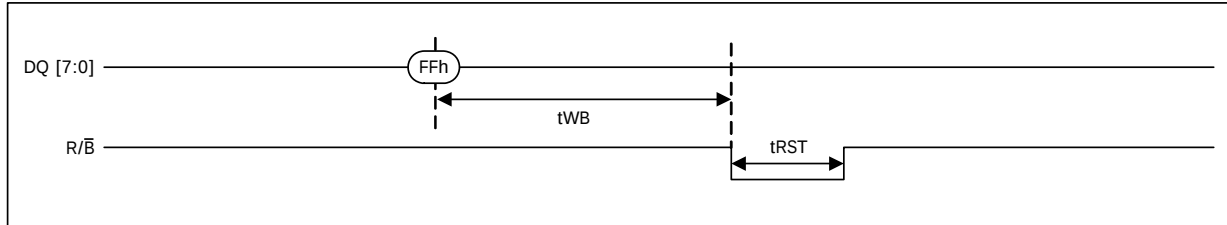
Figure 5-15. Read Status Sequence



5.1.10. Reset Operation

Toggle DDR NAND offers a reset function by command FFh. When the device is in 'Busy' state during any operation, the Reset operation will abort these operations except during power-on when Reset shall not be issued until R/B is set to one (i.e. ready). The contents of memory cells being programmed are no longer valid, as the data will be partially programmed or erased. Although the device is already in process of reset operation, a new reset command will be accepted. Figure of "Reset Sequence" defines the Reset behavior and timings.

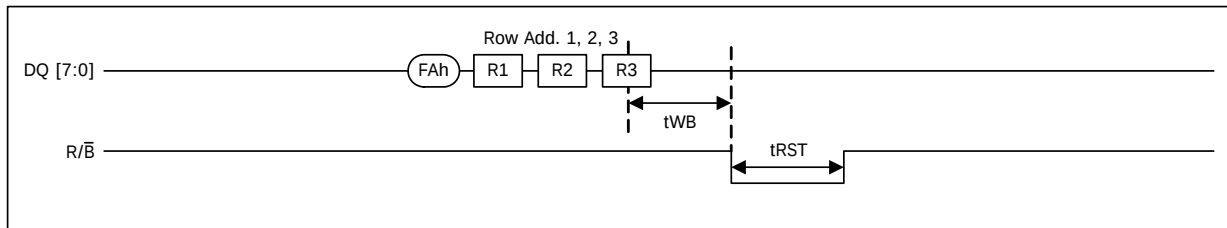
Figure 5-16. Reset Sequence



5.1.11. Reset LUN operation

A certain LUN within a target can be reset by command FAh followed by row addresses. Row addresses are required to set a LUN to be reset. Figure of "Single chip Reset Sequence" defines the Reset LUN behavior and timings.

Figure 5-17. Single chip Reset Sequence



5.2. Extended OPERATION

5.2.1. Extended Command Sets

Table of “*Extended Command Sets*” defines the Extended Command Sets. Primary and Secondary Commands are also categorized in the table. Primary commands are recommended to use when a particular function is implemented, while Secondary commands are for alternative implementation for backward compatibility.

Table 5-13. Extended Command Sets

Function	Primary or secondary	1st Set	Address Cycles for 1st Set	2nd Set	Address Cycles for 2nd Set
Two-Plane Page Read	Primary	00h--32h	5	00h--30h	5
Two-Plane Page Read	Secondary	60h--	3	--30h	-
Two-Plane Random Cache Read	Primary	00h--32h	5	00h--31h	5
Two-Plane Random Data Output ¹⁾	Primary	00h---05h	5	--E0h	2
Two-Plane Page Program	Primary	80h--11h ²⁾	5	81h--10h	5
Two-Plane Block Erase	Primary	60h	3	--D0h	-
Two-Plane Read for Copy-Back	Primary	00h--32h	5	00h--35h	5
Two-Plane Read for Copy-Back	Secondary	60h	5	--35h	-
Two-Plane Copy-Back Program	Primary	85h--11h	5	81h--10h	5
Device Identification Table Read	Primary	ECh	1	-	-
Read status enhanced	Primary	78h	3	-	-
Read LUN#0 Status	Secondary	F1h	-	-	-
Read LUN#1 Status	Secondary	F2h	-	-	-
Read LUN#2 Status	Secondary	F3h	-	-	-
Read LUN#3 Status	Secondary	F4h	-	-	-

NOTE :

- 1) Multi-Plane Random Data out must be used after Multi-Plane Page Read.
- 2) Any command between 11h and 80h/81h/85h is prohibited except 70h/78h/F1h/F2h/F3h/F4h and FFh.

5.2.2. Two-Plane Page Read Operation

The Two-Plane Page Read operation is an extension of the Page Read operation. The device supporting Two-plane page read operation also allows multi Random data-output from each plane(i.e. Two-Plane Random Data Output) once multi-pages from each plane are loaded to page registers. With the primary command, R/B returns to ready in a short time(i.e. tDBSY) after the first command 32h since it does not load data from a selected page, and the selected page data of each plane are transferred to the cache registers via page registers in less than tR after command 30h. When setting page addresses of each plane, the page addresses shall be identical although block addresses differ.

The Two-plane addresses shall be set in a consecutive manner from a lower address plane to a higher address plane and the same plane address shall not be set twice within a set of address setting sequence for the Two-Plane Page Read Operation. Starting plane address shall be Plane0.

Once the data is loaded into the cache registers, the data on the first plane can be read out by issuing the Two-Plane Random Data Output command. The data on the other plane can be also read out using the identical command sequences. Figure of “*Example Timing with Two-plane Page Read (Primary)*” and Figure of “*Example Timing with Two-plane Page Read (Secondary)*”

define Two-plane Page Read and Two-Plane Random Data Output behavior and timings.

Figure 5-18. Example Timing with Two-plane Page Read (Primary)

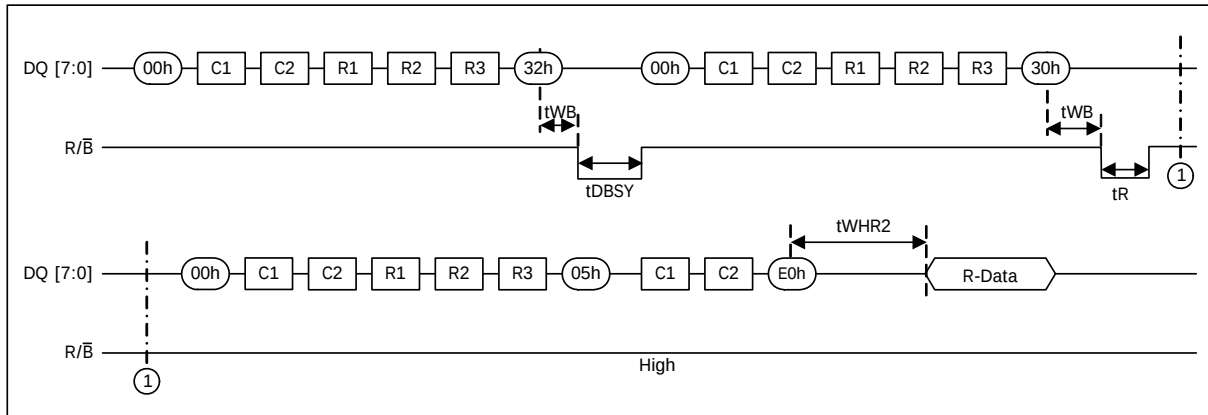
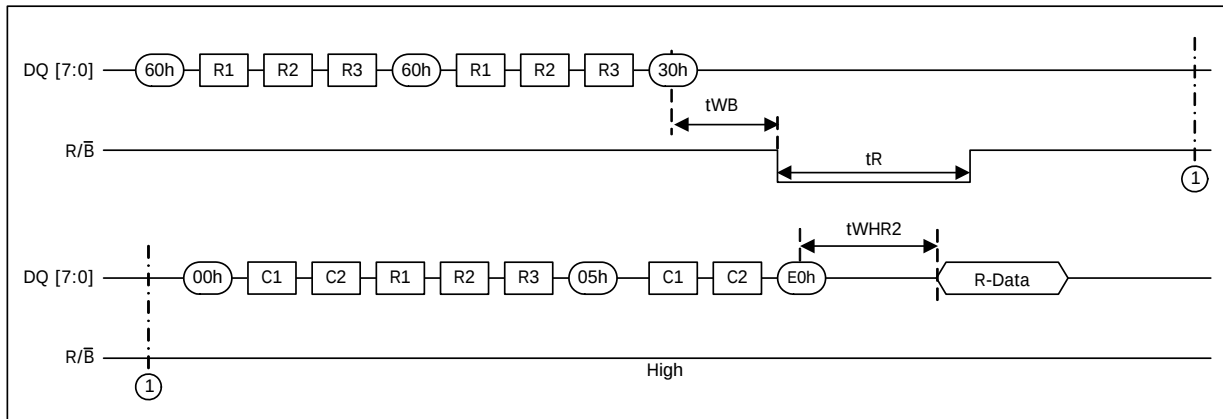


Figure 5-19. Example Timing with Two-plane Page Read (Secondary)



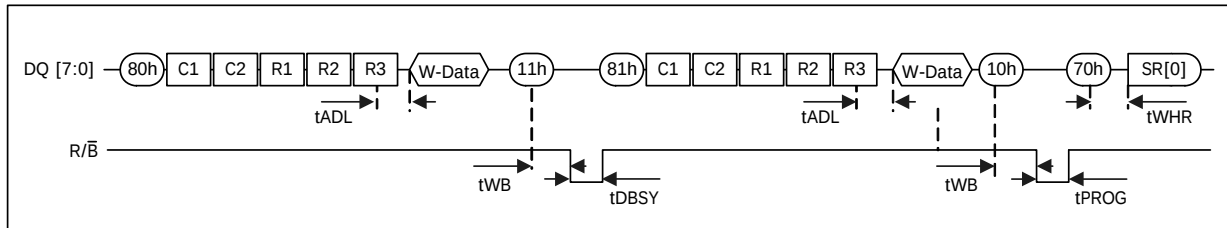
5.2.3. Two-Plane Page Program Operation

Two-Plane Page Program function extends an effective programmable page size using two planes.

When a host moves on a plane for loading another data, command 11h for the second command is used. After 11h command, R/B returns high (i.e. ready) in a short period of time since it is not actual programming operation. At the last page loading, command 81h is issued before loading data and command 10h after data loading is issued for the second command. After command 10h, all loaded data in each plane starts to be programmed to Flash array simultaneously.

Two-plane addresses shall be set in a consecutive manner from a lower address plane to a higher address plane and the same plane address shall not be set twice within a set of address setting sequence for the Two-Plane Page Program Operation. Starting plane address shall be plane0 and the number of planes which are set for Two-Plane Page Program Operation shall be even. Figure of "Example Timing with Two-plane Page Program" defines Two-Plane Page Program behavior and timings.

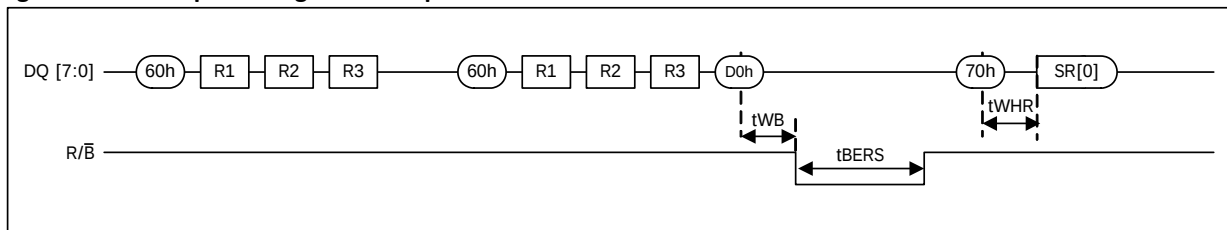
Figure 5-20. Example Timing with Two-plane Page Program



5.2.4. Two-Plane Block Erase

Two-Plane Block Erase allows users to erase multiple blocks comprising a block of each plane simultaneously. Two-plane addresses shall be set in a consecutive manner from a lower address plane to a higher address plane and the same plane address shall not be set twice within a set of address setting sequence for the Two-Plane Block Erase Operation. Starting plane address shall be plane0 and the number of planes which are set for Two-Plane Block Erase Operation shall be even. Figure of “*Example Timing with Two-plane Block Erase*” defines Two-Plane Block Erase behavior and timings.

Figure 5-21. Example Timing with Two-plane Block Erase



5.2.5. Two-Plane Copy-Back Program Operation

The Two-Plane Copy-Back Program is an extension of the Copy-back Program. Two-Plane Copy-Back Program operation is executed two sets of commands, Two-Plane Read for Copy-Back and Two-Plane Copy-Back Program. The Two-plane addresses shall be set in a consecutive manner from a lower address plane to a higher address plane and the same plane address shall not be set twice within a set of address setting sequence for the Two-Plane Copy-Back Program Operation. Starting plane address shall be plane0 and the number of planes which are set for Two-Plane Copy-Back Program Operation shall be even. The read Data shall be copied back to a page in the same plane. Figure of “*Example Timing with Two-plane Copy-Back Program (Primary)*” and Figure of “*Example Timing with Two-plane Copy-Back Program (Secondary)*” defines Two-Plane Copy-Back Program behavior and timings.

Figure 5-22. Example Timing with Two-plane Copy-Back Program (Primary)

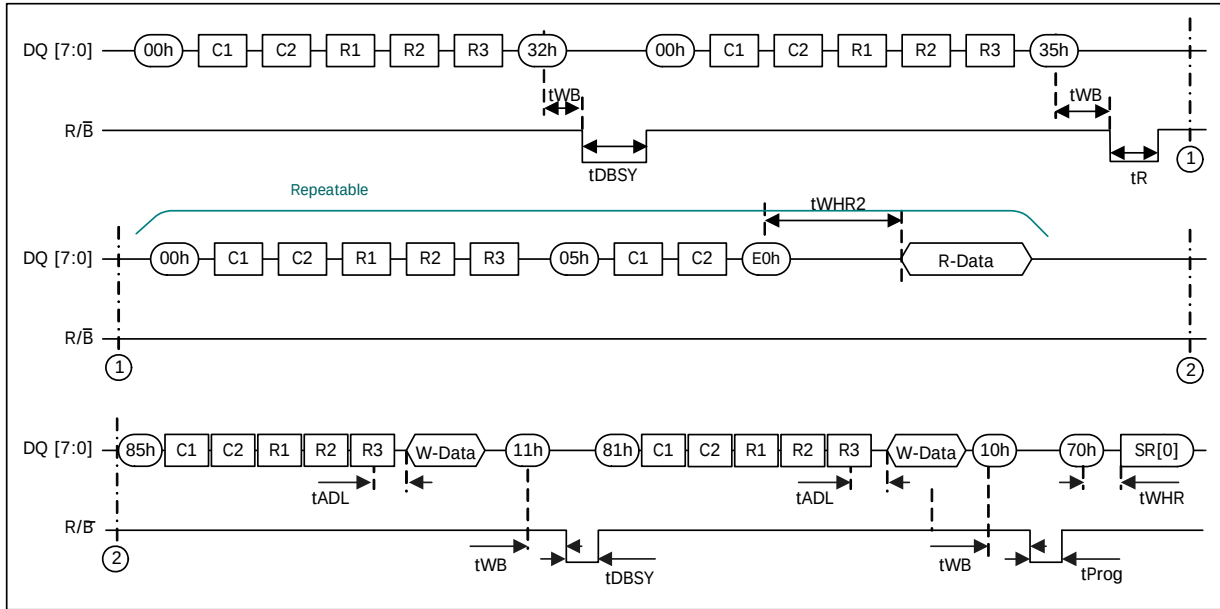
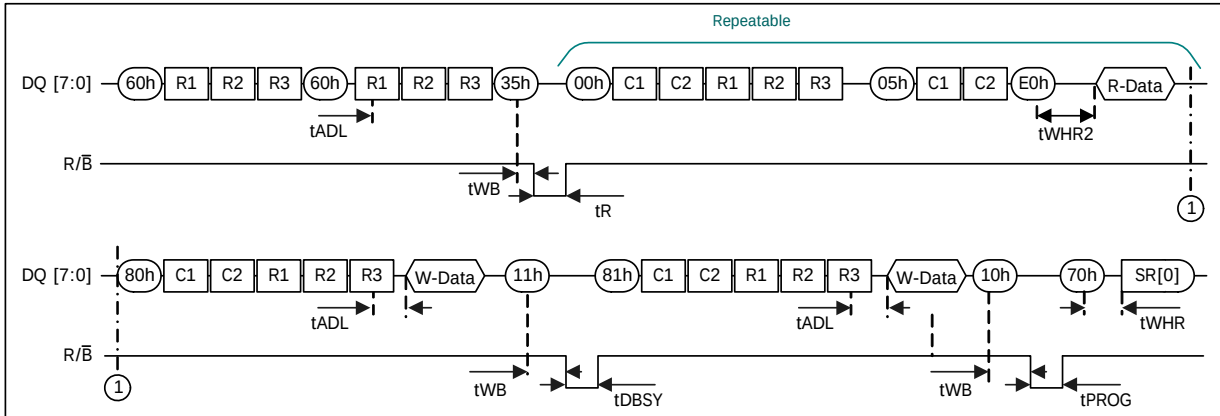


Figure 5-23. Example Timing with Two-plane Copy-Back Program (Secondary)

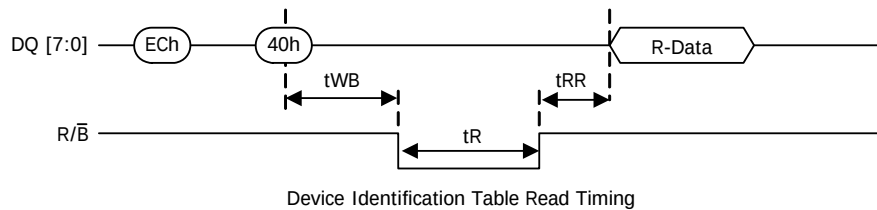


5.2.6. Device Identification Table Read Operation

The device shall return a JEDEC standard formatted parameter page during the data out phase of the READ PARAMETER PAGE command when address 40h is inputted. The READ PARAMETER PAGE command is a ECh value for the command cycle and a 40h value for the address cycle, and the bytes of the parameter page are returned in the data output (DOUT) cycles.

After the command ECh address 40h is received by the NAND device, it will go busy for a period of time (t_R in the figure) after which, the parameter page can be read from the device. The length and contents of the parameter page is to be determined. The timing associated with the bus cycles for the READ PARAMETER PAGE command is defined elsewhere in the JEDEC standard. The READ ID command is used by the controller to identify the device that is attached. This command is used by the controller to gather information about the target flash device. Figure of “*Device Identification Table Read Timing*” defines the behavior and timings.

Figure 5-24. Device Identification Table Read Timing



5.2.7. Device Identification Table Definition

Table for “*Parameter Page Definitions*” defines the parameter page data structure. For parameters that span multiple bytes, the least significant byte of the parameter corresponds to the first byte.

Values are reported in the parameter page in units of bytes when referring to items related to the size of data access (as in an 8-bit data access device). For example, the target will return how many data bytes are in a page. All optional parameters that are not implemented shall be cleared to 00h by the target.

Table 5-14. Parameter Page Definitions

Command	O/M	Description
Revision information and features block		
0-3	M	Parameter page signature Byte 0: "J" (= 4Ah) Byte 1: "E" (= 45h) Byte 2: "S" (= 53h) Byte 3: "D" (= 44h)
4-5	M	Revision number 2-15 Reserved (0) 1 1 = supports revision 1.0 0 Reserved (0)
6-7	M	Features supported 0-15 Reserved (0) <To be defined based on feature discussions.>
8-10	M	Optional commands supported 0-23 Reserved (0) <To be defined based on command set discussions.>
11-31	M	Reserved (0)
Manufacturer information block		
32-43	M	Device manufacturer (12 ASCII characters)
44-63	M	Device model (20 ASCII characters)
64-69	M	JEDEC manufacturer ID
70-71	M	Reserved (0)
72-79		Reserved (0)
Memory organization block		
80-83	M	Number of data bytes per page
84-85	M	Number of spare bytes per page
86-89		Reserved (0)
90-91		Reserved (0)
92-95	M	Number of pages per block
96-99	M	Number of blocks per logical unit (LUN)
100	M	Number of logical units (LUNs)
Memory organization block		

Command	O/M	Description
101	M	Number of address cycles 4-7 Column address cycles 0-3 Row address cycles
102	M	Number of bits per cell
103	M	Number of programs per page
104	M	Multi-plane addressing 4-7 Reserved (0) 0-3 Number of plane address bits
105	M	Multi-plane operation attributes 3-7 Reserved (0) 2 Address restrictions for cache operations 1 1= read cache supported 0 1 = program cache supported
106-143	M	Reserved (0)
Electrical parameters block		
144-145	O	0-15 Reserved (0)
146-147	O	Toggle DDR speed grade 8-15 Reserved (0) 7 1 = supports 5 ns speed grade (~200 MHz) 6 1 = supports 6 ns speed grade (~166 MHz) 5 1 = supports 7.5 ns speed grade (~133 MHz) 4 1 = supports 10 ns speed grade (~100 MHz) 3 1 = supports 12 ns speed grade (~83 MHz) 2 1 = supports 15 ns speed grade (~66 MHz) 1 1 = supports 25 ns speed grade (40 MHz) 0 1 = supports 30 ns speed grade (~33 MHz)
148-149	O	0-15 Reserved (0)
150	O	0-7 Reserved (0)
151	O	Toggle DDR features 0-7 Reserved (0)
152	O	0-7 Reserved (0)
153-154	M	tPROG Maximum page program time (us)
155-156	M	tBERS Maximum block erase time (us)
157-158	M	tR Maximum page read time (us)
159-160	O	tR Maximum multi-plane page read time (us)
161-162	O	tCCS Minimum change column setup time (ns)
163-164		Reserved
165-166		Reserved
167-168	O	Reserved
169	M	Driver strength support 3-7 Reserved (0) 2 1 = supports Overdrive 2 driver strength 1 1 = supports Overdrive 1 driver strength 0 1 = supports driver strength settings

Command	O/M	Description
170-207		Reserved (0)
ECC and endurance block		
208-242		Reserved
243-271		Reserved (0)
272-419		Reserved (0)
Vendor specific block		
420-421	M	Vendor specific Revision number
422		Stack Info.
423-509		Vendor specific
CRC for Parameter Page		
512-1023		Value of bytes 0-511
1024-1535		Value of bytes 0-511
1536+		Additional redundant parameter pages

Byte 0-3: Parameter page signature

This field contains the parameter page signature.

When two or more bytes of the signature are valid, then it denotes that a valid copy of the parameter page is present.

Byte 0 shall be set to 4Ah.

Byte 1 shall be set to 45h.

Byte 2 shall be set to 53h.

Byte 3 shall be set to 44h.

Byte 4-5: Revision number

This field indicates the revisions of the standard that the target complies to.

The target may support multiple revisions of the standard.

This is a bit field where each defined bit corresponds to a particular specification revision that the target may support.

Bit 0 shall be cleared to zero.

Bit 1 when set to one indicates that the target supports revision 1.0.

Bits 2-15 are reserved and shall be cleared to zero.

Byte 6-7: Features supported

This field indicates the optional features that the target supports.

<TBD based on feature discussions.>

Byte 8-10: Optional commands supported

This field indicates the optional commands that the target supports.

<TBD based on command discussions.>

Byte 32-43: Device manufacturer

This field contains the manufacturer of the device.

The content of this field is an ASCII character string of twelve bytes.

The device shall pad the character string with spaces (20h), if necessary, to ensure that the string is the proper length.

There is no standard for how the manufacturer represents their name in the ASCII string.

If the host requires use of a standard manufacturer ID, it should use the JEDEC manufacturer ID.

Byte 44-63: Device model

This field contains the model number of the device.
The content of this field is an ASCII character string of twenty bytes.
The device shall pad the character string with spaces (20h), if necessary, to ensure that the string is the proper length.

Byte 64-69: JEDEC manufacturer ID

This field contains the JEDEC manufacturer ID for the manufacturer of the device.

Byte 70-71: Reserved.

Byte 80-83: Number of data bytes per page

This field contains the number of data bytes per page.
The value reported in this field shall be a power of two.
The minimum value that shall be reported is 512 bytes.

Byte 84-85: Number of spare bytes per page

This field contains the number of spare bytes per page. There are no restrictions on the value.

<TBD: Recommendations should be developed based on ECC strength needed.>

Byte 86-89: Reserved.

Byte 90-91: Reserved.

Byte 92-95: Number of pages per block

This field contains the number of pages per block.

<TBD: Should reference address format.>

Byte 96-99: Number of blocks per logical unit

This field contains the number of blocks per logical unit.
There are no restrictions on this value.

<TBD: Should reference address format.>

Byte 100: Number of logical units (LUNs)

This field indicates the number of logical units the target supports.
Logical unit numbers are sequential, beginning with a LUN address of 0.
This field shall be greater than zero.

<TBD: Should reference address format.>

Byte 101: Number of Address Cycles

This field indicates the number of address cycles used for row and column addresses.
The reported number of address cycles shall be used by the host in operations that require row and/or column addresses (e.g. Page Program).

Bits 0-3 indicate the number of address cycles used for the row address. This field shall be greater than zero.

Bits 4-7 indicate the number of address cycles used for the column address. This field shall be greater than zero.

NOTE :

Throughout this standard examples are shown with 2-byte column addresses and 3-byte row addresses. However, the host is responsible for providing the number of column and row address cycles in each of these sequences based on the values in this field.

Byte 102: Number of bits per cell

This field indicates the number of bits per cell in the Flash array.
This field shall be greater than zero.

Byte 103: Number of programs per page

This field indicates the maximum number of times a portion of a page may be programmed without an erase operation. After the number of programming operations specified have been performed, the host shall issue an erase operation to that block before further program operations to the affected page. This field shall be greater than zero.

Byte 104: Multi-plane addressing

This field describes parameters for multi-plane addressing.
Bits 0-3 indicate the number of bits that are used for plane addresses.
This value shall be greater than 0h when multi-plane operations are supported.
Bits 4-7 are reserved.

Byte 105: Multi-plane operation attributes

This field describes attributes for multi-plane operations.
This byte is mandatory when multi-plane operations are supported as indicated in the Features supported field.

Bit 0 indicates whether program cache is supported with multi-plane programs.
If set to one then program cache is supported for multi-plane program operations.
If cleared to zero then program cache is not supported for multi-plane program operations.
Note that program cache shall not be used with multi-plane copyback program operations.
See bit 2 for restrictions on the plane addresses that may be used.

Bit 1 indicates whether read cache is supported with multi-plane reads.
If set to one then read cache is supported for multi-plane read operations.
If cleared to zero then read cache is not supported for multi-plane read operations.
Note that read cache shall not be used with multi-plane copyback read operations.

Bit 2 indicates whether plane addresses may change during either:
a) a program cache sequence between 15h commands, or b) a read cache sequence between 31h commands.
If set to one and bit 0 is set to one, then the host may change the number and value of plane addresses in the program cache sequence.
If set to one and bit 1 is set to one, then the host may change the number and value of plane addresses in the read cache sequence.
If cleared to zero and bit 0 is set to one, then for each program cache operation the plane addresses and number of plane addresses issued to the LUN shall be the same.
If cleared to zero and bit 1 is set to one, then for each read cache operation the plane addresses and number of plane addresses issued to the LUN shall be the same.
Bits 3-7 are reserved.

Byte 144-145: Reserved.

Byte 146-147: Toggle DDR speed grade

This field indicates the Toggle DDR speed grades supported. The target shall support an inclusive range of speed grades.

Bit 0 when set to one indicates that the target supports the 30 ns speed grade (~33 MHz).
Bit 1 when set to one indicates that the target supports the 25 ns speed grade (40 MHz).
Bit 2 when set to one indicates that the target supports the 15 ns speed grade (~66 MHz).
Bit 3 when set to one indicates that the target supports the 12 ns speed grade (~83 MHz).
Bit 4 when set to one indicates that the target supports the 10 ns speed grade (~100 MHz).
Bit 5 when set to one indicates that the target supports the 7.5 ns speed grade (~133 MHz).
Bit 6 when set to one indicates that the target supports the 6 ns speed grade (~166 MHz).
Bit 7 when set to one indicates that the target supports the 5 ns speed grade (~200 MHz).
Bits 8-15 are reserved and shall be cleared to zero.

Byte 148-149: Reserved

Byte 150: Reserved

Byte 151: Toggle DDR features

This field describes features and attributes for Toggle DDR operation.
This byte is mandatory when the Toggle DDR data interface is supported.
Bits 0-7 are reserved.

Byte 152: Reserved

Byte 153-154: Maximum page program time

This field indicates the maximum page program time (tPROG) in microseconds.

Byte 155-156: Maximum block erase time

This field indicates the maximum block erase time (tBERS) in microseconds.

Byte 157-158: Maximum page read time

This field indicates the maximum page read time (tR) in microseconds.

Byte 159-160: Maximum multi-plane page read time

This field indicates the maximum page read time (tR) for multi-plane page reads in microseconds.
Multi-plane page read times may be longer than single page read times.
This field shall be supported if the target supports multi-plane reads as indicated in the Features supported field.

Byte 161-162: Minimum change column setup time.

This field indicates the minimum change column setup time (tCCS) in nanoseconds.
This parameter is used for the asynchronous SDR and synchronous DDR data interfaces.
After issuing a Change Read Column command, the host shall not read data until a minimum of tCCS time has elapsed.
After issuing a Change Write Column command including all column address cycles, the host shall not write data until a minimum of tCCS time has elapsed.
The value of tCCS shall always be longer than or equal to tWHR and tADL when the Toggle DDR or synchronous DDR data interface is supported.

Byte 163-164: Reserved.

Byte 165-166: Reserved.

Byte 167-168: Reserved.

Byte 169: Driver strength support

This field describes if the target supports configurable driver strengths and its associated features.

Bit 0 when set to one indicates that the target supports configurable driver strength settings as defined in Table TBD.
If this bit is set to one, then the device shall support both the Nominal and Underdrive settings.
If this bit is set to one, then the device shall power-on with a driver strength at the Nominal value defined in Table TBD.
If this bit is cleared to zero, then the driver strength at power-on is undefined.
This bit shall be set to one for devices that support the synchronous DDR or Toggle DDR data interface.

Bit 1 when set to one indicates that the target supports the Overdrive 1 setting in Table TBD for use in the I/O Driver Strength setting.
This bit shall be set to one for devices that support the synchronous DDR or Toggle DDR data interface.

Bit 2 when set to one indicates that the target supports the Overdrive 2 setting in Table TBD for use in the I/O Driver Strength setting.
This bit shall be set to one for devices that support the synchronous DDR or Toggle DDR data interface.
Bits 3-7 are reserved.

Byte 208-242: Reserved.

Byte 420-421: Vendor specific Revision number

This field indicates a vendor specific revision number.

This field should be used by vendors to indicate the supported layout for the vendor specific parameter page area and the vendor specific feature addresses. The format of this field is vendor specific.

Byte 422: Stack Info.

Number of silicon stack in package.

Byte 423-509: Vendor specific

This field is reserved for vendor specific use.

Byte 510-511: Integrity CRC

The Integrity CRC (Cyclic Redundancy Check) field is used to verify that the contents of the parameter page were transferred correctly to the host.

The CRC of the parameter page is a word (16-bit) field. The CRC calculation covers all of data between byte 0 and byte 509 of the parameter page inclusive.

The CRC shall be calculated on byte (8-bit) quantities starting with byte 0 in the parameter page.

The bits in the 8-bit quantity are processed from the most significant bit (bit 7) to the least significant bit (bit 0).

The CRC shall be calculated using the following 16-bit generator polynomial:

$$G(X) = X^{16} + X^{15} + X^2 + 1$$

This polynomial in hex may be represented as 8005h.

The CRC value shall be initialized with a value of 4F4Eh before the calculation begins. There is no XOR applied to the final CRC value after it is calculated. There is no reversal of the data bytes or the CRC calculated value.

Byte 512-1023: Redundant Parameter Page 1

This field shall contain the values of bytes 0-511 of the parameter page. Byte 512 is the value of byte 0.

The redundant parameter page is used when the integrity CRC indicates that there was an error in bytes 0-511.

The redundant parameter page shall be stored in non-volatile media; the target shall not create these bytes by retransmitting the first 512 bytes.

Byte 1024-1535: Redundant Parameter Page 2

This field shall contain the values of bytes 0-511 of the parameter page. Byte 1024 is the value of byte 0.

The redundant parameter page is used when the integrity CRC indicates that there was an error in bytes 0-511 and in the first redundant parameter page.

The redundant parameter page shall be stored in non-volatile media; the target shall not create these bytes by retransmitting the first 512 bytes.

Byte 1536+: Additional Redundant Parameter Pages

Bytes at offset 1536 and above may contain additional redundant copies of the parameter page.

There is no limit to the number of redundant parameter pages that the target may provide.

The target may provide additional copies to guard against the case where all three mandatory copies have invalid CRC checks.

The host should determine whether an additional parameter page is present by checking the first Dword.

If at least two out of four bytes match the parameter page signature, then an additional parameter page is present.

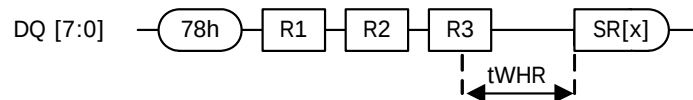
5.2.8. Read Status Enhanced

Read Status Enhanced function is used to check status of selected plane of a LUN. Thus, the function requires row address setting steps before reading status value. Table of “*Read Status Enhanced Definition*” defines status values of each operation and Figure of “*Read Status Sequence*” defines Read Status Enhanced behavior and timings.

Table 5-15. Read Status Enhanced Definition

	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7
Definition of value	Pass : “0” Fail : “1”	Pass : “0” Fail : “1”	Reserved	Reserved	Reserved	Busy : “0” Ready : “1”	Busy : “0” Ready : “1”	Protected : “0” Not Protected : “1”
Block Erase	Pass/Fail	Not Use	Not Use	Not Use	Not Use	Not Use	Busy/Ready	Write Protect
Page Program	Pass/Fail	Not Use	Not Use	Not Use	Not Use	Not Use	Busy/Ready	Write Protect
Read	Not Use	Not Use	Not Use	Not Use	Not Use	Not Use	Busy/Ready	Write Protect

Figure 5-25 Read Status Sequence



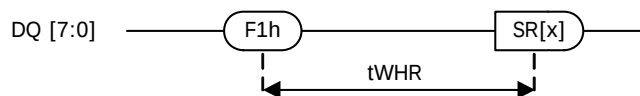
5.2.11. Read LUN #0 Status Operation

Read LUN#0 Status provides status value of LUN#0 without address setting. The function retrieves plane0 and plane1 status only. Table for “*Read LUN#0 Status Definition*” defines the status values and Figure of “*Read LUN#0 Status Sequence*” defines Read LUN#0 Status behavior and timings.

Table 5-16. Read LUN#0 Status Definition

	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7
Definition of value	Pass : “0” Fail : “1”	Pass : “0” Fail : “1”	Pass : “0” Fail : “1”	Pass : “0” Fail : “1”	Pass : “0” Fail : “1”	Busy : “0” Ready : “1”	Busy : “0” Ready : “1”	Protected : “0” Not Protected : “1”
Block Erase	Pass/Fail for LUN#0	Pass/Fail for Plane#0	Pass/Fail for Plane#1	Not Use	Not Use	Not Use	Busy/Ready	Write Protect
Page Program	Pass/Fail for LUN#0	Pass/Fail for Plane#0	Pass/Fail for Plane#1	Not Use	Not Use	Not Use	Busy/Ready	Write Protect
Read	Not Use	Not Use	Not Use	Not Use	Not Use	Not Use	Busy/Ready	Write Protect

Figure 5-26. Read LUN#0 Status Sequence



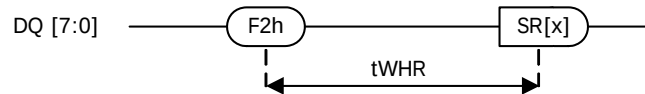
5.2.12. Read LUN #1 Status Operation

Read LUN#1 Status provides status value of LUN#1 without address setting. The function retrieves plane0 and plane1 status only. Table for “*Read LUN#1 Status Definition*” defines the status values and Figure of “*Read LUN#1 Status Sequence*” defines Read LUN#1 Status behavior and timings.

Table 5-17. Read LUN#1 Status Definition

	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7
Definition of value	Pass : "0" Fail : "1"	Pass : "0" Fail : "1"	Pass : "0" Fail : "1"	Pass : "0" Fail : "1"	Pass : "0" Fail : "1"	Busy : "0" Ready : "1"	Busy : "0" Ready : "1"	Protected : "0" Not Protected : "1"
Block Erase	Pass/Fail for LUN#1	Pass/Fail for Plane#0	Pass/Fail for Plane#1	Not Use	Not Use	Not Use	Busy/Ready	Write Protect
Page Program	Pass/Fail for LUN#1	Pass/Fail for Plane#0	Pass/Fail for Plane#1	Not Use	Not Use	Not Use	Busy/Ready	Write Protect
Read	Not Use	Not Use	Not Use	Not Use	Not Use	Not Use	Busy/Ready	Write Protect

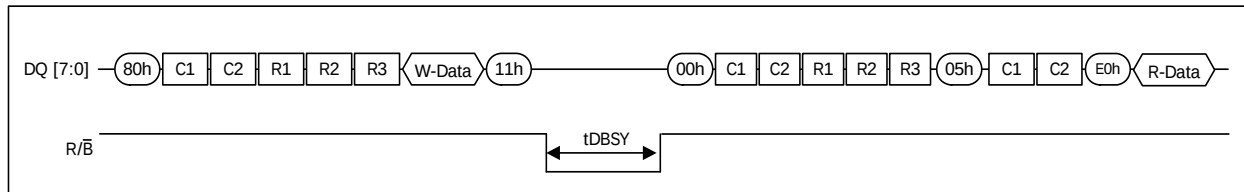
Figure 5-27. Read LUN#1 Sataus Sequence



5.2.13. Register Read Out Mode 1

At program operation, loaded data to the register can be read out before program confirm command(10h). The sequence is as follows. To continue program operation after reading data, Copy-back Program operation (i.e. 85h- Address(5cycle) - Data - 10h) is required.

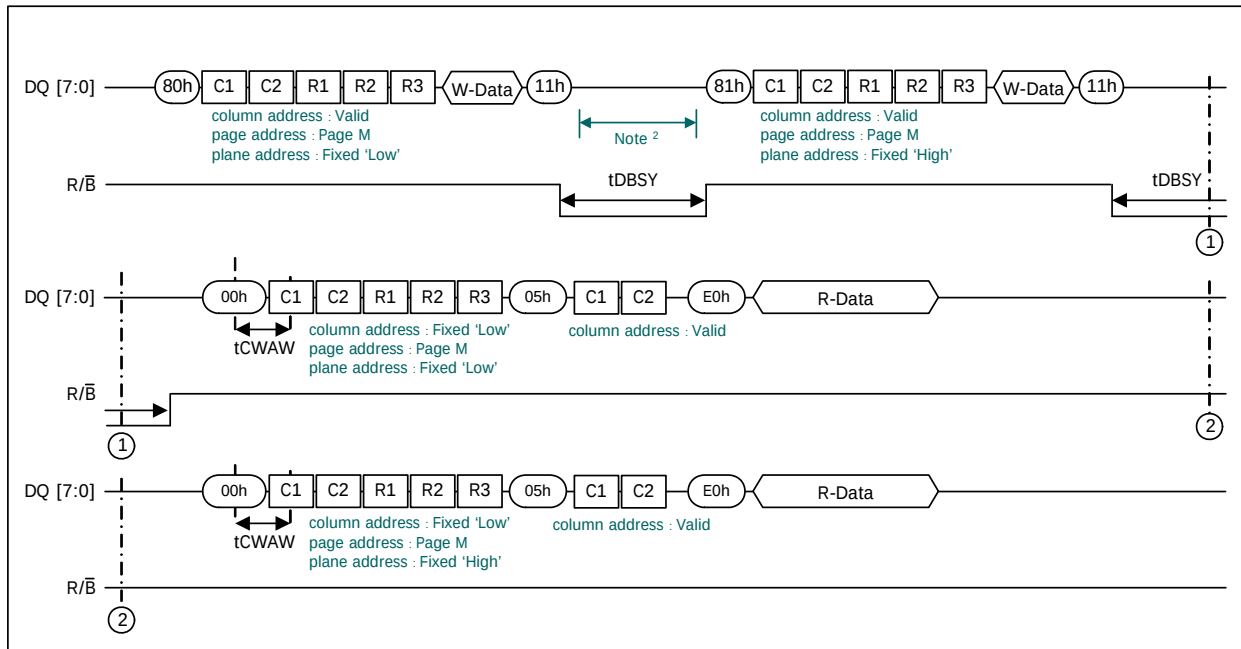
Figure 5-28. Register Read Out



5.2.14. Two-Plane Register Read Out Mode 1

At Two-Plane program operation, loaded data to the register can be read out before program confirm command(10h). The sequence is as follows. To continue Two-Plane program operation after reading data, Two-Plane Random Data Copy-back Program sequence is required.

Figure 5-29. Two-Plane Register Read Out Mode 1



NOTE :

1) Any command between 11h and 81h is prohibited except 70h/78h/F1h and FFh.

5.3. Interleaving Operation

When multiple LUNs share a common CE, it provides interleaving operation between LUNs.

At first, the host issues a operation command to one of the LSB chips, say (LUN #0). Due to DDP device goes into busy state. During this time, MSB chip (LUN #1) is in ready state. So it can execute the operation command issued by the host.

After the execution of operation by LSB chip (LUN #0), it can execute another operation regardless of MSB chip (LUN #1). Before that the host needs to check the status of LSB chip (LUN #0) by issuing 78h/F1h command. Only when the status of LSB chip (LUN #0) becomes ready status, host can issue another operation command. If LSB chip (LUN #0) is in busy state, the host has to wait for LSB chip (LUN #0) to get into ready state.

Similarly, MSB chip (LUN #1) can execute another operation after the completion of the previous operation. The host can monitor the status of MSB chip (LUN #1) by issuing 78h/F2h command. When MSB chip (LUN #1) goes ready state, host can issue another operation command to MSB chip (LUN #1).

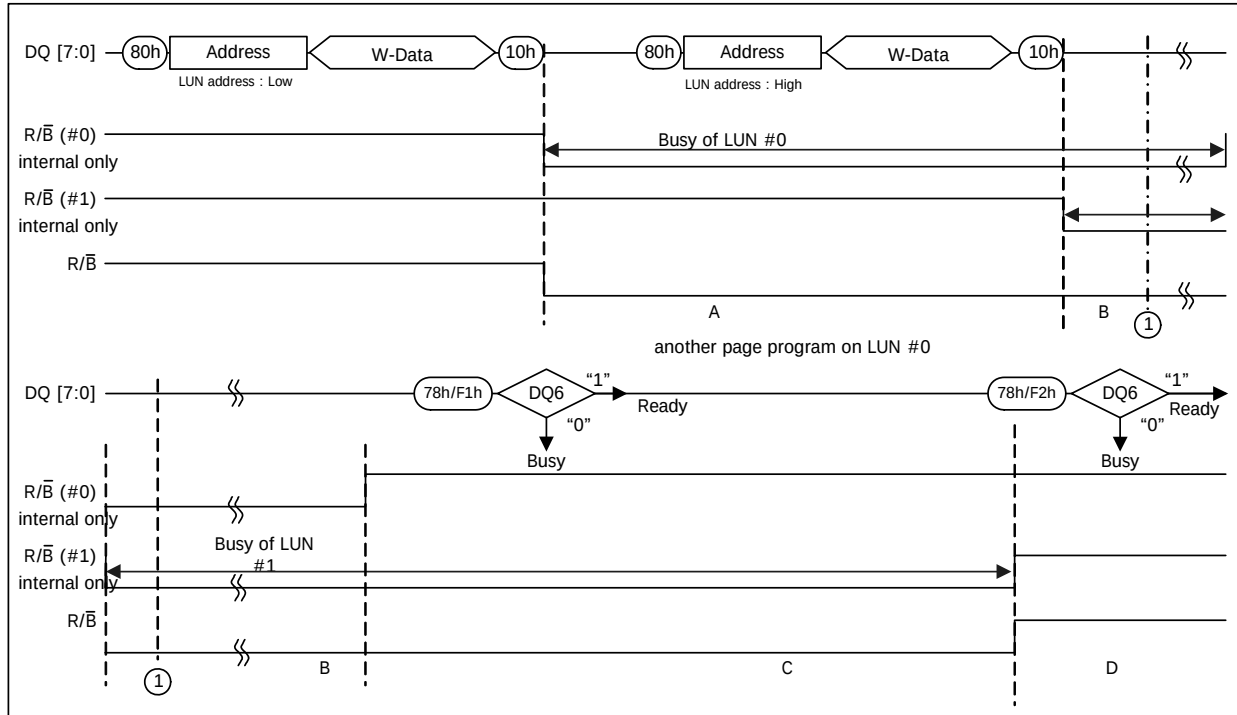
This interleaving operation helps the system improve the system throughput.

NOTE :

During interleave operations, 70h command is prohibited and 78h command is used to check the status.

5.3.1. Interleaving Page Program

Figure 5-30. Interleaving Page Program

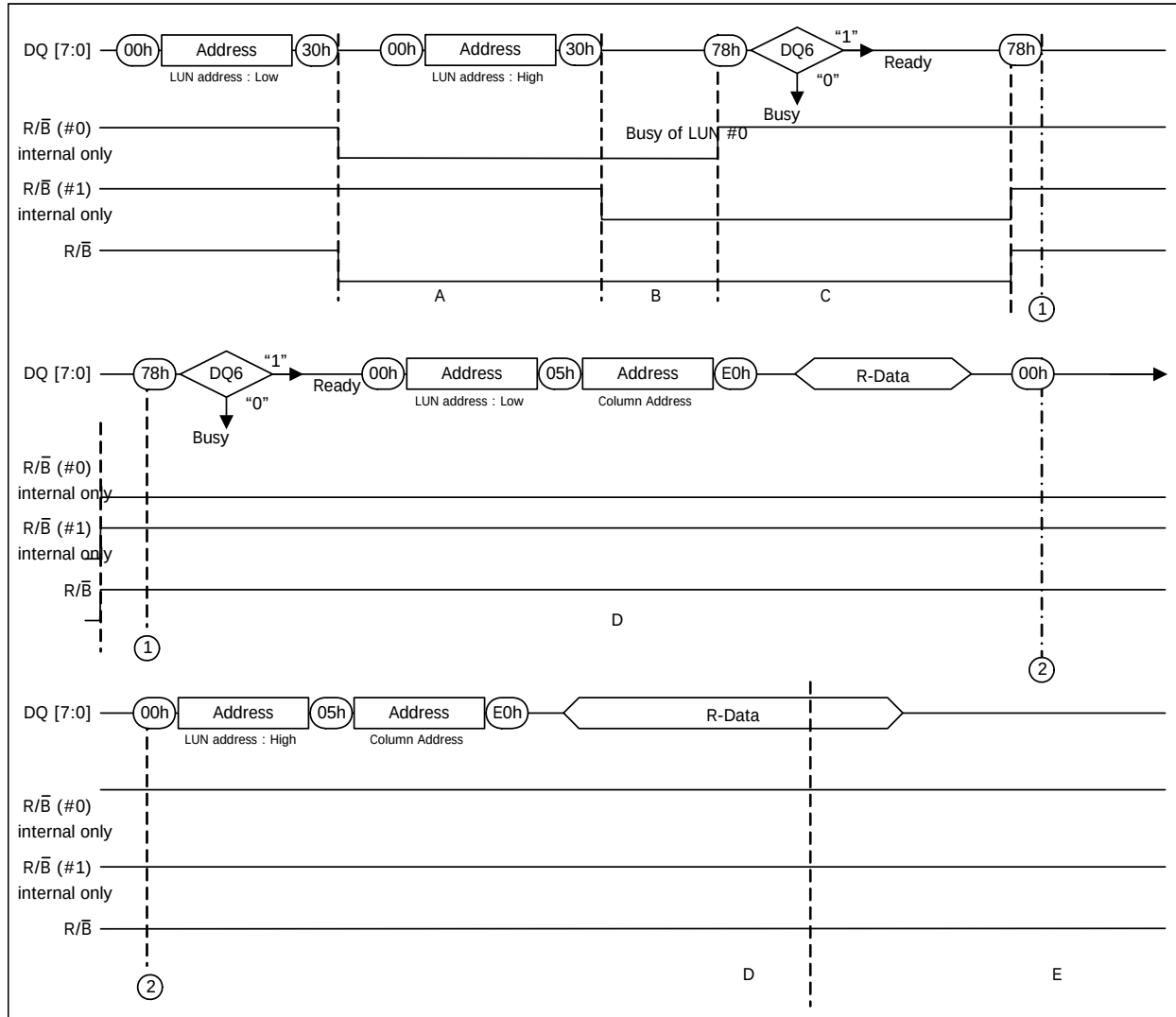


- State A :** LUN #0 is executing page program operation and LUN #1 is in ready state. So the host can issue page program command to LUN #1.
- State B :** Both LUN #0 and LUN #1 are executing page program operation.
- State C :** Page program on LUN #0 is completed, but page program on LUN #1 is still ongoing. And the system should issue 78h/F1h command to detect the status of LUN #0. If LUN #0 is ready, status I/O6 is "1" and the system can issue another page program command to LUN #0.
- State D :** Both of LUN #0 and LUN #1 are ready.

Depending on the above process, the system can operate page program on LUN #0 and LUN #1 alternately.

5.3.2. Interleaving Page Read

Figure 5-31. Interleaving Page Read



State A : LUN #0 is executing page read operation, and LUN #1 is in ready state.

So the host can issue page read command to LUN #1.

State B : Both LUN #0 and LUN #1 are executing page read operation.

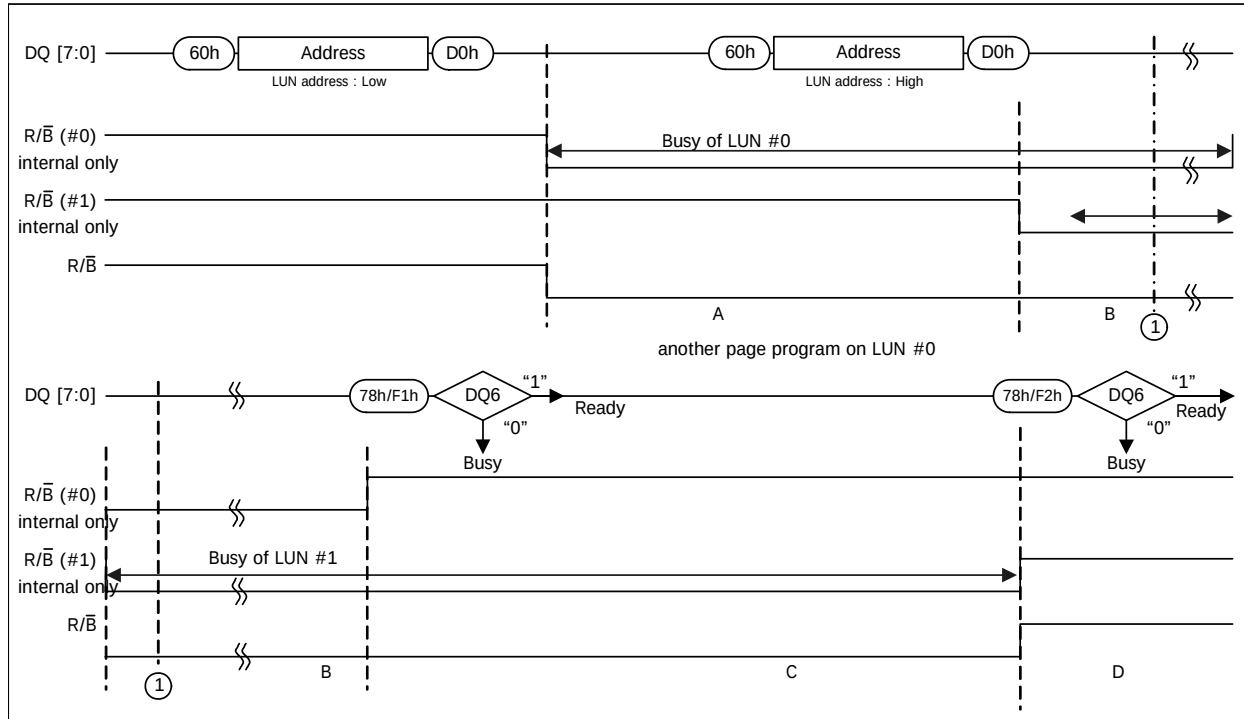
State C : Page read on LUN #0 is completed and LUN #1 is still executing page read operation.

State D : Before the host read the data, the host should check the Ready/Busy status for both LUNs by 78h commands.

State E : LUN #0 and LUN #1 are ready.

5.3.3. Interleaving Block Erase

Figure 5-32. Interleaving Block Erase



State A : LUN #0 is executing block erase operation, and LUN #1 is in ready state. So the host can issue block erase command to LUN #1.

State B : Both LUN #0 and LUN #1 are executing block erase operation.

State C : Block erase on LUN #0 is terminated, but block erase on LUN #1 is still operating. And the system should issue F1h command to detect the status of LUN #0.

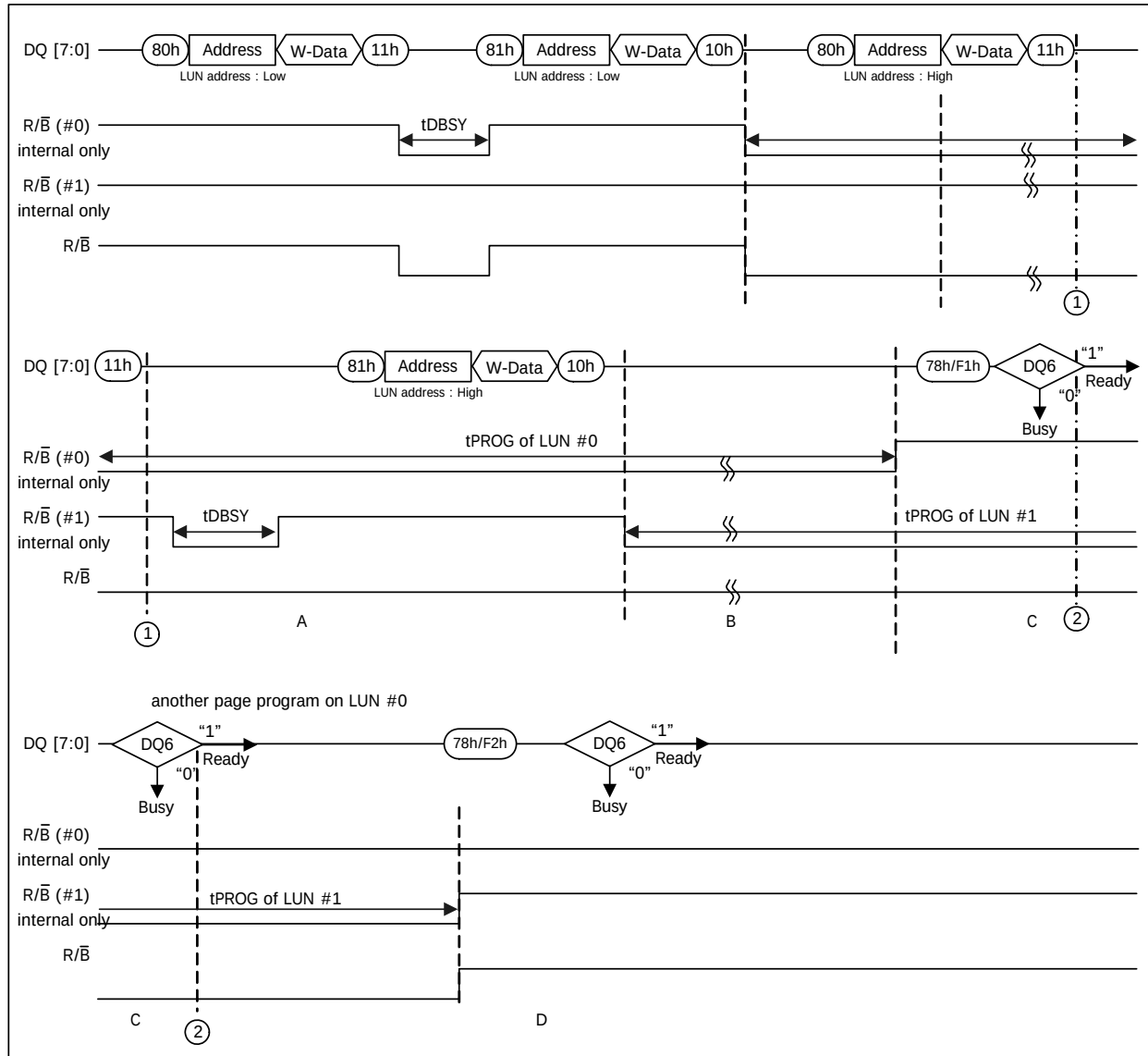
If LUN #0 is ready, status I/O6 is "1" and the system can issue another block erase command to LUN #0.

State D : LUN #0 and LUN #1 are ready.

Depending on the above process, the system can operate block erase on LUN #0 and LUN #1 alternately.

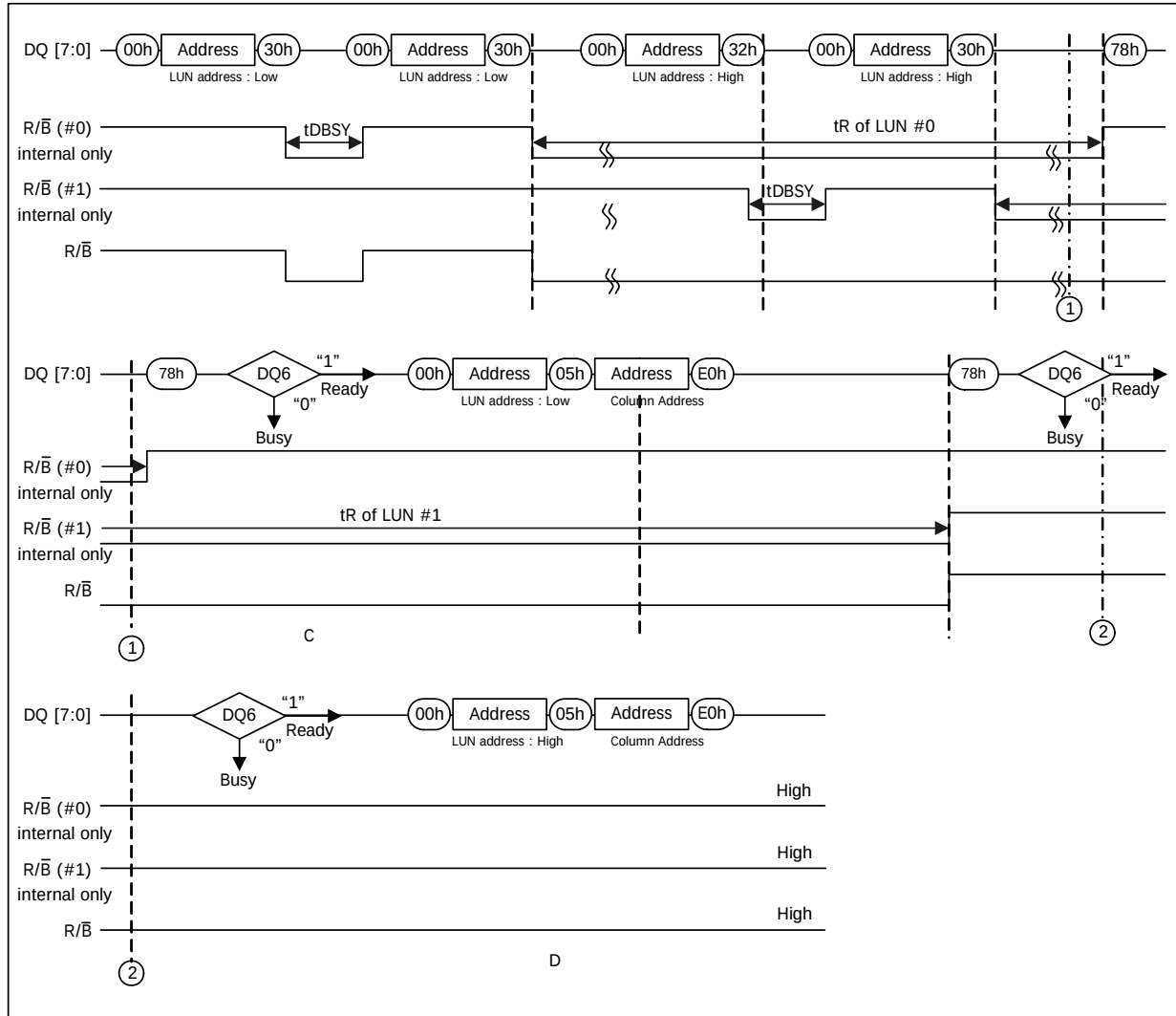
5.3.4. Interleaving Two-Plane Page Program

Figure 5-33. Interleaving Two-Plane Page Program



5.3.5. Interleaving Two-Plane page Read

Figure 5-34. Interleaving Two-Plane page Read



State A : LUN #0 is executing Multi-plane page read operation, and LUN #1 is in ready state.

So the host can issue Multi-plane page read command to LUN #1.

State B : Both LUN #0 and LUN #1 are executing Multi-plane page read operation.

State C : Multi-plane page read on LUN #0 is completed and LUN #0 is ready for the next operation.

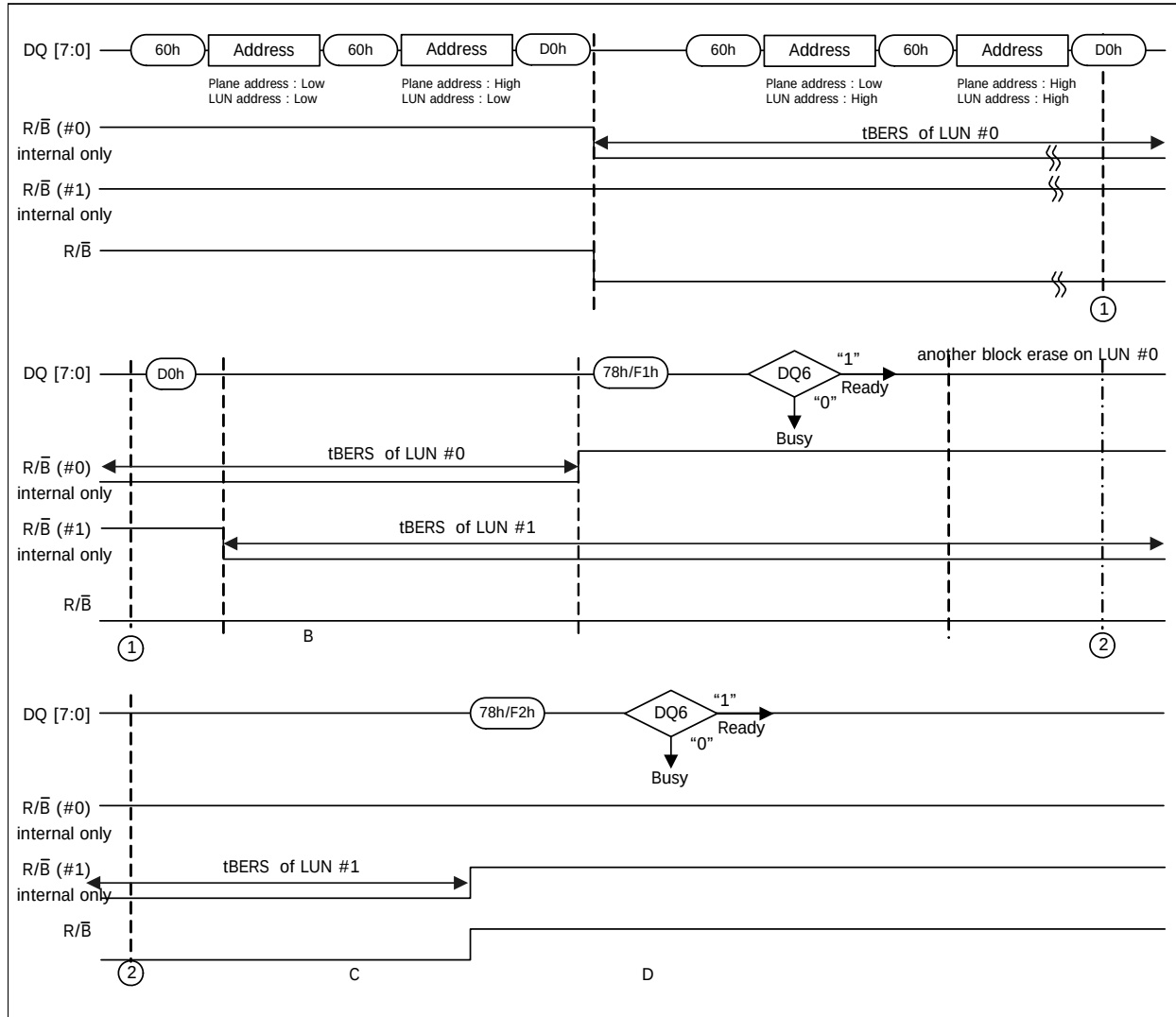
LUN #1 is still executing Multi-plane page read operation.

State D : Both LUN #0 and LUN #1 are ready.

Depending on the above process, the system can operate two-plane page read on LUN #0 and LUN #1 alternately.

5.3.6. Interleaving Multi-Plane Block Erase

Figure 5-35. Interleaving Multi-Plane Block Erase



State A : LUN #0 is executing Multi-plane block erase operation, and LUN #1 is in ready state.

So the host can issue Multi-plane block erase command to LUN #1.

State B : Both LUN #0 and LUN #1 are executing Multi-plane block erase operation.

State C : Multi-plane block erase on LUN #0 is completed and LUN #0 is ready for the next operation.

LUN #1 is still executing Multi-plane block erase operation.

State D : Both LUN #0 and LUN #1 are ready.

NOTE :

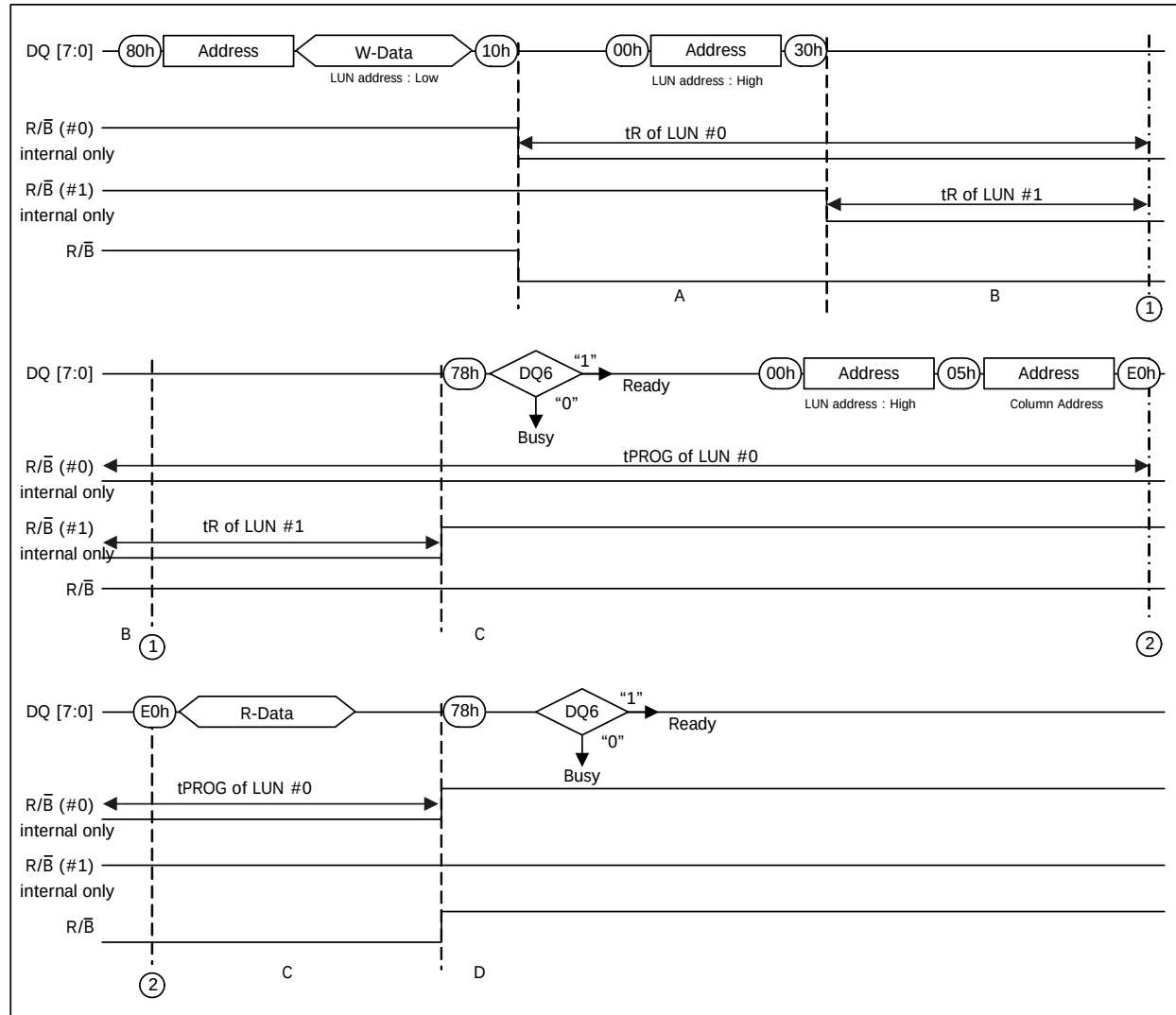
*F1h command is required to check the status of LUN #0 to issue the next block erase command to LUN #0.

F2h command is required to check the status of LUN #1 to issue the next block erase command to LUN #1.

According to the above process, the system can operate two-plane block erase on LUN #0 and LUN #1 alternately.

5.3.7. Interleaving Read to Page Program

Figure 5-36. Interleaving Read to Page Program



State A : LUN #0 is executing page program operation, and LUN #1 is in ready state. So the host can issue read command to LUN #1.

State B : Both LUN #0 is executing page program operation and LUN #1 is executing read operation.

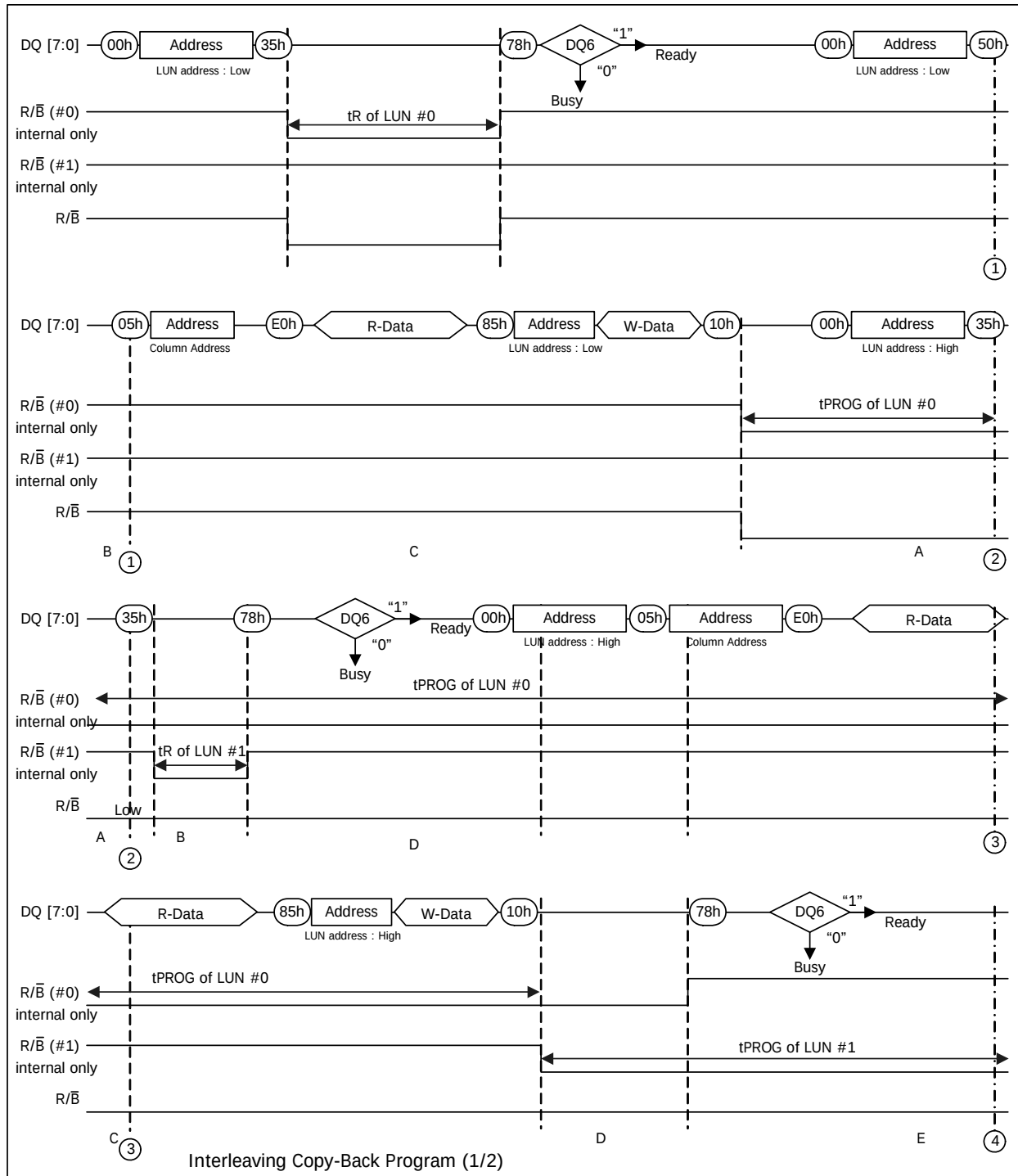
State C : Read operation on LUN #1 is completed and LUN #1 is ready for the next operation. LUN #0 is still executing page program operation.

State D : Both LUN #0 and LUN #1 are ready.

As the above process, the system can operate Interleave read to page program on LUN #0 and LUN #1 alternatively.

5.3.8. Interleaving Copy-Back Program (1/2)

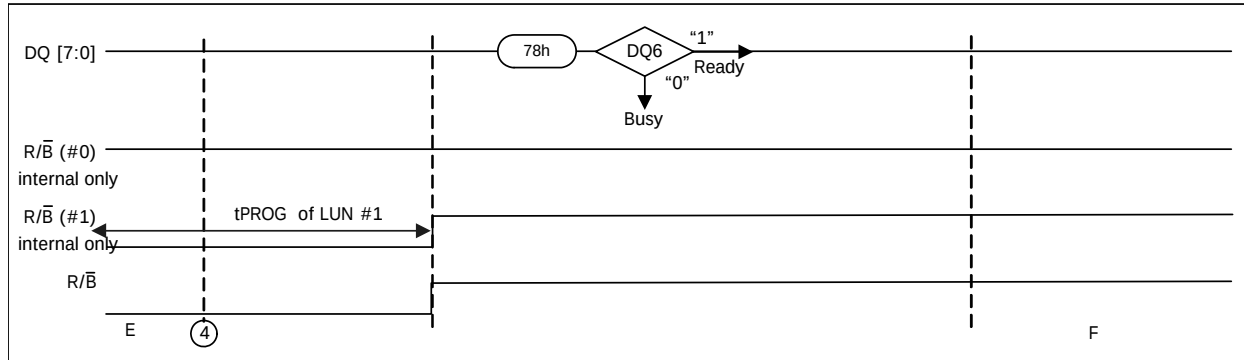
Figure 5-37. Interleaving Copy-Back Program (1/2)



Interleaving Copy-Back Program (1/2)

5.3.9. Interleaving Copy-Back Program (2/2)

Figure 5-38. Interleaving Copy-Back Program (2/2)



State A : LUN #0 is executing copy-back program operation, and LUN #1 is in ready state.

So the host can issue read for copy-back command to LUN #1.

State B : LUN #0 is executing copy-back program operation and LUN #1 is executing read for copy-back operation.

State C : Read for copy-back operation on LUN #1 is completed and LUN #1 is ready for the next operation.

LUN #0 is still executing copy-back program operation.

State D : Both LUN #0 and LUN #1 are executing copy-back program operation.

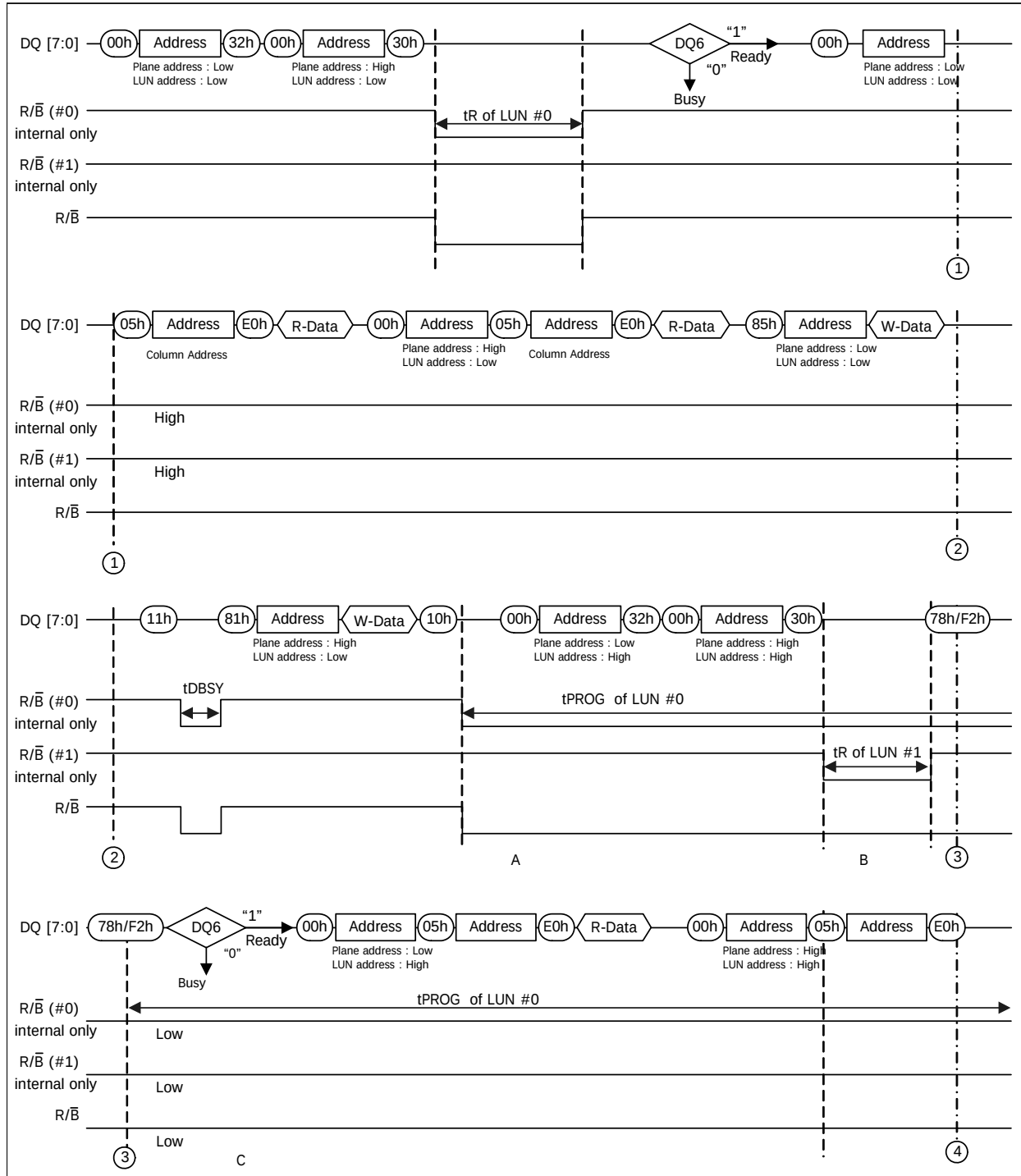
State E : LUN #1 is still executing a copy-back program operation, and LUN #0 is in ready for the next operation.

State F : Both LUN #0 and LUN #1 are ready.

Depending on the above process, the system can operate Interleave copy-back program on LUN #0 and LUN #1 alternatively.

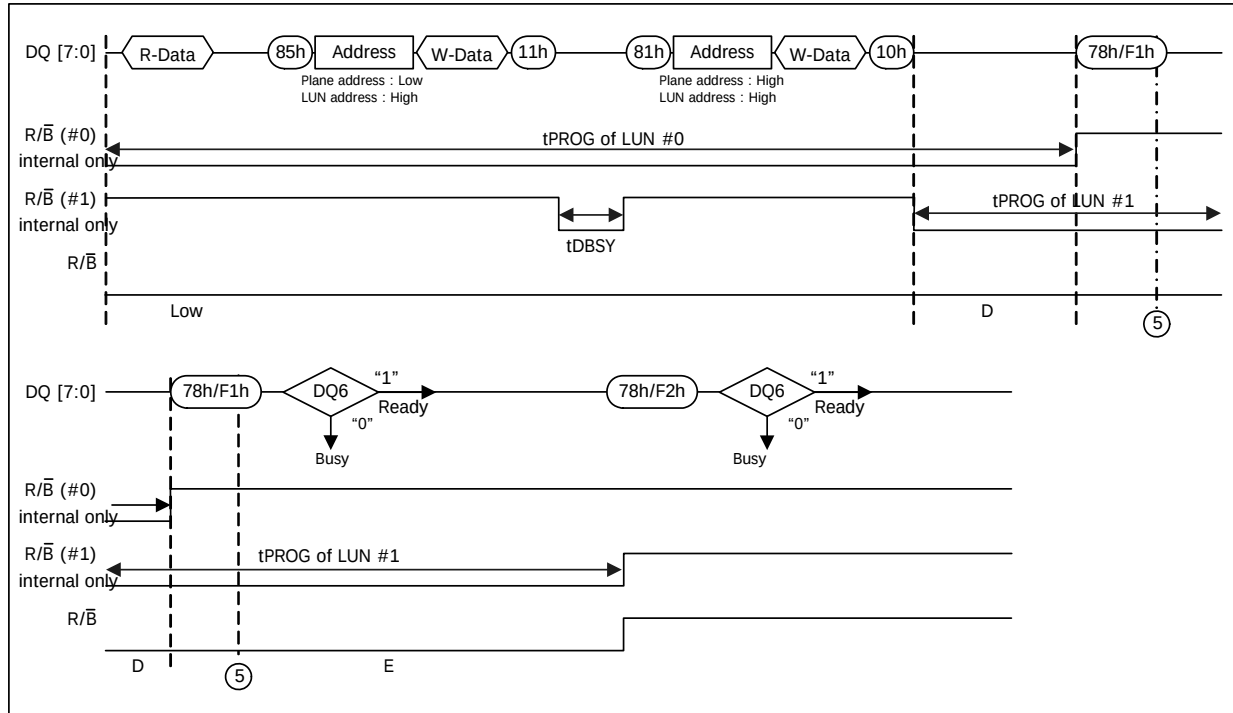
5.3.10. Interleaving Two-Plane Copy Back Program(1/2)

Figure 5-39. Interleaving Two-Plane Copy Back Program(1/2)



5.3.11. Interleaving Two-Plane Copy Back Program(2/2)

Figure 5-40. Interleaving Two-Plane Copy Back Program(2/2)



State A : LUN #0 is executing Multi-plane copy-back program operation, and LUN #1 is in ready state.

So the host can issue Multi-plane read for copy-back command to LUN #1.

State B : LUN #0 is executing Multi-plane copy-back program operation and LUN #1 is executing Multi-plane read for copy-back operation.

State C : Multi-plane read for copy-back operation on LUN #1 is completed and LUN #1 is ready for the next operation.
LUN #0 is still executing Multi-plane copy-back program operation.

State D : Both LUN #0 and LUN #1 are executing Multi-plane copy-back program operation.

State E : LUN #1 is still executing a Multi-plane copy-back program operation, and LUN #0 is in ready for the next operation.

State F : Both LUN #0 and LUN #1 are ready.

NOTE :

*F1h command is required to check the status of LUN #0 to issue the next command to LUN #0.

F2h command is required to check the status of LUN #1 to issue the next command to LUN #1.

Depending on the above process, the system can operate Interleave Multi-plane copy-back program on LUN #0 and LUN #1 alternatively.

5.4. Ready/Busy

The device has a R/B output that provides a hardware method of indicating the completion of a page program, erase and random read completion. The R/B pin is normally high but transitions to low after program or erase command is written to the command register or random read is started after address loading. It returns to high when the internal controller has finished the operation. The pin is an open-drain driver thereby allowing two or more R/B outputs to be Or-tied. Because pull-up resistor value is related to $t_r(R/B)$ and current drain during busy(i_{busy}), an appropriate value can be obtained with the following reference chart. Its value can be determined by the following guidance.

Figure 5-41. Rp vs tr ,tf & Rp vs ibusy

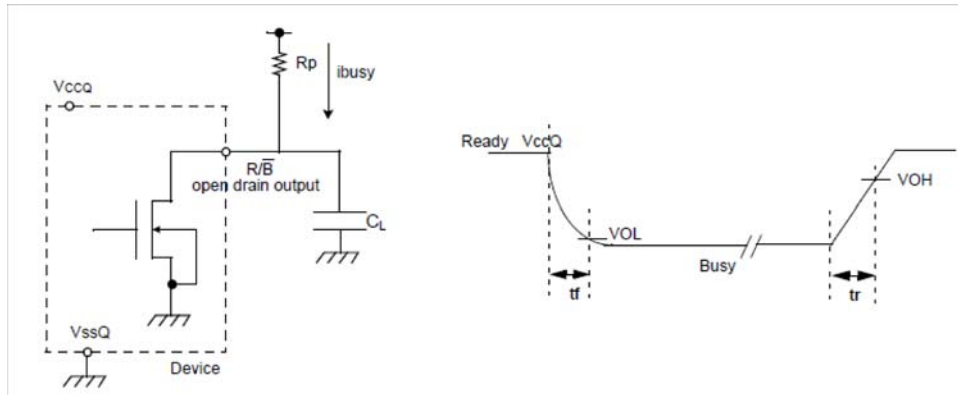
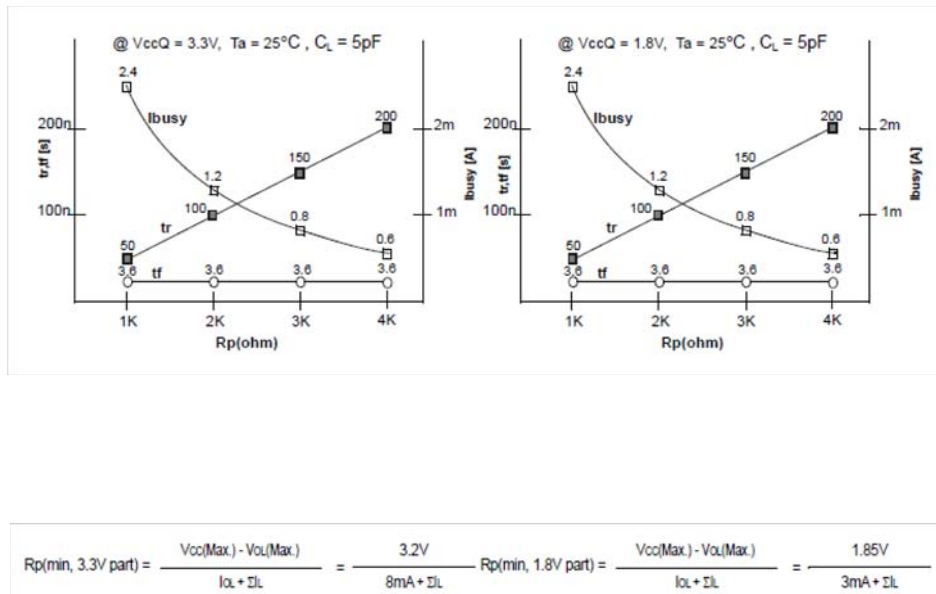


Figure 5-42 Rp value guidance



where I_L is the sum of the input currents of all devices tied to the $\overline{R/B}$ pin.
 $R_{p(\max)}$ is determined by maximum permissible limit of t_r

128Gb Based NAND Flash Addendum

1. Command Description

1.1. Set Parameter

1.1.1. Command Sequence

Command Cycle	N x Parameter Input Cycle	Command Cycle	Required Time before Next Operation
36h	N x (address cycle + data input cycle)	16h	tSPARAM

1.1.2. Timing Diagram

Figure 1. Set Parameter for SDR

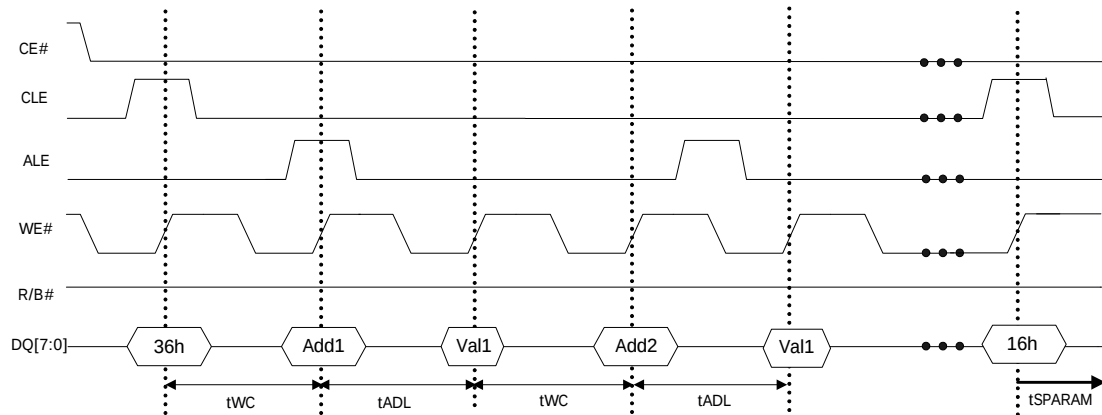
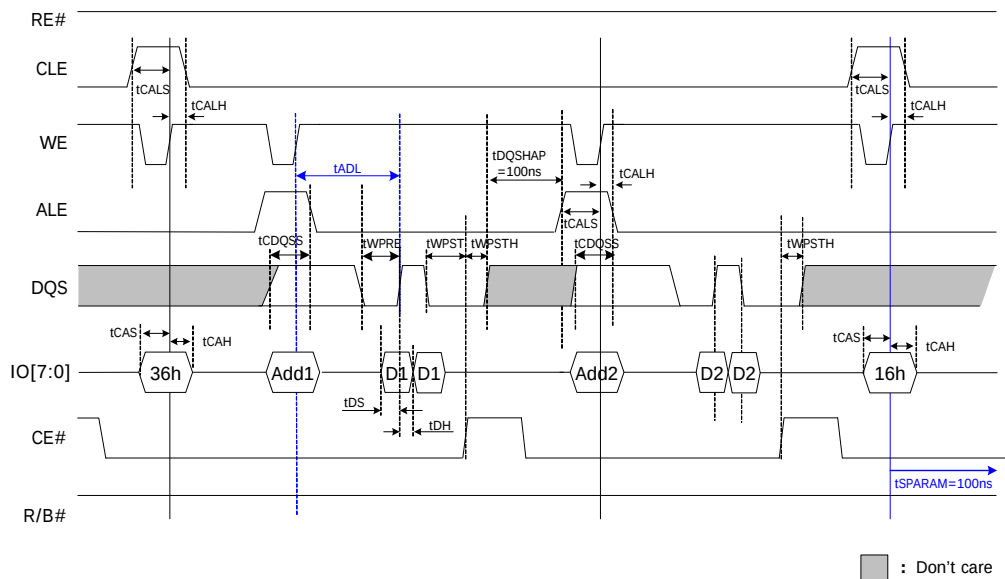


Figure 2. Set Parameter for NV-DDR2



1.1.3. Rules and Limitations

- In case the same parameter is given more than once, only the last update is valid.
- (Backward Compatibility) It is possible to insert a dummy address latch cycle after the Set Parameter setup command (36h).

1.2. Get Parameter

1.2.1. Command Sequence

Command Cycle	Required Time before First Parameter Output Cycle	N x Parameter Output Cycle
37h	tRPARAM	N x (address cycle + data output cycle)

Figure 3. Get Parameter for SDR

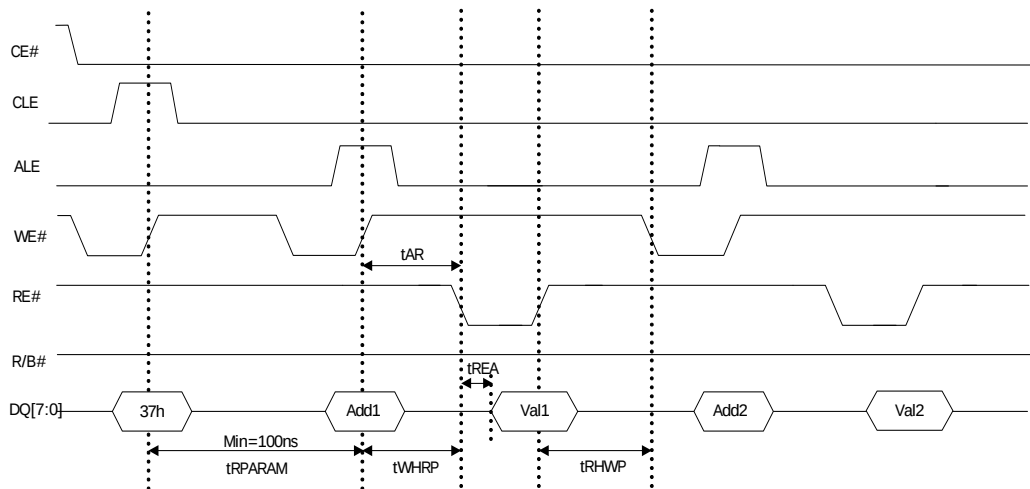
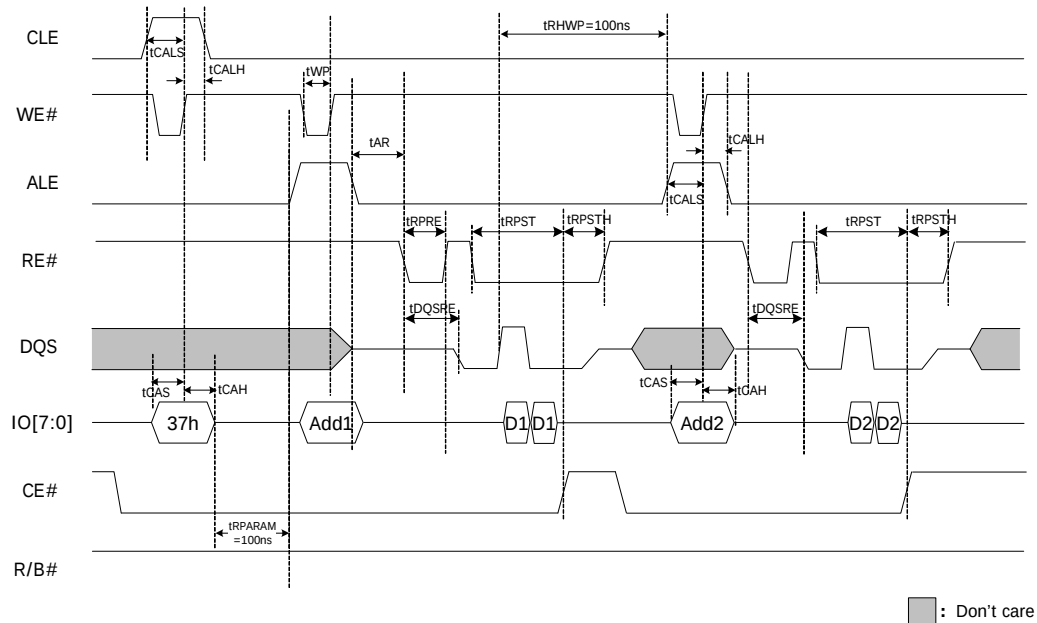


Figure 4. Get Parameter for NV-DDR2



1.2.3. Rules and Limitations

- When get parameter command is performed immediately after set parameter command, the parameters which are represented by shadow register are invalid. The parameters are valid after running internal algorithm.
- When the device is in internal busy state, the device is not allowed to execute get parameter command.